9 General Aptitude: Quantitative Aptitude (494)

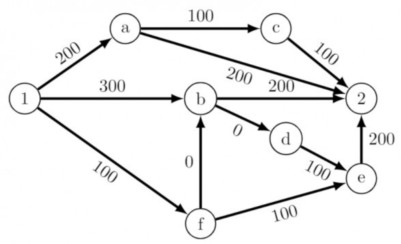
Syllabus: Numerical computation, Numerical estimation, Numerical reasoning and data interpretation

Mark Distribution in Previous GATE

Year	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum	Average	Maximum
1 Mark Count	1	2	4	3	1	2	1	2	1	2	4
2 Marks Count	3	3	2	2	2	2	3	2	2	2.38	3
Total Marks	7	8	8	7	5	6	7	6	**5**	**6.75**	**8**

9.0.1 GATE CSE 2020 | Question: GA-5

There are multiple routes to reach from node 1 to node 2, as shown in the network.



The cost of travel on an edge between two nodes is given in rupees. Nodes 'a', 'b', 'c', 'd', 'e', and 'f' are toll booths. The toll price at toll booths marked 'a' and 'e' is Rs. 200, and is Rs. 100 for the other toll booths. Which is the cheapest route from node 1 to node 2?

A. $1-a-c-2$ B. $1-f-b-2$ C. $1-b-2$ D. $1-f-e-2$

gatecse-2020 quantitative-aptitude graph-theory one-mark

Answer key☟

9.0.2 GATE CSE 2025 | Set 2 | GA Question: 3

If $P\,e^{-x} = Q\,e^{-x}$ for all real values of x, which one of the following statements is true?

A. $P = Q = 0$ B. $P = Q = 1$ C. $P = 1; Q = -1$ D. $\frac{P}{Q} = 0$

gatecse2025-set2 quantitative-aptitude mathematical-logic one-mark

Answer key☟

9.1 Absolute Value (7)

9.1.1 Absolute Value: GATE CSE 2014 Set 2 | Question: GA-8

If x is real and $|x^2 - 2x + 3| = 11$, then possible values of $|-x^3 + x^2 - x|$ include

A. $2, 4$ B. $2, 14$ C. $4, 52$ D. $14, 52$

gatecse-2014-set2 quantitative-aptitude normal absolute-value

Answer key☟

9.1.2 Absolute Value: GATE CSE 2017 Set 1 | Question: GA-8

The expression $\frac{(x+y)-|x-y|}{2}$ is equal to :

A. The maximum of x and y
B. The minimum of x and y
C. 1
D. None of the above

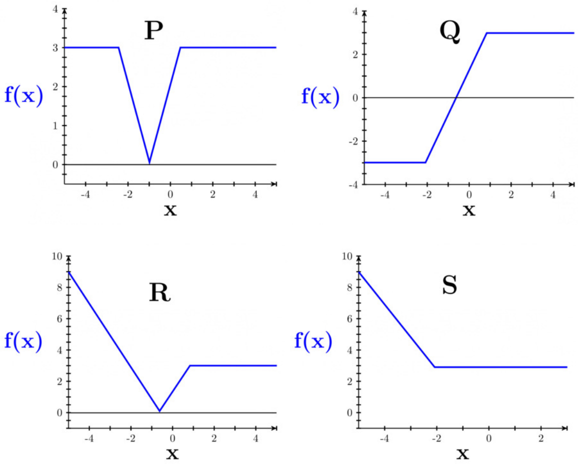
gatecse-2017-set1 general-aptitude quantitative-aptitude maxima-minima absolute-value

Answer key☟

9.1.3 Absolute Value: GATE IN 2023 | GA Question: 9

Which one of the given figures P, Q, R and S represents the graph of the following function?

$$f(x) = ||x+2| - |x-1||$$



A. P
B. Q
C. R
D. S

gatein-2023 quantitative-aptitude absolute-value

Answer key

9.1.4 Absolute Value: GATE2011 AG: GA-7

Given that $f(y) = \frac{|y|}{y}$, and q is non-zero real number, the value of $|f(q) - f(-q)|$ is

A. 0
B. -1
C. 1
D. 2

general-aptitude quantitative-aptitude gate2011-ag absolute-value

Answer key

9.1.5 Absolute Value: GATE2013 AE: GA-8

If $|-2X + 9| = 3$ then the possible value of $|-X| - X^2$ would be:

A. 30
B. -30
C. -42
D. 42

gate2013-ae quantitative-aptitude absolute-value

Answer key

9.1.6 Absolute Value: GATE2013 CE: GA-7

If $|4X - 7| = 5$ then the values of $2|X| - |-X|$ is:

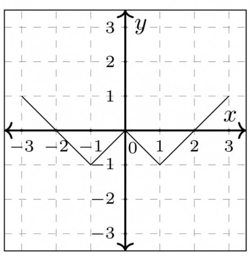
A. $2, \left(\frac{1}{3}\right)$
B. $\left(\frac{1}{2}\right), 3$
C. $\left(\frac{3}{2}\right), 9$
D. $\left(\frac{2}{3}\right), 9$

gate2013-ce quantitative-aptitude absolute-value

Answer key

9.1.7 Absolute Value: GATE2018 ME-1: GA-9

Which of the following functions describe the graph shown in the below figure?



A. $y = ||x| + 1| - 2$
B. $y = ||x| - 1| - 1$

C. $y = || x | + 1 | - 1$
D. $y = || x - 1 | - 1|$

gate2018-me-1 general-aptitude quantitative-aptitude functions absolute-value

Answer key☟

9.2 Age Relation (1)

9.2.1 Age Relation: GATE2018 CE-1: GA-3

Hema's age is 5 years more than twice Hari's age. Suresh's age is 13 years less than 10 times Hari's age. If Suresh is 3 times as old as Hema, how old is Hema?

A. 14 B. 17 C. 18 D. 19

gate2018-ce-1 general-aptitude quantitative-aptitude age-relation

Answer key☟

9.3 Algebra (6)

9.3.1 Algebra: GATE Civil 2021 Set 1 | GA Question: 4

$\bigoplus$ and $\bigodot$ are two operators on numbers p and q such that $p \bigoplus q = \dfrac{p^2 + q^2}{pq}$ and $p \bigodot q = \dfrac{p^2}{q}$;

If $x \bigoplus y = 2 \bigodot 2$, then $x =$

A. $\frac{y}{2}$ B. y C. $\frac{3y}{2}$ D. $2y$

gatecivil-2021-set1 quantitative-aptitude algebra

Answer key☟

9.3.2 Algebra: GATE Civil 2021 Set 2 | GA Question: 4

$\oplus$ and $\odot$ are two operators on numbers p and q such that $p \odot q = p - q$, and $p \oplus q = p \times q$

Then, $(9 \odot (6 \oplus 7)) \odot (7 \oplus (6 \odot 5)) =$

A. 40 B. -26 C. -33 D. -40

gatecivil-2021-set2 quantitative-aptitude algebra easy

Answer key☟

9.3.3 Algebra: GATE ECE 2021 | GA Question: 2

p and q are positive integers and $\dfrac{p}{q} + \dfrac{q}{p} = 3$, then, $\dfrac{p^2}{q^2} + \dfrac{q^2}{p^2} =$

A. 3 B. 7 C. 9 D. 11

gateec-2021 quantitative-aptitude algebra

Answer key☟

9.3.4 Algebra: GATE ECE 2024 | GA Question: 5

The greatest prime factor of $\left(3^{199} - 3^{196}\right)$ is

A. 13 B. 17 C. 3 D. 11

gateece-2024 quantitative-aptitude algebra

Answer key☟

9.3.5 Algebra: GATE2011 MN: GA-61

If $\dfrac{(2y+1)}{(y+2)} < 1$, then which of the following alternatives gives the CORRECT range of y ?

A. $-2 < y < 2$
B. $-2 < y < 1$
C. $-3 < y < 1$
D. $-4 < y < 1$

quantitative-aptitude gate2011-mn algebra

Answer key☟

9.3.6 Algebra: GATE2018 CE-2: GA-3

$\underbrace{a+a+a+\cdots+a}_{\text{n times}} = a^2b$ and $\underbrace{b+b+b+\cdots+b}_{\text{m times}} = ab^2$, where a, b, n, m are natural numbers. What is the value of $\left(\underbrace{m+m+m+\cdots+m}_{\text{n times}}\right)\left(\underbrace{n+n+n+\cdots+n}_{\text{m times}}\right)$?

A. $2a^2b^2$
B. a^4b^4
C. $ab(a+b)$
D. a^2+b^2

gate2018-ce-2 algebra quantitative-aptitude

Answer key☟

9.4 Alligation Mixture (3)

9.4.1 Alligation Mixture: GATE CH 2023 | GA Question: 7

To construct a wall, sand and cement are mixed in the ratio of $3:1$. The cost of sand and that of cement are in the ratio of $1:2$.

If the total cost of sand and cement to construct the wall is 1000 rupees, then what is the cost (in rupees) of cement used?

A. 400
B. 600
C. 800
D. 200

gatech-2023 quantitative-aptitude alligation-mixture

Answer key☟

9.4.2 Alligation Mixture: GATE CSE 2011 | Question: 65

A container originally contains 10 litres of pure spirit. From this container, 1 litre of spirit replaced with 1 litre of water. Subsequently, 1 litre of the mixture is again replaced with 1 litre of water and this process is repeated one more time. How much spirit is now left in the container?

A. 7.58 litres
B. 7.84 litres
C. 7 litres
D. 7.29 litres

gatecse-2011 quantitative-aptitude normal alligation-mixture

Answer key☟

9.4.3 Alligation Mixture: GATE2010 TF: GA-9

A tank has 100 liters of water. At the end of every hour, the following two operations are performed in sequence: $i)$ water equal to $m\%$ of the current contents of the tank is added to the tank , $ii)$ water equal to $n\%$ of the current contents of the tank is removed from the tank. At the end of 5 hours, the tank contains exactly 100 liters of water . The relation between m and n is :

A. $m = n$
B. $m > n$
C. $m < n$
D. None of the previous

general-aptitude quantitative-aptitude gate2010-tf alligation-mixture

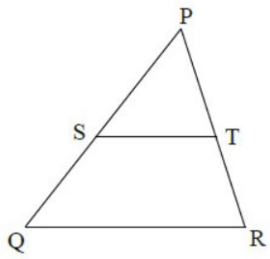
Answer key☟

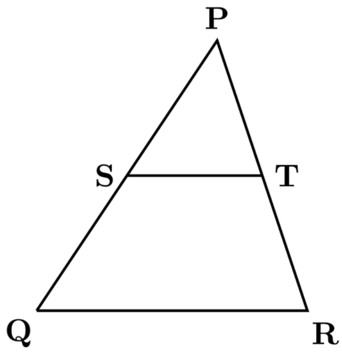
9.5 Area (2)

9.5.1 Area: GATE CSE 2025 | Set 1 | GA Question: 7

In the diagram, the lines QR and ST are parallel to each other. The shortest distance between these two lines is half the shortest distance between the point P and line QR. What is the ratio of the area of the triangle PST to the area of the trapezium SQRT?

Note: The figure shown is representative.





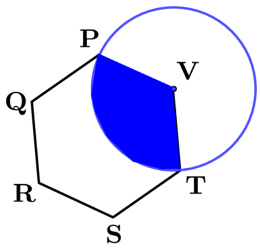
A. $\frac{1}{3}$ B. $\frac{1}{4}$ C. $\frac{2}{5}$ D. $\frac{1}{2}$

gatecse2025-set1 quantitative-aptitude geometry area two-marks

Answer key☟

9.5.2 Area: GATE Civil 2023 Set 1 | GA Question: 3

In the given figure, PQRSTV is a regular hexagon with each side of length 5 cm. A circle is drawn with its centre at V such that it passes through P. What is the area $\left(\text{in cm}^2\right)$ of the shaded region? (The diagram is representative)



A. $\frac{25\pi}{3}$ B. $\frac{20\pi}{3}$ C. 6π D. 7π

gatecivil-2023-set1 quantitative-aptitude geometry area

Answer key☟

9.6 Arithmetic Series (9)

9.6.1 Arithmetic Series: GATE CSE 2013 | Question: 58

What will be the maximum sum of $44, 42, 40, \ldots$?

A. 502 B. 504 C. 506 D. 500

gatecse-2013 quantitative-aptitude easy arithmetic-series

Answer key☟

9.6.2 Arithmetic Series: GATE CSE 2015 Set 2 | Question: GA-6

If the list of letters $P,\ R,\ S,\ T,\ U$ is an arithmetic sequence, which of the following are also in arithmetic sequence?

I. $2P, 2R, 2S, 2T, 2U$
II. $P-3, R-3, S-3, T-3, U-3$
III. P^2, R^2, S^2, T^2, U^2

A. I only
B. I and II
C. II and III
D. I and III

gatecse-2015-set2 quantitative-aptitude normal arithmetic-series

Answer key

9.6.3 Arithmetic Series: GATE Chemical 2020 | GA Question: 5

The difference between the sum of the first $2n$ natural numbers and the sum of the first n odd natural numbers is ______

A. n^2-n
B. n^2+n
C. $2n^2-n$
D. $2n^2+n$

gate2020-ch quantitative-aptitude arithmetic-series

Answer key

9.6.4 Arithmetic Series: GATE Civil 2020 Set 1 | GA Question: 8

Insert seven numbers between 2 and 34, such that the resulting sequence including 2 and 34 is an arithmetic progression. The sum of these inserted seven numbers is ______.

A. 120
B. 124
C. 126
D. 130

gate2020-ce-1 quantitative-aptitude arithmetic-series

Answer key

9.6.5 Arithmetic Series: GATE Civil 2024 Set 1 | GA Question: 2

In a locality, the houses are numbered in the following way:

The house-numbers on one side of a road are consecutive odd integers starting from 301, while the house-numbers on the other side of the road are consecutive even numbers starting from 302. The total number of houses is the same on both sides of the road.

If the difference of the sum of the house-numbers between the two sides of the road is 27, then the number of houses on each side of the road is

A. 27
B. 52
C. 54
D. 26

gatecivil-2024-set1 quantitative-aptitude arithmetic-series

Answer key

9.6.6 Arithmetic Series: GATE Civil 2024 Set 2 | GA Question: 4

If the sum of the first 20 consecutive positive odd numbers is divided by 20^2, the result is

A. 1
B. 20
C. 2
D. $1/2$

gatecivil-2024-set2 quantitative-aptitude arithmetic-series

Answer key

9.6.7 Arithmetic Series: GATE Mechanical 2020 Set 1 | GA Question: 8

The sum of the first n terms in the sequence $8,\ 88,\ 888,\ 8888, \ldots$ is ________.

A.
$$\frac{81}{80}(10^n-1)+\frac{9}{8}n$$

B.
$\frac{81}{80}(10^n-1)-\frac{9}{8}n$

C.
$\frac{80}{81}(10^n-1)+\frac{8}{9}n$

D. $\frac{80}{81}(10^n-1)-\frac{8}{9}n$

gateme-2020-set1 quantitative-aptitude arithmetic-series

Answer key☟

9.6.8 Arithmetic Series: GATE2011 AG: GA-6

The sum of n terms of the series $4+44+444+\ldots\ldots$ is

A. $\frac{4}{81}\left[10^{n+1}-9n-1\right]$

B. $\frac{4}{81}\left[10^{n-1}-9n-1\right]$

C. $\frac{4}{81}\left[10^{n+1}-9n-10\right]$

D. $\frac{4}{81}\left[10^{n}-9n-10\right]$

general-aptitude quantitative-aptitude gate2011-ag arithmetic-series

Answer key☟

9.6.9 Arithmetic Series: GATE2019 EE: GA-6

How many integers are there between 100 and 1000 all of whose digits are even?

A. 60 B. 80 C. 100 D. 90

gate2019-ee general-aptitude quantitative-aptitude arithmetic-series

Answer key☟

9.7 Average (3)

9.7.1 Average: GATE CSE 2025 | Set 1 | GA Question: 3

The average marks obtained by a class in an examination were calculated as 30.8. However, while checking the marks entered, the teacher found that the marks of one student were entered incorrectly as 24 instead of 42. After correcting the marks, the average becomes 31.4. How many students does the class have?

A. 25 B. 28 C. 30 D. 32

gatecse2025-set1 quantitative-aptitude average one-mark

Answer key☟

9.7.2 Average: GATE Mechanical 2022 Set 1 | GA Question: 2

The average of the monthly salaries of M, N and S is ₹4000. The average of the monthly salaries of N, S and P is ₹5000. The monthly salary of P is ₹6000.

What is the monthly salary of M as a percentage of the monthly salary of P?

A. 50% B. 75% C. 100% D. 125%

gateme-2022-set1 quantitative-aptitude average percentage

Answer key

9.7.3 Average: GATE Mechanical 2022 Set 2 | GA Question: 7

For the past m days, the average daily production at a company was 100 units per day.
If today's production of 180 units changes the average to 110 units per day, What is the value of m?

A. 18 B. 10 C. 7 D. 5

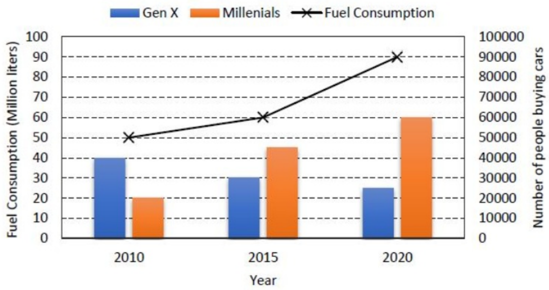
gateme-2022-set2 quantitative-aptitude average

Answer key

9.8 Bar Graph (18)

9.8.1 Bar Graph: GATE 2024 Electrical | GA Question: 8

The chart below shows the data of the number of cars bought by Millennials and Gen X people in a country from the year 2010 to 2020 as well as the yearly fuel consumption of the country (in Million liters).



Considering the data presented in the chart, which one of the following options is true?

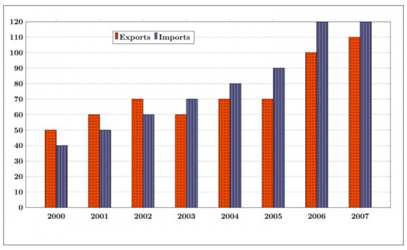
A. The percentage increase in fuel consumption from 2010 to 2015 is more than the percentage increase in fuel consumption from 2015 to 2020.
B. The increase in the number of Millennial car buyers from 2015 to 2020 is less than the decrease in the number of Gen X car buyers from 2010 to 2015.
C. The increase in the number of Millennial car buyers from 2010 to 2015 is more than the decrease in the number of Gen X car buyers from 2010 to 2015.
D. The decrease in the number of Gen X car buyers from 2015 to 2020 is more than the increase in the number of Millennial car buyers from 2010 to 2015.

gate2024-ee data-interpretation bar-graph quantitative-aptitude

Answer key

9.8.2 Bar Graph: GATE CSE 2015 Set 3 | Question: GA-10

The exports and imports (in crores of $Rs.$) of a country from the year 2000 to 2007 are given in the following bar chart. In which year is the combined percentage increase in imports and exports the highest?

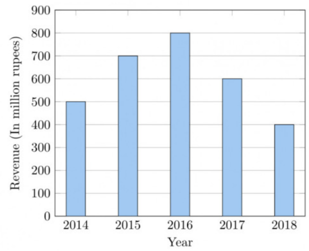


gatecse-2015-set3 quantitative-aptitude data-interpretation normal numerical-answers bar-graph

Answer key

9.8.3 Bar Graph: GATE CSE 2020 | Question: GA-10

The total revenue of a company during $2014-2018$ is shown in the bar graph. If the total expenditure of the company in each year is 500 million rupees, then the aggregate profit or loss (in percentage) on the total expenditure of the company during $2014-2018$ is ___________.

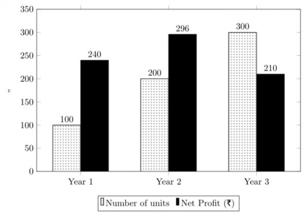


A. 16.67% profit
B. 16.67% loss
C. 20% profit
D. 20% loss

gatecse-2020 quantitative-aptitude data-interpretation bar-graph two-marks

Answer key

9.8.4 Bar Graph: GATE CSE 2021 Set 2 | GA Question: 9



The number of units of a product sold in three different years and the respective net profits are presented in the figure above. The cost/unit in Year 3 was ₹ 1, which was half the cost/unit in Year 2. The cost/unit in Year 3 was one-third of the cost/unit in Year 1. Taxes were paid on the selling price at 10%, 13%, and 15% respectively for the three years. Net profit is calculated as the difference between the selling price and the sum of cost and taxes paid in that year.

The ratio of the selling price in Year 2 to the selling price in Year 3 is _________.

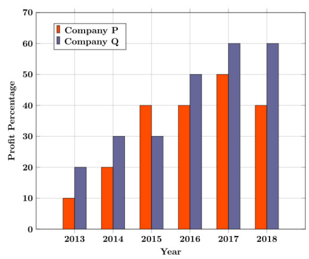
A. $4:3$
B. $1:1$
C. $3:4$
D. $1:2$

gatecse-2021-set2 quantitative-aptitude data-interpretation bar-graph two-marks

Answer key

9.8.5 Bar Graph: GATE Chemical 2020 | GA Question: 10

The profit shares of two companies P and Q are shown in the figure. If the two companies have invested a fixed and equal amount every year, then the ratio of the total revenue of company P to the revenue of company Q, during $2013-2018$ is ________.

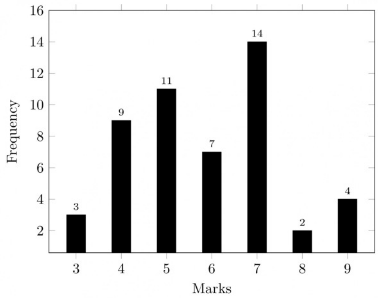


A. $15:17$
B. $16:17$
C. $17:15$
D. $17:16$

gate2020-ch quantitative-aptitude data-interpretation bar-graph

Answer key

9.8.6 Bar Graph: GATE Civil 2022 Set 1 | GA Question: 8



The above frequency chart shows the frequency distribution of marks obtained by a set of students in an exam.

From the data presented above, which one of the following is CORRECT?

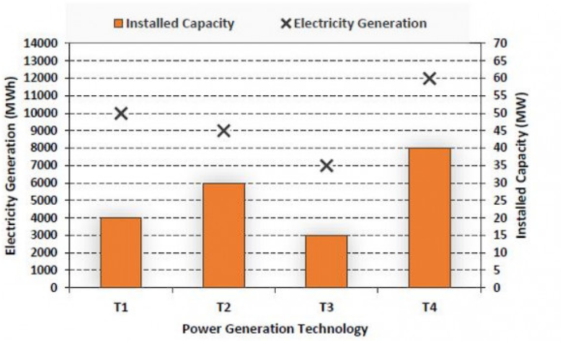
A. mean > mode > median
B. mode > median> mean
C. mode > mean > median
D. median > mode > mean

gatecivil-2022-set1 quantitative-aptitude data-interpretation bar-graph statistics

Answer key☟

9.8.7 Bar Graph: GATE Civil 2024 Set 1 | GA Question: 8

The chart given below compares the Installed Capacity (MW) of four power generation technologies, $\text{T1}, \text{T2}, \text{T3}$, and T4, and their Electricity Generation (MWh) in a time of 1000 hours (h).



The Capacity Factor of a power generation technology is:

$$\text{Capacity Factor } = \frac{\text{Electricity Generation (MWh)}}{\text{Installed Capacity (MW) } \times 1000(\text{ h})}$$

Which one of the given technologies has the highest Capacity Factor?

A. T1
B. T2
C. T3
D. T4

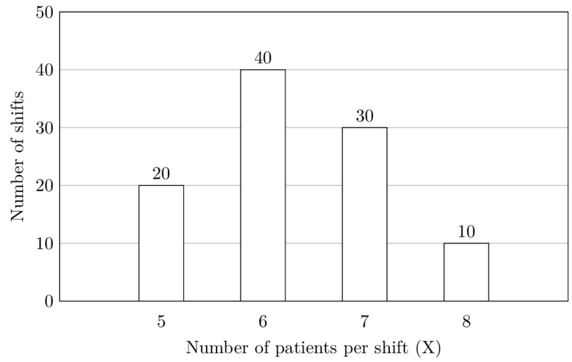
gatecivil-2024-set1 quantitative-aptitude bar-graph data-interpretation

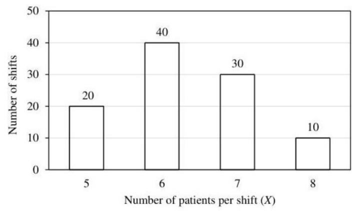
Answer key☟

9.8.8 Bar Graph: GATE DA 2025 | GA Question: 10

The number of patients per shift (X) consulting Dr. Gita in her past 100 shifts is shown in the figure. If the amount she earns is ₹ $1000(X-0.2)$, what is the average amount (in ₹) she has earned per shift in the past 100 shifts?

Note: The figure shown is representative.





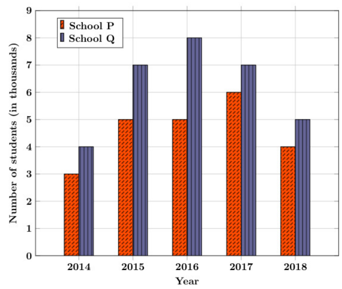
A. $6,100$ B. $6,300$ C. $6,000$ D. $6,500$

gateda-2025 quantitative-aptitude bar-graph data-interpretation two-marks

Answer key

9.8.9 Bar Graph: GATE ECE 2020 | GA Question: 10

The following figure shows the data of students enrolled in 5 years $(2014 \text{ to } 2018)$ for two schools P and Q. During this period, the ratio of the average number of the students enrolled in school P to the average of the difference of the number of students enrolled in schools P and Q is _______.

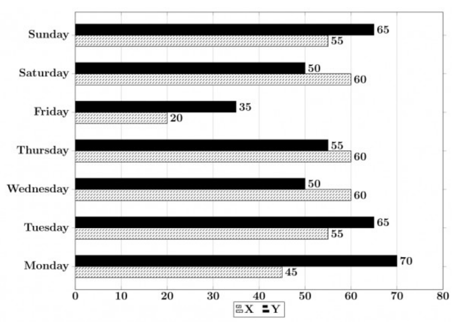


A. $8:23$ B. $23:8$ C. $23:31$ D. $31:23$

gate2020-ec quantitative-aptitude data-interpretation bar-graph

Answer key

9.8.10 Bar Graph: GATE ECE 2021 | GA Question: 9



The number of minutes spent by two students, X and Y, exercising every day in a given week are shown in the bar chart above.

The number of days in the given week in which one of the students spent a minimum of 10% more than the other student, on a given day, is

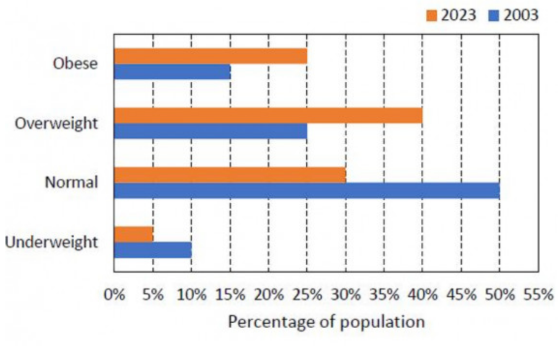
A. 4 B. 5 C. 6 D. 7

gateec-2021 quantitative-aptitude data-interpretation bar-graph

Answer key

9.8.11 Bar Graph: GATE ECE 2024 | GA Question: 8

The bar chart shows the data for the percentage of population falling into different categories based on Body Mass Index (BMI) in 2003 and 2023.



Based on the data provided, which one of the following options is INCORRECT?

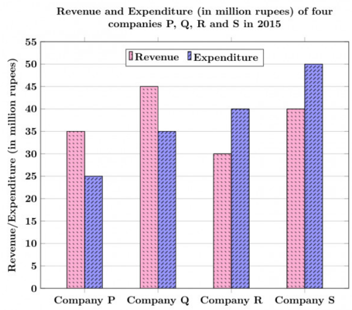
A. The ratio of the percentage of population falling into overweight category to the percentage of population falling into normal category has increased in 20 years.
B. The ratio of the percentage of population falling into underweight category to the percentage of population falling into normal category has decreased in 20 years.
C. The ratio of the percentage of population falling into obese category to the percentage of population falling into normal category has decreased in 20 years.
D. The percentage of population falling into normal category has decreased in 20 years.

gateece-2024 data-interpretation bar-graph quantitative-aptitude

Answer key

9.8.12 Bar Graph: GATE Electrical 2020 | GA Question: 10

The revenue and expenditure of four different companies $\mathrm{P}, \mathrm{Q}, \mathrm{R}$ and S in 2015 are shown in the figure. If the revenue of company Q in 2015 was 20% more than that in 2014, and company Q had earned a profit of 10% on expenditure in 2014, then its expenditure (in million rupees) in 2014 was _______.

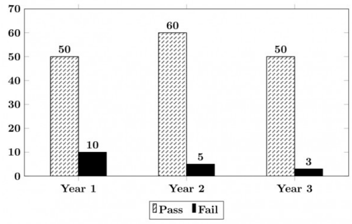


A. 32.7 B. 33.7 C. 34.1 D. 35.1

gate2020-ee quantitative-aptitude data-interpretation bar-graph

Answer key

9.8.13 Bar Graph: GATE Electrical 2021 | GA Question: 9



The number of students passing or failing in an exam for a particular subject is presented in the bar chart above. Students who pass the exam cannot appear for the exam again. Students who fail the exam in the first attempt must appear for the exam in the following year. Students always pass the exam in their second attempt.

The number of students who took the exam for the first time in the year 2 and the year 3 respectively, are ______________.

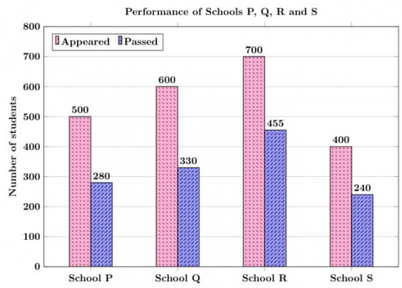
A. 65 and 53 B. 60 and 50 C. 55 and 53 D. 55 and 48

gateee-2021 quantitative-aptitude data-interpretation bar-graph

Answer key☟

9.8.14 Bar Graph: GATE Mechanical 2020 Set 1 | GA Question: 10

The bar graph shows the data of the students who appeared and passed in an examination for four schools P, Q, R, and S. The average of success rates (in percentage) of these four schools is _______.



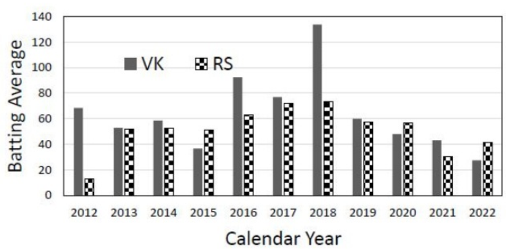
A. 58.5% B. 58.8% C. 59.0% D. 59.3%

gateme-2020-set1 quantitative-aptitude data-interpretation bar-graph

Answer key☟

9.8.15 Bar Graph: GATE Mechanical 2024 | GA Question: 8

The bar chart gives the batting averages of VK and RS for 11 calendar years from 2012 to 2022. Considering that 2015 and 2019 are world cup years, which one of the following options is true?



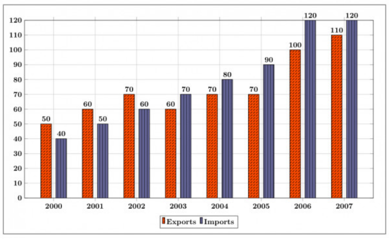
A. RS has a higher yearly batting average than that of VK in every world cup year.
B. VK has a higher yearly batting average than that of RS in every world cup year.
C. $\text{VK}'\text{s}$ yearly batting average is consistently higher than that of RS between the two world cup years.
D. $\text{RS}'\text{s}$ yearly batting average is consistently higher than that of VK in the last three years.

gateme-2024 quantitative-aptitude bar-graph data-interpretation

Answer key☟

9.8.16 Bar Graph: GATE2014 EC-1: GA-9

The exports and imports (in crores of Rs.) of a country from 2000 to 2007 are given in the following bar chart. If the trade deficit is defined as excess of imports over exports, in which year is the trade deficit $1/5$th of the exports?



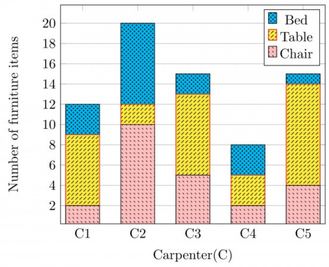
A. 2005 B. 2004 C. 2007 D. 2006

gate2014-ec-1 quantitative-aptitude data-interpretation bar-graph normal

Answer key☟

9.8.17 Bar Graph: GATE2017 CE-1: GA-10

The bar graph below shows the output of five carpenters over one month. each of whom made different items of furniture: chairs, tables, and beds.



Consider the following statements.

i. The number of beds made by carpenter $C2$ is exactly the same as the number of tables made by carpenter $C3$
ii. The total number of chairs made by all carpenters is less than the total number of tables.

Which one of the following is true?

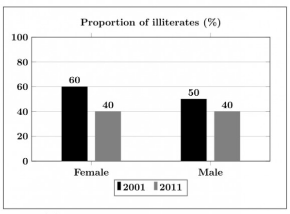
A. Only i B. Only ii C. Both i and ii D. Neither i nor ii

gate2017-ce-1 general-aptitude quantitative-aptitude data-interpretation bar-graph

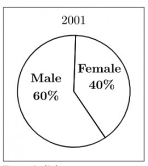
Answer key☟

9.8.18 Bar Graph: GATE2019 EC: GA-7

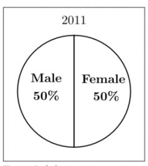
The bar graph in panel (a) shows the proportion of male and female illiterates in 2001 and 2011. The proportions of males and females in 2001 and 2011 are given in Panel (b) and (c), respectively. The total population did not change during this period. The percentage increase in the total number of literates from 2001 to 2011 is ______.



Panel (a)



Panel (b)



Panel (c)

A. 30.43 B. 33.43 C. 34.43 D. 35.43

gate2019-ec quantitative-aptitude data-interpretation bar-graph

Answer key☟

9.9 Calendar (2)

9.9.1 Calendar: GATE Civil 2020 Set 2 | GA Question: 7

For the year 2019, which of the previous year's calendar can be used?

A. 2011 B. 2012 C. 2013 D. 2014

gate2020-ce-2 quantitative-aptitude calendar

Answer key☟

9.9.2 Calendar: GATE Mechanical 2022 Set 2 | GA Question: 4

A person was born on the fifth Monday of February in a particular year.
Which one of the following statements is correct based on the above information?

A. The 2^{nd} February of that year is a Tuesday
B. There will be five Sundays in the month of February in that year
C. The 1^{st} February of that year is a Sunday
D. All Mondays of February in that year have even dates

gateme-2022-set2 quantitative-aptitude calendar

Answer key☟

9.10 Cartesian Coordinates (10)

9.10.1 Cartesian Coordinates: GATE CSE 2020 | Question: GA-9

Two straight lines are drawn perpendicular to each other in $X-Y$ plane. If α and β are the acute angles the straight lines make with the X- axis, then $\alpha+\beta$ is ________.

A. 60° B. 90° C. 120° D. 180°

gatecse-2020 quantitative-aptitude geometry cartesian-coordinates two-marks

Answer key☟

9.10.2 Cartesian Coordinates: GATE CSE 2022 | GA Question: 2

A function $y(x)$ is defined in the interval $[0,1]$ on the $x-$ axis as

$$y(x)=\begin{cases}2 & \text{if} \quad 0\leq x<\frac{1}{3}\\ 3 & \text{if} \quad \frac{1}{3}\leq x<\frac{3}{4}\\ 1 & \text{if} \quad \frac{3}{4}\leq x\leq 1\end{cases}$$

Which one of the following is the area under the curve for the interval $[0,1]$ on the $x-$ axis?

A. $\frac{5}{6}$ B. $\frac{6}{5}$ C. $\frac{13}{6}$ D. $\frac{6}{13}$

gatecse-2022 quantitative-aptitude cartesian-coordinates area one-mark

Answer key☟

9.10.3 Cartesian Coordinates: GATE Civil 2022 Set 1 | GA Question: 2

Two straight lines pass through the origin $(x_0,y_0)=(0,0)$. One of them passes through the point $(x_1,y_1)=(1,3)$ and the other passes through the point $(x_2,y_2)=(1,2)$.
What is the area enclosed between the straight lines in the interval $[0,1]$ on the x-axis?

A. 0.5 B. 1.0 C. 1.5 D. 2.0

gatecivil-2022-set1 quantitative-aptitude cartesian-coordinates area

Answer key

9.10.4 Cartesian Coordinates: GATE ECE 2022 | GA Question: 8

Four points $\mathrm{P}(0,1), \mathrm{Q}(0,-3), \mathrm{R}(-2,-1)$, and $\mathrm{S}(2,-1)$ represent the vertices of a quadrilateral. What is the area enclosed by the quadrilateral?

A. 4 B. $4\sqrt{2}$ C. 8 D. $8\sqrt{2}$

gateece-2022 quantitative-aptitude cartesian-coordinates area

Answer key

9.10.5 Cartesian Coordinates: GATE Mechanical 2022 Set 2 | GA Question: 3

If $f(x) = 2\ \ln\left(\sqrt{e^x}\right)$, what is the area bounded by $f(x)$ for the interval $[0,2]$ on the $x-$axis?

A. $\frac{1}{2}$ B. 1 C. 2 D. 4

gateme-2022-set2 quantitative-aptitude cartesian-coordinates area

Answer key

9.10.6 Cartesian Coordinates: GATE2012 AE: GA-9

Two points $(4, p)$ and $(0, q)$ lie on a straight line having a slope of $3/4$. The value of $(p\text{–}q)$ is

A. -3 B. 0 C. 3 D. 4

gate2012-ae quantitative-aptitude cartesian-coordinates geometry

Answer key

9.10.7 Cartesian Coordinates: GATE2014 AE: GA-4

If $y = 5x^2 + 3$, then the tangent at $x = 0$, $y = 3$

A. passes through $x = 0, y = 0$
B. has a slope of $+1$
C. is parallel to the x-axis
D. has a slope of -1

gate2014-ae quantitative-aptitude geometry cartesian-coordinates

Answer key

9.10.8 Cartesian Coordinates: GATE2016 EC-3: GA-10

A straight line is fit to a data set (ln x, y). This line intercepts the abscissa at ln $x = 0.1$ and has a slope of -0.02. What is the value of y at $x = 5$ from the fit?

A. -0.030 B. -0.014 C. 0.014 D. 0.030

gate2016-ec-3 quantitative-aptitude cartesian-coordinates

Answer key

9.10.9 Cartesian Coordinates: GATE2016 EC-3: GA-9

Find the area bounded by the lines $3x + 2y = 14, 2x - 3y = 5$ in the first quadrant.

A. 14.95 B. 15.25 C. 15.70 D. 20.35

gate2016-ec-3 cartesian-coordinates geometry normal

Answer key

9.10.10 Cartesian Coordinates: GATE2017 ME-1: GA-8

Let S_1 be the plane figure consisting of the points (x, y) given by the inequalities $|x - 1| \leq 2$ and $|y + 2| \leq 3$. Let S_2 be the plane figure given by the inequalities $x - y \geq -2$, $y \geq 1$, and $x \leq 3$. Let S be the union of S_1 and S_2. The area of S is.

A. 26 B. 28 C. 32 D. 34

gate2017-me-1 general-aptitude quantitative-aptitude geometry cartesian-coordinates

Answer key

9.11.1 Circle: GATE CSE 2018 | Question: GA-3

The area of a square is d. What is the area of the circle which has the diagonal of the square as its diameter?

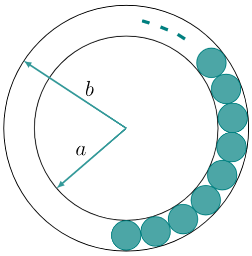
A. πd
B. πd^2
C. $\frac{1}{4}\pi d^2$
D. $\frac{1}{2}\pi d$

gatecse-2018 quantitative-aptitude geometry circle normal one-mark

Answer key

9.11.2 Circle: GATE CSE 2020 | Question: GA-8

The figure below shows an annular ring with outer and inner as b and a, respectively. The annular space has been painted in the form of blue colour circles touching the outer and inner periphery of annular space. If maximum n number of circles can be painted, then the unpainted area available in annular space is _____.

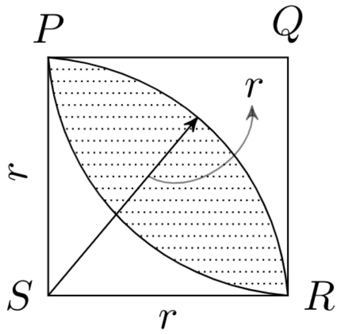


A. $\pi[(b^2-a^2)-\frac{n}{4}(b-a)^2]$
B. $\pi[(b^2-a^2)-n(b-a)^2]$
C. $\pi[(b^2-a^2)+\frac{n}{4}(b-a)^2]$
D. $\pi[(b^2-a^2)+n(b-a)^2]$

gatecse-2020 quantitative-aptitude geometry circle area two-marks

Answer key

9.11.3 Circle: GATE Civil 2021 Set 2 | GA Question: 9



In the figure shown above, PQRS is a square. The shaded portion is formed by the intersection of sectors of circles with radius equal to the side of the square and centers at S and Q.

The probability that any point picked randomly within the square falls in the shaded area is ____________

A. $4-\frac{\pi}{2}$
B. $\frac{1}{2}$
C. $\frac{\pi}{2}-1$
D. $\frac{\pi}{4}$

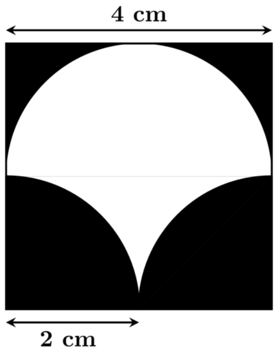
gatecivil-2021-set2 quantitative-aptitude geometry circle

Answer key

9.11.4 Circle: GATE Civil 2023 Set 1 | GA Question: 10

A square of side length 4 cm is given. The boundary of the shaded region is defined by one semi-circle on the top and two circular arcs at the bottom, each of radius 2 cm, as shown.

The area of the shaded region is__________cm^2.



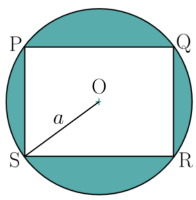
A. 8 B. 4 C. 12 D. 10

gatecivil-2023-set1 quantitative-aptitude geometry circle

Answer key☟

9.11.5 Circle: GATE ECE 2020 | GA Question: 8

A circle with centre O is shown in the figure. A rectangle PQRS of maximum possible area is inscribed in the circle. If the radius of the circle is a, then the area of the shaded portion is _______.



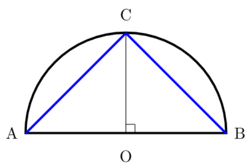
A. $\pi a^2 - a^2$ B. $\pi a^2 - \sqrt{2}a^2$ C. $\pi a^2 - 2a^2$ D. $\pi a^2 - 3a^2$

gate2020-ec quantitative-aptitude geometry circle area

Answer key☟

9.11.6 Circle: GATE Electrical 2020 | GA Question: 9

Given a semicircle with O as the centre, as shown in the figure, the ratio $\dfrac{\overline{AC}+\overline{CB}}{\overline{AB}}$ is _______, where $\overline{AC}$, $\overline{CB}$ and $\overline{AB}$ are chords.



A. $\sqrt{2}$ B. $\sqrt{3}$ C. 2 D. 3

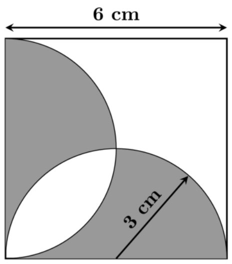
gate2020-ee quantitative-aptitude geometry circle

Answer key☟

9.11.7 Circle: GATE Electrical 2023 | GA Question: 10

A square with sides of length $6\,\mathrm{cm}$ is given. The boundary of the shaded region is defined by two semi-circles whose diameters are the sides of the square, as shown.

The area of the shaded region is_______ cm^2.



A. 6π B. 18 C. 20 D. 9π

gate2023-ee quantitative-aptitude geometry circle

Answer key

9.12 Clock Time (10)

9.12.1 Clock Time: GATE CSE 2014 Set 2 | Question: GA-10

At what time between 6 a. m. and 7 a. m. will the minute hand and hour hand of a clock make an angle closest to 60°?

A. $6:22$ a.m. B. $6:27$ a.m. C. $6:38$ a.m. D. $6:45$ a.m.

gatecse-2014-set2 quantitative-aptitude normal clock-time

Answer key

9.12.2 Clock Time: GATE Civil 2024 Set 1 | GA Question: 5

On a given day, how many times will the second-hand and the minute-hand of a clock cross each other during the clock time $12:05:00$ hours to $12:55:00$ hours?

A. 51 B. 49 C. 50 D. 55

gatecivil-2024-set1 quantitative-aptitude clock-time

Answer key

9.12.3 Clock Time: GATE ECE 2020 | GA Question: 7

It is quarter past three in your watch. The angle between the hour hand and the minute hand is ________.

A. 0° B. 7.5° C. 15° D. 22.5°

gate2020-ec quantitative-aptitude clock-time

Answer key

9.12.4 Clock Time: GATE Mechanical 2021 Set 2 | GA Question: 3

A digital watch X beeps every 30 seconds while watch Y beeps every 32 seconds. They beeped together at $10 \text{ AM}.$

The immediate next time that they will beep together is ____

A. 10.08 AM B. 10.42 AM C. 11.00 AM D. 10.00 PM

gateme-2021-set2 quantitative-aptitude clock-time

Answer key

9.12.5 Clock Time: GATE Mechanical 2022 Set 1 | GA Question: 10

In a 12-hour clock that runs correctly, how many times do the second, minute, and hour hands of the clock coincide, in a 12-hour duration from 3 PM in a day to 3 AM the next day?

A. 11 B. 12 C. 144 D. 2

gateme-2022-set1 quantitative-aptitude clock-time

Answer key

9.12.6 Clock Time: GATE Mechanical 2023 | GA Question: 4

The minute-hand and second-hand of a clock cross each other ___________ times between $09:15:00$ AM and $09:45:00$ AM on a day.

A. 30 B. 15 C. 29 D. 31

gateme-2023 quantitative-aptitude clock-time

Answer key

9.12.7 Clock Time: GATE2016 EC-2: GA-8

Two and quarter hours back, when seen in a mirror, the reflection of a wall clock without number markings seemed to show $1:30$. What is the actual current time shown by the clock?

A. $8:15$ B. $11:15$ C. $12:15$ D. $12:45$

gate2016-ec-2 clock-time

Answer key

9.12.8 Clock Time: GATE2018 CE-2: GA-7

A faulty wall clock is known to gain 15 minutes every 24 hours. It is synchronized to the correct time at 9 AM on 11th July. What will be the correct time to the nearest minute when the clock shows 2 PM on 15th July of the same year?

A. $12:45$ PM B. $12:58$ PM C. $1:00$ PM D. $2:00$ PM

gate2018-ce-2 general-aptitude quantitative-aptitude clock-time normal

Answer key

9.12.9 Clock Time: GATE2019 EC: GA-9

Two design consultants, P and Q, started working from 8 AM for a client. The client budgeted a total of USD 3000 for the consultants. P stopped working when the hour hand moved by 210 degrees on the clock. Q stopped working when the hour hand moved by 240 degrees. P took two tea breaks of 15 minutes each during her shift, but took no lunch break. Q took only one lunch break for 20 minutes, but no tea breaks. The market rate for consultants is USD 200 per hour and breaks are not paid. After paying the consultants, the client shall have USD _______ remaining in the budget.

A. 000.00 B. 166.67 C. 300.00 D. 433.33

gate2019-ec general-aptitude quantitative-aptitude clock-time

Answer key

9.12.10 Clock Time: GATE2019 ME-1: GA-3

A worker noticed that the hour hand on the factory clock had moved by 225 degrees during her stay at the factory. For how long did she stay in the factory?

A. 3.75 hours B. 4 hours and 15 mins C. 8.5 hours D. 7.5 hours

gate2019-me-1 general-aptitude quantitative-aptitude clock-time

Answer key

9.13 Combinatory (12)

9.13.1 Combinatory: GATE CSE 2010 | Question: 65

Given digits $2, 2, 3, 3, 3, 4, 4, 4, 4$ how many distinct 4 digit numbers greater than 3000 can be formed?

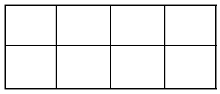
A. 50 B. 51 C. 52 D. 54

gatecse-2010 quantitative-aptitude combinatory normal

Answer key

9.13.2 Combinatory: GATE CSE 2016 Set 2 | Question: GA-09

In a 2×4 rectangle grid shown below, each cell is rectangle. How many rectangles can be observed in the grid?



A. 21 B. 27 C. 30 D. 36

gatecse-2016-set2 quantitative-aptitude normal combinatory

Answer key☟

9.13.3 Combinatory: GATE CSE 2017 Set 1 | Question: GA-9

Arun, Gulab, Neel and Shweta must choose one shirt each from a pile of four shirts coloured red, pink, blue and white respectively. Arun dislikes the colour red and Shweta dislikes the colour white. Gulab and Neel like all the colours. In how many different ways can they choose the shirts so that no one has a shirt with a colour he or she dislikes?

A. 21 B. 18 C. 16 D. 14

gatecse-2017-set1 combinatory quantitative-aptitude

Answer key☟

9.13.4 Combinatory: GATE2011 MN: GA-59

In how many ways 3 scholarships can be awarded to 4 applicants, when each applicant can receive any number of scholarships?

A. 4 B. 12 C. 64 D. 81

quantitative-aptitude gate2011-mn combinatory

Answer key☟

9.13.5 Combinatory: GATE2012 AR: GA-5

Ten teams participate in a tournament. Every team plays each of the other teams twice. The total number of matches to be played is

A. 20 B. 45 C. 60 D. 90

gate2012-ar quantitative-aptitude combinatory

Answer key☟

9.13.6 Combinatory: GATE2014 EC-4: GA-10

A five digit number is formed using the digits $1, 3, 5, 7$ and 9 without repeating any of them. What is the sum of all such possible five digit numbers?

A. 6666660 B. 6666600 C. 6666666 D. 6666606

gate2014-ec-4 quantitative-aptitude normal combinatory

Answer key☟

9.13.7 Combinatory: GATE2015 CE-2: GA-8

How many four digit numbers can be formed with the 10 digits $0, 1, 2, \dots, 9$ if no number can start with 0 and if repetitions are not allowed?

gate2015-ce-2 quantitative-aptitude combinatory

Answer key☟

9.13.8 Combinatory: GATE2015 ME-3: GA-5

Five teams have to compete in a league, with every team playing every other team exactly once, before going to the next round. How many matches will have to be held to complete the league round of matches?

A. 20 B. 10 C. 8 D. 5

gate2015-me-3 quantitative-aptitude combinatory

Answer key

9.13.9 Combinatory: GATE2017 EC-2: GA-9

The number of 3-digit numbers such that the digit 1 is never to the immediate right of 2 is

A. 781 B. 791 C. 881 D. 891

gate2017-ec-2 quantitative-aptitude combinatory

Answer key

9.13.10 Combinatory: GATE2017 ME-2: GA-8

There are 4 women P, Q, R, S and 5 men V, W, X, Y, Z in a group. We are required to form pairs each consisting of one woman and one man. P is not to be paired with Z, and Y must necessarily be paired with someone. In how many ways can 4 such pairs be formed?

A. 74 B. 76 C. 78 D. 80

gate2017-me-2 quantitative-aptitude combinatory

Answer key

9.13.11 Combinatory: GATE2018 CE-2: GA-4

A three-member committee has to be formed from a group of 9 people. How many such distinct committees can be formed?

A. 27 B. 72 C. 81 D. 84

gate2018-ce-2 quantitative-aptitude combinatory

Answer key

9.13.12 Combinatory: GATE2019 EC: GA-4

Five different books (P, Q, R, S, T) are to be arranged on a shelf. The books R and S are to be arranged first and second, respectively from the right side of the shelf. The number of different orders in which P, Q and T may be arranged is ______.

A. 2 B. 6 C. 12 D. 120

gate2019-ec quantitative-aptitude combinatory

Answer key

9.14 Compound Interest (5)

9.14.1 Compound Interest: GATE DS&AI 2024 | GA Question: 8

Person 1 and Person 2 invest in three mutual funds A, B, and C. The amounts they invest in each of these mutual funds are given in the table.

	Mutual fund A	Mutual fund B	Mutual fund C
Person1	$10,000$	$20,000$	$20,000$
Person2	$20,000$	$15,000$	$15,000$

At the end of one year, the total amount that Person 1 gets is ₹500 more than Person 2. The annual rate of return for the mutual funds B and C is 15% each. What is the annual rate of return for the mutual fund A?

A. 7.5% B. 10% C. 15% D. 20%

gate-ds-ai-2024 quantitative-aptitude compound-interest two-marks

Answer key

9.14.2 Compound Interest: GATE ECE 2021 | GA Question: 1

The current population of a city is $11,02,500$. If it has been increasing at the rate of 5% per annum, what was its population 2 years ago?

A. $9,92,500$ B. $9,95,006$ C. $10,00,000$ D. $12,51,506$

gateec-2021 quantitative-aptitude compound-interest

Answer key☟

9.14.3 Compound Interest: GATE2010 MN: GA-5

A person invest Rs.1000 at 10% annual compound interest for 2 years. At the end of two years, the whole amount is invested at an annual simple interest of 12% for 5 years. The total value of the investment finally is :

A. 1776 B. 1760 C. 1920 D. 1936

general-aptitude quantitative-aptitude gate2010-mn compound-interest

Answer key☟

9.14.4 Compound Interest: GATE2014 AG: GA-5

The population of a new city is 5 million and is growing at 20% annually. How many years would it take to double at this growth rate?

A. $3-4$ years B. $4-5$ years C. $5-6$ years D. $6-7$ years

gate2014-ag quantitative-aptitude compound-interest normal

Answer key☟

9.14.5 Compound Interest: GATE2018 EC: GA-6

Leila aspires to buy a car worth $Rs. 10,00,000$ after 5 years. What is the minimum amount in Rupees that she should deposit now in a bank which offers 10% annual rate of interest, if the interest was compounded annually?

A. $5,00,000$ B. $6,21,000$ C. $6,66,667$ D. $7,50,000$

gate2018-ec general-aptitude quantitative-aptitude compound-interest normal

Answer key☟

9.15 Conditional Probability (6)

9.15.1 Conditional Probability: GATE CSE 2012 | Question: 63

An automobile plant contracted to buy shock absorbers from two suppliers X and Y . X supplies 60% and Y supplies 40% of the shock absorbers. All shock absorbers are subjected to a quality test. The ones that pass the quality test are considered reliable. Of $X's$ shock absorbers, 96% are reliable. Of $Y's$ shock absorbers, 72% are reliable.

The probability that a randomly chosen shock absorber, which is found to be reliable, is made by Y is

A. 0.288 B. 0.334 C. 0.667 D. 0.720

gatecse-2012 quantitative-aptitude probability normal conditional-probability

Answer key☟

9.15.2 Conditional Probability: GATE Mechanical 2021 Set 2 | GA Question: 7

A box contains 15 blue balls and 45 black balls. If 2 balls are selected randomly, without replacement, the probability of an outcome in which the first selected is a blue ball and the second selected is a black ball, is _____

A. $\frac{3}{16}$

B. $\frac{45}{236}$

C. $\frac{1}{4}$

D. $\frac{3}{4}$

gateme-2021-set2 quantitative-aptitude probability conditional-probability

Answer key

9.15.3 Conditional Probability: GATE2013 AE: GA-10

In a factory, two machines $M1$ and $M2$ manufacture 60% and 40% of the autocomponents respectively. Out of the total production, 2% of $M1$ and 3% of $M2$ are found to be defective. If a randomly drawn autocomponent from the combined lot is found defective, what is the probability that it was manufactured by $M2$?

A. 0.35 B. 0.45 C. 0.5 D. 0.4

gate2013-ae quantitative-aptitude conditional-probability

Answer key

9.15.4 Conditional Probability: GATE2014 AG: GA-10

10% of the population in a town is HIV^+. A new diagnostic kit for HIV detection is available; this kit correctly identifies HIV^+ individuals 95% of the time, and HIV^- individuals 89% of the time. A particular patient is tested using this kit and is found to be positive. The probability that the individual is actually positive is ______.

gate2014-ag quantitative-aptitude probability conditional-probability normal numerical-answers

Answer key

9.15.5 Conditional Probability: GATE2014 EC-1: GA-10

You are given three coins: one has heads on both faces, the second has tails on both faces, and the third has a head on one face and a tail on the other. You choose a coin at random and toss it, and it comes up heads. The probability that the other face is tails is

A. $\frac{1}{4}$ B. $\frac{1}{3}$ C. $\frac{1}{2}$ D. $\frac{2}{3}$

gate2014-ec-1 quantitative-aptitude probability conditional-probability

Answer key

9.15.6 Conditional Probability: GATE2015 ME-3: GA-10

A coin is tossed thrice. Let X be the event that head occurs in each of the first two tosses. Let Y be the event that a tail occurs on the third toss. Let Z be the event that two tails occur in three tosses.
Based on the above information, which one of the following statements is TRUE?

A. X and Y are not independent
B. Y and Z are dependent
C. Y and Z are independent
D. X and Z are independent

gate2015-me-3 conditional-probability probability quantitative-aptitude

Answer key

9.16 Cones (1)

9.16.1 Cones: GATE ECE 2021 | GA Question: 8

Consider a square sheet of side 1 unit. In the first step, it is cut along the main diagonal to get two triangles. In the next step. one of the cut triangles is revolved about its short edge to form a solid cone. The volume of the resulting cone, in cubic units, is ____________

A. $\frac{\pi}{3}$ B. $\frac{2\pi}{3}$ C. $\frac{3\pi}{2}$ D. 3π

gateec-2021 quantitative-aptitude mensuration cones

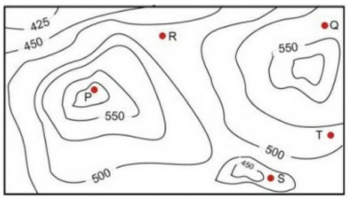
Answer key

9.17 Contour Plots (4)

9.17.1 Contour Plots: GATE CSE 2017 Set 1 | Question: GA-10

A contour line joins locations having the same height above the mean sea level. The following is a contour plot of a geographical region. Contour lines are shown at 25 m intervals in this plot. If in a flood, the water level rises to

525 m, which of the villages P, Q, R, S, T get submerged?



A. P, Q B. P, Q, T C. R, S, T D. Q, R, S

gatecse-2017-set1 general-aptitude quantitative-aptitude data-interpretation normal contour-plots

Answer key

9.17.2 Contour Plots: GATE CSE 2017 Set 2 | Question: GA-10

An air pressure contour line joins locations in a region having the same atmospheric pressure. The following is an air pressure contour plot of a geographical region. Contour lines are shown at 0.05 bar intervals in this plot.

If the possibility of a thunderstorm is given by how fast air pressure rises or drops over a region, which of the following regions is most likely to have a thunderstorm?

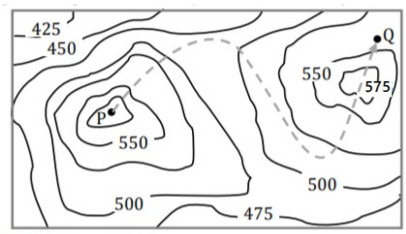
A. P B. Q C. R D. S

gatecse-2017-set2 quantitative-aptitude data-interpretation normal contour-plots

Answer key

9.17.3 Contour Plots: GATE2017 EC-1: GA-10

A contour line joins locations having the same height above the mean sea level. The following is a contour plot of a geographical region. Contour lines are shown at 25 m intervals in this plot.



The path from P to Q is best described by

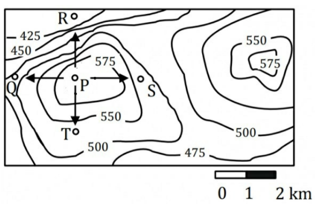
A. Up-Down-Up-Down B. Down-Up-Down-Up C. Down-Up-Down D. Up-Down-Up

gate2017-ec-1 general-aptitude quantitative-aptitude data-interpretation contour-plots

Answer key

9.17.4 Contour Plots: GATE2017 EC-2: GA-10

A contour line joins locations having the same height above the mean sea level. The following is a contour plot of a geographical region. Contour lines are shown at 25 m intervals in this plot.



Which of the following is the steepest path leaving from P?

A. P to Q　B. P to R　C. P to S　D. P to T

gate2017-ec-2 general-aptitude quantitative-aptitude data-interpretation contour-plots

Answer key☟

9.18 Cost Market Price (6)

9.18.1 Cost Market Price: GATE 2024 Electrical | GA Question: 4

A shopkeeper buys shirts from a producer and sells them at 20% profit. A customer has to pay ₹ $3,186.00$ including 18% taxes, per shirt. At what price did the shopkeeper buy each shirt from the producer?

A. ₹ $2,500.00$　B. ₹ $1,975.40$　C. ₹ $2,250.00$　D. ₹ $2,548.80$

gate2024-ee quantitative-aptitude cost-market-price

Answer key☟

9.18.2 Cost Market Price: GATE CSE 2011 | Question: 63

The variable cost (V) of manufacturing a product varies according to the equation $V = 4q$, where q is the quantity produced. The fixed cost (F) of production of same product reduces with q according to the equation $F = \frac{100}{q}$. How many units should be produced to minimize the total cost $(V + F)$?

A. 5　B. 4　C. 7　D. 6

gatecse-2011 quantitative-aptitude cost-market-price normal

Answer key☟

9.18.3 Cost Market Price: GATE CSE 2012 | Question: 56

The cost function for a product in a firm is given by $5q^2$, where q is the amount of production. The firm can sell the product at a market price of ₹50 per unit. The number of units to be produced by the firm such that the profit is maximized is

A. 5　B. 10　C. 15　D. 25

gatecse-2012 quantitative-aptitude cost-market-price normal

Answer key☟

9.18.4 Cost Market Price: GATE CSE 2019 | Question: GA-4

Ten friends planned to share equally the cost of buying a gift for their teacher. When two of them decided not to contribute, each of the other friends had to pay Rs. 150 more. The cost of the gift was Rs. _____

A. 666　B. 3000　C. 6000　D. 12000

gatecse-2019 general-aptitude quantitative-aptitude cost-market-price one-mark

Answer key☟

9.18.5 Cost Market Price: GATE Electrical 2022 | GA Question: 8

The price of an item is 10% cheaper in an online store S compared to the price at another online store M. Store S charges ₹ 150 for delivery. There are no delivery charges for orders from the store M. A person bought the items from the store S and saved ₹ 100.

What is the price of the item at the online store S (in ₹) if there are no other charges than what is described above?

A. 2500　B. 2250　C. 1750　D. 1500

gate2022-ee quantitative-aptitude cost-market-price

Answer key☟

9.18.6 Cost Market Price: GATE2014 AE: GA-5

A foundry has a fixed daily cost of $\text{Rs } 50,000$ whenever it operates and a variable cost of $\text{RS } 800Q$, where Q is the daily production in tonnes. What is the cost of production in Rs per tonne for a daily production of 100 tonnes.

gate2014-ae quantitative-aptitude cost-market-price numerical-answers

Answer key

9.19 Counting (6)

9.19.1 Counting: GATE Civil 2023 Set 2 | GA Question: 4

There are 4 red, 5 green, and 6 blue balls inside a box. If N number of balls are picked simultaneously, what is the smallest value of N that guarantees there will be at least two balls of the same colour?

One cannot see the colour of the balls until they are picked.

A. 4 B. 15 C. 5 D. 2

gatecivil-2023-set2 quantitative-aptitude counting

Answer key

9.19.2 Counting: GATE Electrical 2020 | GA Question: 5

If $\mathrm{P}, \mathrm{Q}, \mathrm{R}, \mathrm{S}$ are four individuals, how many teams of size exceeding one can be formed, with Q as a member?

A. 5 B. 6 C. 7 D. 8

gate2020-ee quantitative-aptitude counting

Answer key

9.19.3 Counting: GATE Mechanical 2023 | GA Question: 9

How many pairs of sets (S, T) are possible among the subsets of $\{1,2,3,4,5,6\}$ that satisfy the condition that S is a subset of T?

A. 729 B. 728 C. 665 D. 664

gateme-2023 quantitative-aptitude counting

Answer key

9.19.4 Counting: GATE2017 EC-1: GA-9

There are 3 Indians and 3 Chinese in a group of 6 people. How many subgroups of this group can we choose so that every subgroup has at least one Indian?

A. 56 B. 52 C. 48 D. 44

gate2017-ec-1 general-aptitude quantitative-aptitude counting

Answer key

9.19.5 Counting: GATE2018 CH: GA-8

To pass a test, a candidate needs to answer at least 2 out of 3 questions correctly. A total of $6,30,000$ candidates appeared for the test. Question A was correctly answered by $3,30,000$ candidates. Question B was answered correctly by $2,50,000$ candidates. Question C was answered correctly by $2,60,000$ candidates. Both questions A and B were answered correctly by $1,00,000$ candidates. Both questions B and C were answered correctly by $90,000$ candidates. Both questions A and C were answered correctly by $80,000$ candidates. If the number of students answering all questions correctly is the same as the number answering none, how many candidates failed to clear the test?

A. $30,000$ B. $2,70,000$ C. $3,90,000$ D. $4,20,000$

gate2018-ch general-aptitude quantitative-aptitude counting

Answer key

9.19.6 Counting: GATE2018 EE: GA-6

An e-mail password must contain three characters. The password has to contain one numeral from 0 to 9, one upper case and one lower case character from the English alphabet. How many distinct passwords are possible?

A. $6,760$ B. $13,520$ C. $40,560$ D. $1,05,456$

gate2018-ee general-aptitude quantitative-aptitude normal combinatory counting

Answer key

9.20 Cube (2)

9.20.1 Cube: GATE CSE 2016 Set 1 | Question: GA05

A cube is built using 64 cubic blocks of side one unit. After it is built, one cubic block is removed from every corner of the cube. The resulting surface area of the body (in square units) after the removal is ________.

a. 56
b. 64
c. 72
d. 96

gatecse-2016-set1 quantitative-aptitude geometry normal cube

Answer key

9.20.2 Cube: GATE CSE 2021 Set 2 | GA Question: 3

If θ is the angle, in degrees, between the longest diagonal of the cube and any one of the edges of the cube, then, $\cos\theta =$

A. $\frac{1}{2}$
B. $\frac{1}{\sqrt{3}}$
C. $\frac{1}{\sqrt{2}}$
D. $\frac{\sqrt{3}}{2}$

gatecse-2021-set2 quantitative-aptitude mensuration cube one-mark

Answer key

9.21 Currency Notes (1)

9.21.1 Currency Notes: GATE2012 CY: GA-10

Raju has 14 currency notes in his pocket consisting of only Rs. 20 notes and Rs. 10 notes. The total money value of the notes is Rs. 230. The number of Rs. 10 notes that Raju has is

A. 5
B. 6
C. 9
D. 10

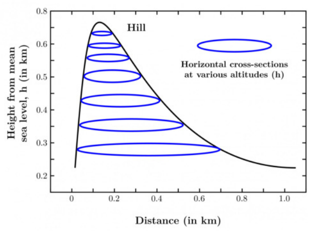
gate2012-cy quantitative-aptitude numerical-computation currency-notes

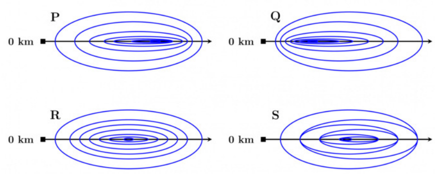
Answer key

9.22 Curves (2)

9.22.1 Curves: GATE IN 2023 | GA Question: 5

In the given diagram, ovals are marked at different heights (h) of a hill. Which one of the following options $\mathbf{P}, \mathbf{Q}, \mathbf{R}$, and $\mathbf{S}$ depicts the top view of the hill?





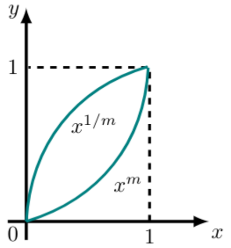
A. $\mathbf{P}$
B. $\mathbf{Q}$
C. $\mathbf{R}$
D. $\mathbf{S}$

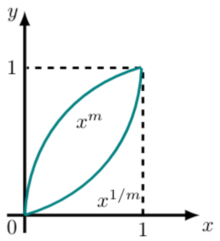
gatein-2023 quantitative-aptitude curves

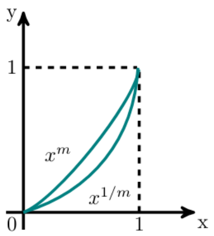
Answer key

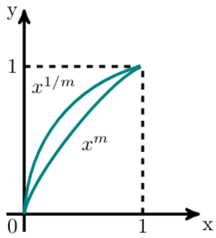
9.22.2 Curves: GATE Mechanical 2020 Set 1 | GA Question: 9

Select the graph that schematically represents BOTH $y = x^m$ and $y = x^{1/m}$ properly in the interval $0 \leq x \leq 1$, for integer values of m, where $m > 1$.

A. 

B. 

C. 

D. 

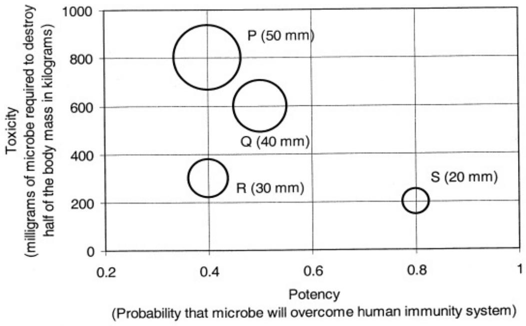
gateme-2020-set1 quantitative-aptitude functions curves

Answer key

9.23 Data Interpretation (5)

9.23.1 Data Interpretation: GATE CSE 2011 | Question: 62

P, Q, R and S are four types of dangerous microbes recently found in a human habitat. The area of each circle with its diameter printed in brackets represents the growth of a single microbe surviving human immunity system within 24 hours of entering the body. The danger to human beings varies proportionately with the toxicity, potency and growth attributed to a microbe shown in the figure below:



A pharmaceutical company is contemplating the development of a vaccine against the most dangerous microbe. Which microbe should the company target in its first attempt?

A. P
B. Q
C. R
D. S

gatecse-2011 quantitative-aptitude data-interpretation normal

Answer key☟

9.23.2 Data Interpretation: GATE DA 2025 | GA Question: 7

Weight of a person can be expressed as a function of their age. The function usually varies from person to person. Suppose this function is identical for two brothers, and it monotonically increases till the age of 50 years and then it monotonically decreases. Let a_1 and a_2 (in years) denote the ages of the brothers and $a_1 < a_2$.

Which one of the following statements is correct about their age on the day when they attain the same weight?

A. $a_1 < a_2 < 50$
B. $a_1 < 50 < a_2$
C. $50 < a_1 < a_2$
D. Either $a_1 = 50$ or $a_2 = 50$

gateda-2025 quantitative-aptitude data-interpretation two-marks

Answer key☟

9.23.3 Data Interpretation: GATE2018 CE-1: GA-5

The temperature T in a room varies as a function of the outside temperature T_0 and the number of persons in the room p, according to the relation $T = K(\theta p + T_0)$, where θ and K are constants. What would be the value of θ given the following data?

T_0	p	T
25	2	32.4
30	5	42.0

A. 0.8
B. 1.0
C. 2.0
D. 10.0

gate2018-ce-1 general-aptitude quantitative-aptitude data-interpretation

Answer key☟

9.23.4 Data Interpretation: GATE2018 EC: GA-9

A cab was involved in a hit and run accident at night. You are given the following data about the cabs in the city and the accident.

i. 85% of cabs in the city are green and the remaining cabs are blue.
ii. A witness identified the cab involved in the accident as blue.
iii. It is known that a witness can correctly identify the cab colour only 80% of the time.

Which of the following options is closest to the probability that the accident was caused by a blue cab?

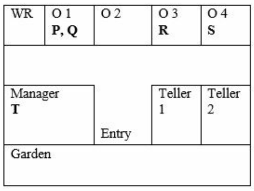
A. 12%
B. 15%
C. 41%
D. 80%

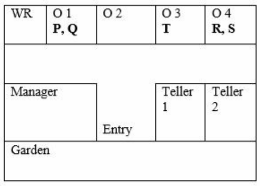
gate2018-ec general-aptitude quantitative-aptitude normal data-interpretation

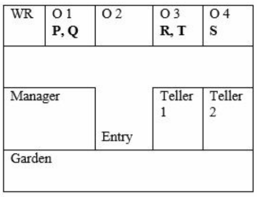
Answer key☟

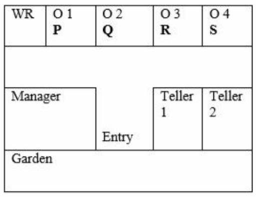
9.23.5 Data Interpretation: GATE2019 EC: GA-10

Five people P, Q, R, S and T work in a bank. P and Q don't like each other but have to share an office till T gets a promotion and moves to the big office next to the garden. R, who is currently sharing an office with T wants to move to the adjacent office with S, the handsome new intern. Given the floor plan, what is the current location of Q, R and T?
(O=Office, WR=Washroom)

A. 

B. 

C. 

D. 

gate2019-ec general-aptitude data-interpretation

Answer key

9.24 Digital Image Processing (1)

9.24.1 Digital Image Processing: GATE DA 2025 | GA Question: 3

A 4×4 digital image has pixel intensities (U) as shown in the figure. The number of pixels with $U \leq 4$ is:

0	1	0	2
4	7	3	3
5	5	4	4
6	7	3	2

A. 3 B. 8 C. 11 D. 9

gateda-2025 quantitative-aptitude digital-image-processing easy one-mark

Answer key

9.25 Factors (4)

9.25.1 Factors: GATE CSE 2013 | Question: 62

Out of all the 2-digit integers between 1 and 100, a 2-digit number has to be selected at random. What is the probability that the selected number is not divisible by 7 ?

A. $\left(\frac{13}{90}\right)$ B. $\left(\frac{12}{90}\right)$ C. $\left(\frac{78}{90}\right)$ D. $\left(\frac{77}{90}\right)$

gatecse-2013 quantitative-aptitude easy probability factors

Answer key

9.25.2 Factors: GATE CSE 2018 | Question: GA-4

What would be the smallest natural number which when divided either by 20 or by 42 or by 76 leaves a remainder of 7 in each case?

A. 3047 B. 6047 C. 7987 D. 63847

gatecse-2018 quantitative-aptitude factors one-mark

Answer key

9.25.3 Factors: GATE2010 MN: GA-8

Consider the set of integers $\{1, 2, 3, \ldots, 5000\}$. The number of integers that is divisible by neither 3 nor 4 is :

A. 1668 B. 2084 C. 2500 D. 2916

general-aptitude quantitative-aptitude gate2010-mn factors

Answer key☟

9.25.4 Factors: GATE2018 EC: GA-3

If the number $715 \blacksquare 423$ is divisible by 3 ($\blacksquare$ denotes the missing digit in the thousandths place), then the smallest whole number in the place of $\blacksquare$ is _________.

A. 0 B. 2 C. 5 D. 6

gate2018-ec general-aptitude quantitative-aptitude easy factors

Answer key☟

9.26 Fractions (4)

9.26.1 Fractions: GATE CSE 2018 | Question: GA-6

In appreciation of the social improvements completed in a town, a wealthy philanthropist decided to gift $Rs\ 750$ to each male senior citizen in the town and $Rs\ 1000$ to each female senior citizen. Altogether, there were 300 senior citizens eligible for this gift. However, only $\dfrac{8}{9}^{\text{th}}$ of the eligible men and $\dfrac{2}{3}^{\text{rd}}$ of the eligible women claimed the gift.How much money (in Rupees) did the philanthropist give away in total?

A. $1,50,000$ B. $2,00,000$ C. $1,75,000$ D. $1,51,000$

gatecse-2018 quantitative-aptitude fractions normal two-marks

Answer key☟

9.26.2 Fractions: GATE Civil 2022 Set 2 | GA Question: 3

Both the numerator and the denominator of $\dfrac{3}{4}$ are increased by a positive integer, x, and those of $\dfrac{15}{17}$ are decreased by the same integer. This operation results in the same value for both the fractions.

What is the value of x?

A. 1 B. 2 C. 3 D. 4

gatecivil-2022-set2 quantitative-aptitude fractions

Answer key☟

9.26.3 Fractions: GATE Civil 2024 Set 1 | GA Question: 3

For positive integers p and q, with $\frac{p}{q} \neq 1, \left(\frac{p}{q}\right)^{\frac{p}{q}} = p^{\left(\frac{p}{q}-1\right)}$. Then,

A. $q^p = p^q$ B. $q^p = p^{2q}$ C. $\sqrt{q} = \sqrt{p}$ D. $\sqrt[p]{q} = \sqrt[q]{p}$

gatecivil-2024-set1 quantitative-aptitude fractions

Answer key☟

9.26.4 Fractions: GATE2016 EC-1: GA-9

If $q^{-a} = \dfrac{1}{r}$ and $r^{-b} = \dfrac{1}{s}$ and $s^{-c} = \dfrac{1}{q}$, the value of abc is ________.

A. $(rqs)^{-1}$ B. 0 C. 1 D. $r + q + s$

gate2016-ec-1 quantitative-aptitude fractions

Answer key☟

9.27 Functions (18)

9.27.1 Functions: GATE CSE 2015 Set 2 | Question: GA-9

If p, q, r, s are distinct integers such that:

$$f(p,q,r,s) = \max(p,q,r,s)$$

$$g(p,q,r,s) = \min(p,q,r,s)$$

$$h(p,q,r,s) = \text{remainder of } \frac{(p\times q)}{(r\times s)} \text{ if } (p\times q) > (r\times s)$$
$$\text{or remainder of } \frac{(r\times s)}{(p\times q)} \text{ if } (r\times s) > (p\times q)$$

Also a function $fgh(p,q,r,s) = f(p,q,r,s) \times g(p,q,r,s) \times h(p,q,r,s)$
Also the same operations are valid with two variable functions of the form $f(p,q)$
What is the value of $fg(h(2,5,7,3), 4,6,8)$?

gatecse-2015-set2 functions normal numerical-answers

Answer key

9.27.2 Functions: GATE CSE 2015 Set 3 | Question: GA-5

A function $f(x)$ is linear and has a value of 29 at $x = -2$ and 39 at $x = 3$. Find its value at $x = 5$.

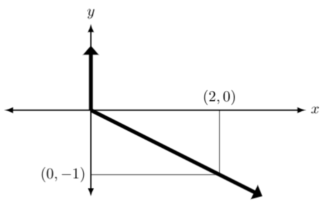
A. 59
B. 45
C. 43
D. 35

gatecse-2015-set3 quantitative-aptitude normal functions

Answer key

9.27.3 Functions: GATE CSE 2015 Set 3 | Question: GA-8

Choose the most appropriate equation for the function drawn as thick line, in the plot below.



A. $x = y - |y|$
B. $x = -(y - |y|)$
C. $x = y + |y|$
D. $x = -(y + |y|)$

gatecse-2015-set3 quantitative-aptitude normal functions

Answer key

9.27.4 Functions: GATE CSE 2023 | GA Question: 7

Consider two functions of time (t),

$$f(t) = 0.01t^2$$
$$g(t) = 4t$$

where $0 < t < \infty$.
Now consider the following two statements:

i. For some $t > 0, g(t) > f(t)$.
ii. There exists a T, such that $f(t) > g(t)$ for all $t > T$.

Which one of the following options is TRUE?

A. only (i) is correct
B. only (ii) is correct
C. both (i) and (ii) are correct
D. neither (i) nor (ii) is correct

gatecse-2023 quantitative-aptitude functions two-marks

Answer key

9.27.5 Functions: GATE CSE 2023 | GA Question: 9

$f(x)$ and $g(y)$ are functions of x and y, respectively, and $f(x) = g(y)$ for all values of x and y. Which one of the following options is necessarily TRUE for a x and y?

A. $f(x) = 0$ and $g(y) = 0$
B. $f(x) = g(y) =$ constant
C. $f(x) \neq$ constant and $g(y) \neq$ constant
D. $f(x) + g(y) = f(x) - g(y)$

gatecse-2023 quantitative-aptitude functions two-marks

Answer key

9.27.6 Functions: GATE Civil 2020 Set 2 | GA Question: 5

If $f(x) = x^2$ for each $x \in (-\infty, \infty)$, then $\dfrac{f\left(f\left(f(x)\right)\right)}{f(x)}$ is equal to _______.

A. $f(x)$
B. $(f(x))^2$
C. $(f(x))^3$
D. $(f(x))^4$

gate2020-ce-2 quantitative-aptitude functions

Answer key

9.27.7 Functions: GATE Civil 2021 Set 1 | GA Question: 9

A function, λ, is defined by

$$\lambda(p,q) = \begin{cases} (p-q)^2, & \text{if } p \geq q, \\ p+q, & \text{if } p < q. \end{cases}$$

The value of the expression $\dfrac{\lambda\left(-(-3+2), (-2+3)\right)}{\left(-(-2+1)\right)}$ is:

A. -1
B. 0
C. $\frac{16}{3}$
D. 16

gatecivil-2021-set1 quantitative-aptitude functions easy

Answer key

9.27.8 Functions: GATE ECE 2020 | GA Question: 5

A superadditive function $f(\cdot)$ satisfies the following property

$$f(x_1 + x_2) \geq f(x_1) + f(x_2)$$

Which of the following functions is a superadditive function for $x > 1$?

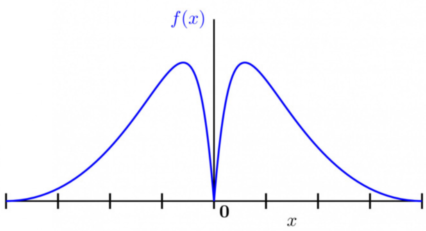
A. e^x
B. $\sqrt{x}$
C. $1/x$
D. e^{-x}

gate2020-ec quantitative-aptitude functions

Answer key

9.27.9 Functions: GATE ECE 2023 | GA Question: 7

Which one of the following options represents the given graph?



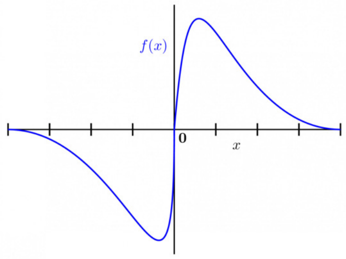
A. $f(x) = x^2 2^{-|x|}$ B. $f(x) = x2^{-|x|}$ C. $f(x) = |x|2^{-x}$ D. $f(x) = x2^{-x}$

gateece-2023 quantitative-aptitude functions

Answer key

9.27.10 Functions: GATE Electrical 2023 | GA Question: 7

Which one of the following options represents the given graph?



A. $f(x) = x^2 2^{-|x|}$
B. $f(x) = x2^{-|x|}$
C. $f(x) = |x|2^{-x}$
D. $f(x) = x2^{-x}$

gate2023-ee quantitative-aptitude functions

Answer key

9.27.11 Functions: GATE IN 2023 | GA Question: 7

Ankita has to climb 5 stairs starting at the ground, while respecting the following rules:

1. At any stage, Ankita can move either one or two stairs up.
2. At any stage, Ankita cannot move to a lower step.

Let $F(N)$ denote the number of possible ways in which Ankita can reach the N^{th} stair. For example, $F(1) = 1, F(2) = 2, F(3) = 3$.

The value of $F(5)$ is__________.

A. 8 B. 7 C. 6 D. 5

gatein-2023 quantitative-aptitude functions

Answer key

9.27.12 Functions: GATE Mechanical 2020 Set 1 | GA Question: 5

Define $[x]$ as the greatest integer less than or equal to x, for each $x \in (-\infty, \infty)$. If $y = [x]$, then area under y for $x \in [1, 4]$ is _______.

A. 1 B. 3 C. 4 D. 6

gateme-2020-set1 quantitative-aptitude functions

Answer key

9.27.13 Functions: GATE Mechanical 2022 Set 2 | GA Question: 8

Consider the following for non-zero positive integers, p and q.

$f(p,q) = \underbrace{p \times p \times p \times \cdots\cdots\cdots \times p = p^q}_{q \text{ terms}}; f(p,1) = p$

$g(p,q) = p^{p^{p^{p^{(\text{up to q terms})}}}}; g(p,1) = p$

Which one of the following options is correct based on the above?

A. $f(2,2) = g(2,2)$
B. $f(g(2,2),2) < f(2,g(2,2))$
C. $g(2,1) \neq f(2,1)$
D. $f(3,2) > g(3,2)$

gateme-2022-set2 quantitative-aptitude functions

Answer key☟

9.27.14 Functions: GATE2010 MN: GA-10

Given the following four functions $f_1(n) = n^{100}$, $f_2(n) = (1.2)^n$, $f_3(n) = 2^{n/2}$, $f_4(n) = 3^{n/3}$ which function will have the largest value for sufficiently large values of n $(i.e.\ n \to \infty)$?

A. f_4
B. f_3
C. f_2
D. f_1

general-aptitude quantitative-aptitude gate2010-mn functions

Answer key☟

9.27.15 Functions: GATE2012 AR: GA-7

Let $f(x) = x - [x]$, where $x \geq 0$ and $[x]$ is the greatest integer not larger than x. Then $f(x)$ is a

A. monotonically increasing function
B. monotonically decreasing function
C. linearly increasing function between two integers
D. linearly decreasing function between two integers

gate2012-ar quantitative-aptitude functions normal

Answer key☟

9.27.16 Functions: GATE2015 EC-3: GA-5

If $x > y > 1$, which of the following must be true?

i. $\ln x > \ln y$
ii. $e^x > e^y$
iii. $y^x > x^y$
iv. $\cos x > \cos y$

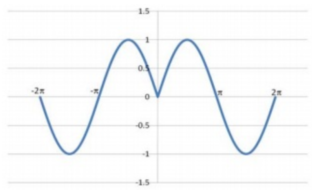
A. (i) and (ii)
B. (i) and (iii)
C. (iii) and (iv)
D. (ii) and (iv)

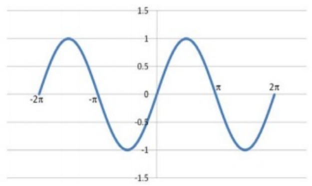
gate2015-ec-3 general-aptitude quantitative-aptitude functions

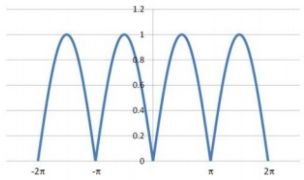
Answer key☟

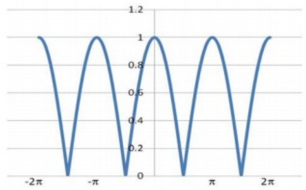
9.27.17 Functions: GATE2016 ME-2: GA-10

Which of the following curves represents the function $y = \ln\left(\left| e^{[|\sin(|x|)|]} \right|\right)$ for $|x| < 2\pi$? Here, x represents the abscissa and y represents the ordinate.

A.


B.


C. 

D. 

gate2016-me-2 functions quantitative-aptitude

Answer key

9.27.18 Functions: GATE2018 EE: GA-5

Functions $F(a, b)$ and $G(a, b)$ are defined as follows:
$F(a, b) = (a - b)^2$ and $G(a, b) = \mid a - b \mid$, where $|x|$ represents the absolute value of x.

What would be the value of $G(F(1, 3), G(1, 3))$?

A. 2 B. 4 C. 6 D. 36

gate2018-ee general-aptitude quantitative-aptitude easy functions

Answer key

9.28 Geometry (29)

9.28.1 Geometry: GATE 2024 Electrical | GA Question: 10

A surveyor has to measure the horizontal distance from her position to a distant reference point C. Using her position as the center, a $200 \mathrm{~m}$ horizontal line segment is drawn with the two endpoints A and B. Points A, B, and C are not collinear. Each of the angles $\angle \mathrm{CAB}$ and $\angle \mathrm{CBA}$ are measured as 87.8°. The distance (in m) of the reference point C from her position is nearest to

A. 2603 B. 2606 C. 2306 D. 2063

gate2024-ee quantitative-aptitude geometry

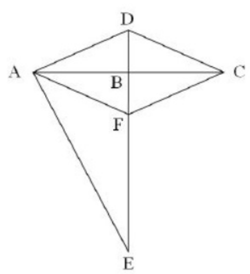
Answer key

9.28.2 Geometry: GATE 2024 Electrical | GA Question: 7

In the following figure,

$$\begin{gathered}\mathrm{CD}=5 \mathrm{~cm}, \mathrm{BE}=10 \mathrm{~cm}, \mathrm{AE}=12 \mathrm{~cm}, \\ \angle \mathrm{DAB}=\angle \mathrm{DCB}, \text { and } \angle \mathrm{DAE}=\angle \mathrm{DBC}=90^{\circ}\end{gathered}$$

Points AFCD create a rhombus.



The length of BF (in cm) is

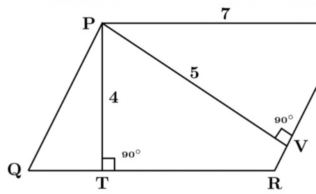
A. 3 B. 2 C. 4 D. 6

gate2024-ee quantitative-aptitude geometry

Answer key☟

9.28.3 Geometry: GATE CH 2023 | GA Question: 3

In the given figure, PQRS is a parallelogram with $\mathrm{PS}=7\ \mathrm{cm}, \mathrm{PT}=4\ \mathrm{cm}$ and $\mathrm{PV}=5\ \mathrm{cm}$. What is the length of RS in cm? (The diagram is representative.)



A. $\frac{20}{7}$ B. $\frac{28}{5}$ C. $\frac{9}{2}$ D. $\frac{35}{4}$

gatech-2023 quantitative-aptitude geometry

Answer key☟

9.28.4 Geometry: GATE CSE 2014 Set 1 | Question: GA-10

When a point inside of a tetrahedron (a solid with four triangular surfaces) is connected by straight lines to its corners, how many (new) internal planes are created with these lines?

gatecse-2014-set1 quantitative-aptitude geometry combinatory normal numerical-answers

Answer key☟

9.28.5 Geometry: GATE CSE 2024 | Set 1 | GA Question: 7

A rectangular paper sheet of dimensions $54\ \mathrm{cm} \times 4\ \mathrm{cm}$ is taken. The two longer edges of the sheet are joined together to create a cylindrical tube. A cube whose surface area is equal to the area of the sheet is also taken.

Then, the ratio of the volume of the cylindrical tube to the volume of the cube is

A. $1/\pi$ B. $2/\pi$ C. $3/\pi$ D. $4/\pi$

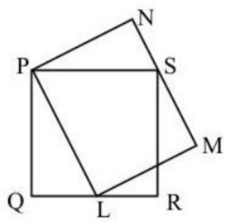
gatecse2024-set1 quantitative-aptitude geometry two-marks

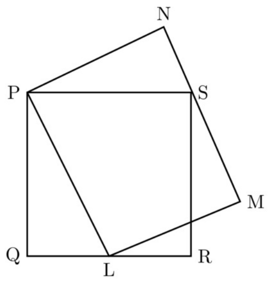
Answer key☟

9.28.6 Geometry: GATE CSE 2025 | Set 2 | GA Question: 9

In the Given figure, PQRS is a square of side 2 cm and PLMN is a rectangle. The corner L of teh rectangle is on the side QR. Side MN of the rectangle passes through the corner S of the square.

What is the area (in cm^2) of the rectangle PLMN?
Note: The figure shown is representative.



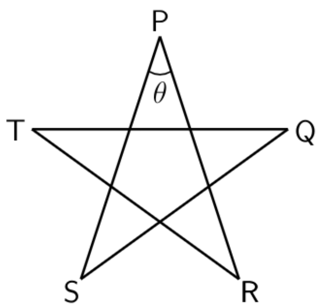


A. $2\sqrt{2}$ B. 2 C. 8 D. 4

gatecse2025-set2 quantitative-aptitude geometry two-marks

Answer key

9.28.7 Geometry: GATE Civil 2021 Set 1 | GA Question: 8



Five line segments of equal lengths, $\mathrm{PR}, \mathrm{PS}, \mathrm{QS}, \mathrm{QT}$ and RT are used to form a star as shown in the figure above. The value of θ, in degrees, is _______________

A. 36 B. 45 C. 72 D. 108

gatecivil-2021-set1 quantitative-aptitude geometry

Answer key

9.28.8 Geometry: GATE Civil 2024 Set 1 | GA Question: 10

During a half-moon phase, the Earth-Moon-Sun form a right triangle. If the Moon-Earth-Sun angle at this half-moon phase is measured to be $89.85\ ^{\circ}$, the ratio of the Earth-Sun and Earth-Moon distances is closest to

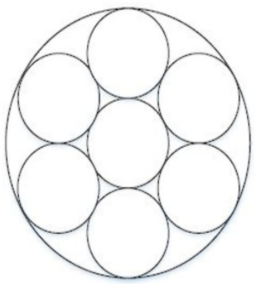
A. 328 B. 382 C. 238 D. 283

gatecivil-2024-set1 quantitative-aptitude geometry

Answer key

9.28.9 Geometry: GATE Civil 2024 Set 2 | GA Question: 7

Seven identical cylindrical chalk-sticks are fitted tightly in a cylindrical container. The figure below shows the arrangement of the chalk-sticks inside the cylinder.



The length of the container is equal to the length of the chalk-sticks. The ratio of the occupied space to the empty space of the container is

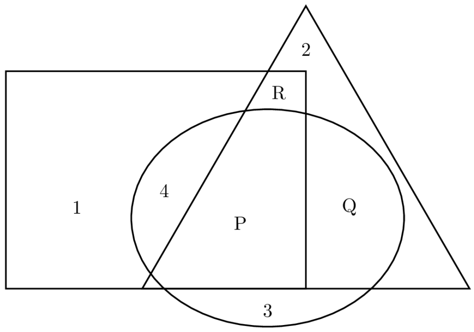
A. $5/2$ B. $7/2$ C. $9/2$ D. 3

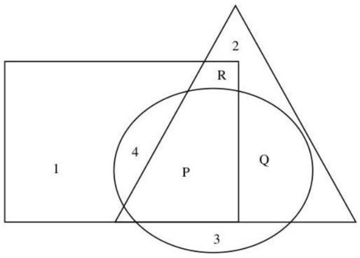
gatecivil-2024-set2 quantitative-aptitude geometry

Answer key☟

9.28.10 Geometry: GATE DA 2025 | GA Question: 4

In the given figure, the numbers associated with the rectangle, triangle, and ellipse are $1, 2$, and 3, respectively. Which one among the given options is the most appropriate combination of P, Q, and R ?





A. $P=6; Q=5; R=3$
B. $P=5; Q=6; R=3$
C. $P=3; Q=6; R=6$
D. $P=5; Q=3; R=6$

gateda-2025 quantitative-aptitude geometry one-mark

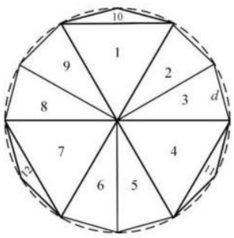
Answer key☟

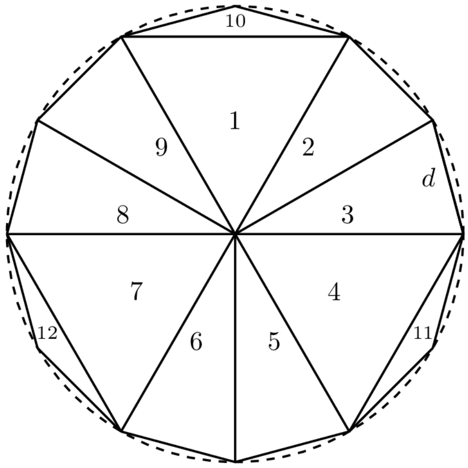
9.28.11 Geometry: GATE DA 2025 | GA Question: 8

A regular dodecagon (12-sided regular polygon) is inscribed in a circle of radius r cm as shown in the figure. The side of the dodecagon is d cm. All the triangles (numbered 1 to 12) in the figure are used to form squares of side r cm and each numbered triangle is used only once to form a square.

The number of squares that can be formed and the number of triangles required to form each square, respectively, are:

Note: The figure shown is representative.





A. $3;4$ B. $4;3$ C. $3;3$ D. $3;2$

gateda-2025 quantitative-aptitude geometry two-marks

Answer key

9.28.12 Geometry: GATE ECE 2022 | GA Question: 3

A trapezium has vertices marked as $\mathrm{P}, \mathrm{Q}, \mathrm{R}$ and S (in that order anticlockwise). The side PQ is parallel to side SR.

Further, it is given that, $\mathrm{PQ} = 11$ cm, $\mathrm{QR} = 4$ cm, $\mathrm{RS} = 6$ cm and $\mathrm{SP} = 3$ cm.

What is the shortest distance between PQ and SR (in cm)?

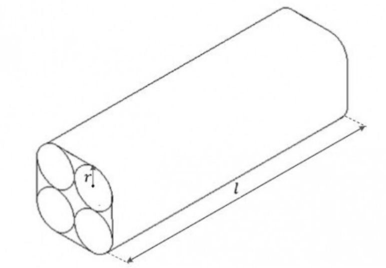
A. 1.80 B. 2.40 C. 4.20 D. 5.76

gateece-2022 quantitative-aptitude geometry

Answer key

9.28.13 Geometry: GATE ECE 2024 | GA Question: 7

Four identical cylindrical chalk-sticks, each of radius $r = 0.5$ cm and length $l = 10$ cm, are bound tightly together using a duct tape as shown in the following figure.



The width of the duct tape is equal to the length of the chalk-stick. The area (in cm^2) of the duct tape required to wrap the bundle of chalk-sticks once, is

A. $20(4+\pi)$ B. $20(8+\pi)$ C. $10(8+\pi)$ D. $10(4+\pi)$

gateece-2024 quantitative-aptitude geometry

Answer key

9.28.14 Geometry: GATE Mechanical 2022 Set 1 | GA Question: 7

A rhombus is formed by joining the midpoints of the sides of a unit square.

What is the diameter of the largest circle that can be inscribed within the rhombus?

A. $\frac{1}{\sqrt{2}}$ B. $\frac{1}{2\sqrt{2}}$ C. $\sqrt{2}$ D. $2\sqrt{2}$

gateme-2022-set1 quantitative-aptitude geometry

Answer key

9.28.15 Geometry: GATE2015 EC-3: GA-8

From a circular sheet of paper of radius 30 cm, a sector of 10% area is removed. If the remaining part is used to make a conical surface, then the ratio of the radius and height of the cone is ______

gate2015-ec-3 geometry quantitative-aptitude normal

Answer key

9.28.16 Geometry: GATE2016 CE-2: GA-9

A square pyramid has a base perimeter x, and the slant height is half of the perimeter. What is the lateral surface area of the pyramid

A. x^2 B. $0.75x^2$ C. $0.50x^2$ D. $0.25x^2$

gate2016-ce-2 geometry quantitative-aptitude

Answer key

9.28.17 Geometry: GATE2016 EC-2: GA-10

A wire of length 340 mm is to be cut into two parts. One of the parts is to be made into a square and the other into a rectangle where sides are in the ratio of $1:2$. What is the length of the side of the square (in mm) such that the combined area of the square and the rectangle is a **MINIMUM**?

A. 30 B. 40 C. 120 D. 180

gate2016-ec-2 geometry quantitative-aptitude

Answer key

9.28.18 Geometry: GATE2016 ME-2: GA-5

A window is made up of a square portion and an equilateral triangle portion above it. The base of the triangular portion coincides with the upper side of the square. If the perimeter of the window is 6 m, the area of the window in m^2 is ___________.

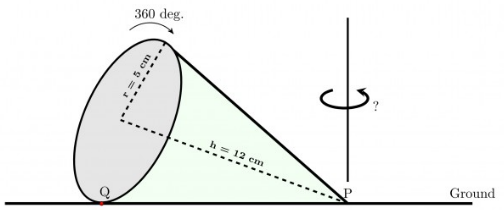
A. 1.43 B. 2.06 C. 2.68 D. 2.88

gate2016-me-2 quantitative-aptitude geometry

Answer key

9.28.19 Geometry: GATE2017 ME-1: GA-3

A right-angled cone (with base radius 5 cm and height 12 cm), as shown in the figure below, is rolled on the ground keeping the point P fixed until the point Q (at the base of the cone, as shown) touches the ground again.



By what angle (in radians) about P does the cone travel?

A. $\frac{5\pi}{12}$ B. $\frac{5\pi}{24}$ C. $\frac{24\pi}{5}$ D. $\frac{10\pi}{13}$

gate2017-me-1 general-aptitude quantitative-aptitude geometry

Answer key

9.28.20 Geometry: GATE2018 CE-1: GA-4

Tower A is $90\ m$ tall and tower B is $140\ m$ tall. They are $100\ m$ apart. A horizontal skywalk connects the floors at $70\ m$ in both the towers. If a taut rope connects the top of tower A to the bottom tower B, at what distance (in meters) from tower A will the rope intersect the skywalk?

gate2018-ce-1 general-aptitude quantitative-aptitude geometry numerical-answers

Answer key☟

9.28.21 Geometry: GATE2018 CH: GA-5

Arrange the following three-dimensional objects in the descending order of their volumes:

i. A cuboid with dimensions $10\ \text{cm}, 8\ \text{cm and } 6\ \text{cm}$
ii. A cube of side $8\ \text{cm}$
iii. A cylinder with base radius $7\ \text{cm}$ and height $7\ \text{cm}$
iv. A sphere of radius $7\ \text{cm}$

A. $\text{i}), \text{ii}), \text{iii}), \text{iv})$
B. $\text{ii}), \text{i}), \text{iv}), \text{iii})$
C. $\text{iii}), \text{ii}), \text{i}), \text{iv})$
D. $\text{iv}), \text{iii}), \text{ii}), \text{i})$

gate2018-ch quantitative-aptitude normal geometry

Answer key☟

9.28.22 Geometry: GATE2018 CH: GA-7

A set of 4 parallel lines intersect with another set of 5 parallel lines. How many parallelograms are formed?

A. 20
B. 48
C. 60
D. 72

gate2018-ch general-aptitude quantitative-aptitude easy geometry

Answer key☟

9.28.23 Geometry: GATE2018 EC: GA-5

A $1.5m$ tall person is standing at a distance of $3m$ from a lamp post. The light from the lamp at the top of the post casts her shadow. The length of the shadow is twice her height. What is the height of the lamp post in meters?

A. 1.5
B. 3
C. 4.5
D. 6

gate2018-ec general-aptitude quantitative-aptitude normal geometry

Answer key☟

9.28.24 Geometry: GATE2018 ME-1: GA-4

A rectangle becomes a square when its length and breadth are reduced by 10 m and 5 m, respectively. During this process, the rectangle loses 650 m^2 of area. What is the area of the original rectangle in square meters?

A. 1125
B. 2250
C. 2924
D. 4500

gate2018-me-1 general-aptitude quantitative-aptitude geometry

Answer key☟

9.28.25 Geometry: GATE2018 ME-2: GA-4

The perimeters of a circle, a square and an equilateral triangle are equal. Which one of the following statements is true?

A. The circle has the largest area
B. The square has the largest area
C. The equilateral triangle has the largest area
D. All the three shapes have the same area

gate2018-me-2 general-aptitude quantitative-aptitude geometry easy

Answer key☟

9.28.26 Geometry: GATE2018 ME-2: GA-7

A wire would enclose an area of 1936 m^2, if it is bent to a square. The wire is cut into two pieces. The longer piece is thrice as long as the shorter piece. The long and the short pieces are bent into a square and a circle, respectively. Which of the following choices is closest to the sum of the areas enclosed by the two pieces in square meters?

A. 1096 B. 1111 C. 1243 D. 2486

gate2018-me-2 general-aptitude quantitative-aptitude geometry

Answer key☟

9.28.27 Geometry: GATE2019 CE-1: GA-3

On a horizontal ground, the base of a straight ladder is 6 m away from the base of a vertical pole. The ladder makes an angle of 45° to the horizontal. If the ladder is resting at a point located at one-fifth of the height of the pole from the bottom, the height of the pole is ______ meters.

A. 15 B. 25 C. 30 D. 35

gate2019-ce-1 general-aptitude quantitative-aptitude geometry

Answer key☟

9.28.28 Geometry: GATE2019 CE-2: GA-3

Suresh wanted to lay a new carpet in his new mansion with an area of 70×55 sq.mts. However an area of 550 sq. mts. had to be left out for flower pots. If the cost carpet is Rs.50 sq. mts. how much money (in Rs.) will be spent by Suresh for the carpet now?

A. $Rs.1,65,000$ B. $Rs.1,92,500$ C. $Rs.2,75,000$ D. $Rs.1,27,500$

gate2019-ce-2 general-aptitude quantitative-aptitude geometry

Answer key☟

9.28.29 Geometry: GATE2019 CE-2: GA-4

A retaining wall with measurements 30 m $\times 12$ m $\times 6$ m was constructed with bricks of dimensions 8 cm $\times 6$ cm $\times 6$ cm. If 60% of the wall consists of bricks, the number of bricks used for the construction is _______ lakhs.

A. 30 B. 40 C. 45 D. 75

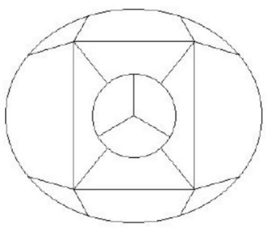
gate2019-ce-2 general-aptitude quantitative-aptitude geometry

Answer key☟

9.29 Graph Coloring (1)

9.29.1 Graph Coloring: GATE DS&AI 2024 | GA Question: 2

The 15 parts of the given figure are to be painted such that no two adjacent parts with shared boundaries (excluding corners) have the same color. The minimum number of colors required is



A. 4 B. 3 C. 5 D. 6

gate-ds-ai-2024 quantitative-aptitude graph-coloring one-mark

Answer key☟

9.30 Inequality (4)

9.30.1 Inequality: GATE CH 2022 | GA Question: 8

Consider the following inequalities.

I. $3p - q < 4$
II. $3q - p < 12$

Which one of the following expressions below satisfies the above two inequalities?

A. $p + q < 8$
B. $p + q = 8$
C. $8 \leq p + q < 16$
D. $p + q \geq 16$

gatech-2022 quantitative-aptitude inequality

Answer key☟

9.30.2 Inequality: GATE ECE 2022 | GA Question: 7

Consider the following inequalities.

A. $2x – 1 > 7$
B. $2x – 9 < 1$

Which one of the following expressions below satisfies the above two inequalities?

A. $x \leq –4$
B. $-4 < x \leq 4$
C. $4 < x < 5$
D. $x \geq 5$

gateece-2022 quantitative-aptitude inequality

Answer key☟

9.30.3 Inequality: GATE Mechanical 2022 Set 2 | GA Question: 2

Which one of the following is a representation (not to scale and in bold) of all values of x satisfying the inequality $2 - 5x \leq -\frac{6x - 5}{3}$ on the real number line?

A. 

B. 

C. 

D. 

gateme-2022-set2 quantitative-aptitude inequality

Answer key☟

9.30.4 Inequality: GATE Mechanical 2023 | GA Question: 7

Consider the following inequalities

$$p^2 - 4q < 4$$
$$3p + 2q < 6$$

where p and q are positive integers.

The value of $(p + q)$ is __________.

A. 2
B. 1
C. 3
D. 4

gateme-2023 quantitative-aptitude inequality

Answer key☟

9.31 LCM HCF (1)

9.31.1 LCM HCF: GATE Electrical 2022 | GA Question: 3

Three bells P, Q and R are rung periodically in a school. P is rung every 20 minutes; Q is rung every 30 minutes and R is rung every 50 minutes.

If all the three bells are rung at $12:00\,\text{PM}$, when will the three bells ring together again the next time?

A. $5:00\,\text{PM}$ B. $5:30\,\text{PM}$ C. $6:00\,\text{PM}$ D. $6:30\,\text{PM}$

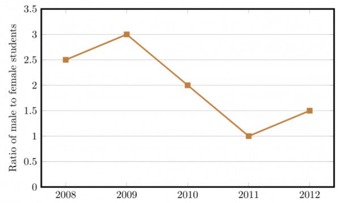
gate2022-ee   quantitative-aptitude   lcm-hcf

Answer key

9.32 Line Graph (12)

9.32.1 Line Graph: GATE CSE 2014 Set 2 | Question: GA-9

The ratio of male to female students in a college for five years is plotted in the following line graph. If the number of female students doubled in 2009, by what percent did the number of male students increase in 2009?

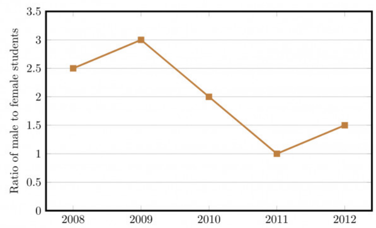


gatecse-2014-set2   quantitative-aptitude   data-interpretation   numerical-answers   normal   line-graph

Answer key

9.32.2 Line Graph: GATE CSE 2014 Set 3 | Question: GA-9

The ratio of male to female students in a college for five years is plotted in the following line graph. If the number of female students in 2011 and 2012 is equal, what is the ratio of male students in 2012 to male students in 2011?

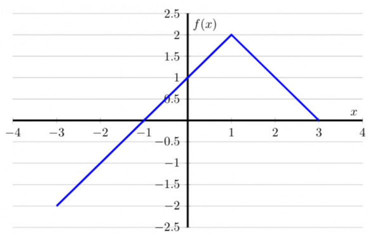


A. $1:1$ B. $2:1$ C. $1.5:1$ D. $2.5:1$

gatecse-2014-set3   quantitative-aptitude   data-interpretation   normal   line-graph

Answer key

9.32.3 Line Graph: GATE CSE 2016 Set 2 | Question: GA-10



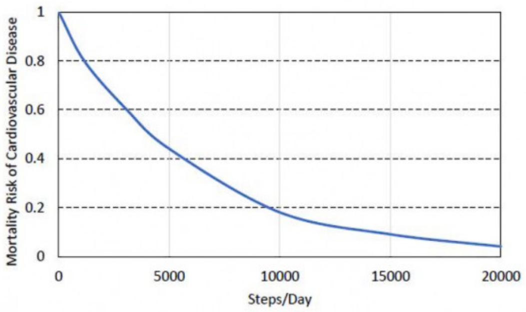
A. $f(x) = 1 - |x - 1|$ B. $f(x) = 1 + |x - 1|$
C. $f(x) = 2 - |x - 1|$ D. $f(x) = 2 + |x - 1|$

gatecse-2016-set2   quantitative-aptitude   data-interpretation   normal   line-graph

Answer key

9.32.4 Line Graph: GATE Civil 2024 Set 2 | GA Question: 8

The plot below shows the relationship between the mortality risk of cardiovascular disease and the number of steps a person walks per day. Based on the data, which one of the following options is true?



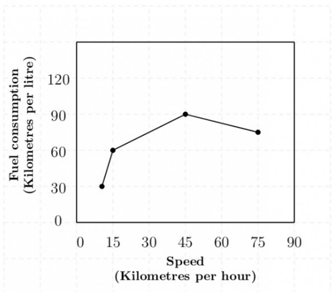
A. The risk reduction on increasing the steps/day from 0 to 10000 is less than the risk reduction on increasing the steps/day from 10000 to 20000.
B. The risk reduction on increasing the steps/day from 0 to 5000 is less than the risk reduction on increasing the steps/day from 15000 to 20000.
C. For any 5000 increment in steps/day the largest risk reduction occurs on going from 0 to 5000.
D. For any 5000 increment in steps/day the largest risk reduction occurs on going from 15000 to 20000.

gatecivil-2024-set2 line-graph quantitative-aptitude data-interpretation

Answer key☟

9.32.5 Line Graph: GATE2011 AG: GA-9

The fuel consumed by a motorcycle during a journey while traveling at various speeds is indicated in the graph below.



The distances covered during four laps of the journey are listed in the table below

Lap	Distance (kilometres)	Average Speed (kilometres per hour)
P	15	15
Q	75	45
R	40	75
S	10	10

From the given data, we can conclude that the fuel consumed per kilometre was least during the lap

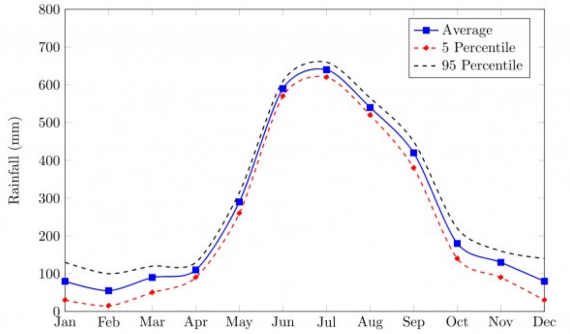
A. P B. Q C. R D. S

general-aptitude quantitative-aptitude gate2011-ag data-interpretation line-graph

Answer key☟

9.32.6 Line Graph: GATE2014 AE: GA-10

The monthly rainfall chart based on 50 years of rainfall in Agra is shown in the following figure.



Which of the following are true? (k percentile is the value such that k percent of the data fall below that value)

i. On average, it rains more in July than in December
ii. Every year, the amount of rainfall in August is more than that in January
iii. July rainfall can be estimated with better confidence than February rainfall
iv. In August, there is at least 500 mm of rainfall

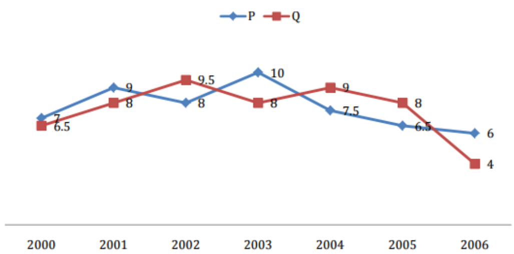
A. (i) and (ii) B. (i) and (iii) C. (ii) and (iii) D. (iii) and (iv)

gate2014-ae quantitative-aptitude data-interpretation line-graph

Answer key

9.32.7 Line Graph: GATE2016 CE-2: GA-6

Two finance companies, P and Q, declared fixed annual rates of interest on the amounts invested with them. The rates of interest offered by these companies may differ from year to year. Year-wise annual rates of interest offered by these companies are shown by the line graph provided below.



If the amounts invested in the companies, P and Q, in 2006 are in the ratio $8:9$, then the amounts received after one year as interests from companies P and Q would be in the ratio:

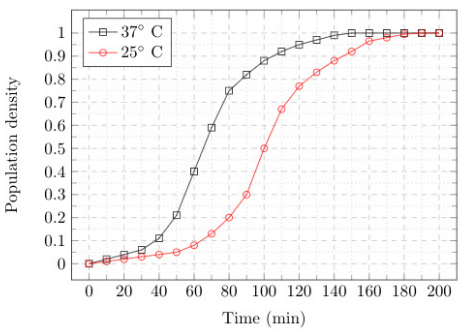
A. $2:3$ B. $3:4$ C. $6:7$ D. $4:3$

gate2016-ce-2 data-interpretation quantitative-aptitude line-graph

Answer key

9.32.8 Line Graph: GATE2017 ME-1: GA-10

The growth of bacteria (lactobacillus) in milk leads to curd formation. A minimum bacterial population density of 0.8 (in suitable units) is needed to form curd. In the graph below, the population density of lactobacillus in 1 litre of milk is plotted as a function of time, at two different temperatures, 25°C and 37°C.



Consider the following statements based on the data shown above:

i. The growth in bacterial population stops earlier at 37°C as compared to 25°C
ii. The time taken for curd formation at 25°C is twice the time taken at 37°C

Which one of the following options is correct?

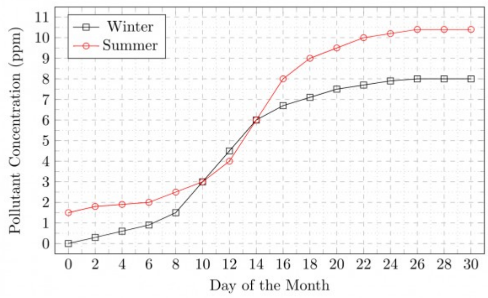
A. Only i
B. Only ii
C. Both i and ii
D. Neither i nor ii

gate2017-me-1 general-aptitude quantitative-aptitude data-interpretation line-graph

Answer key☟

9.32.9 Line Graph: GATE2017 ME-2: GA-10

In the graph below, the concentration of a particular pollutant in a lake is plotted over (alternate) days of a month in winter (average temperature $10^{\circ}C$) and a month in summer (average temperature $30^{\circ}C$).



Consider the following statements based on the data shown above:

i. Over the given months, the difference between the maximum and the minimum pollutant concentrations is the same in both winter and summer
ii. There are at least four days in the summer month such that the pollutant concentrations on those days are within 1 ppm of the pollutant concentrations on the corresponding days in the winter month.

Which one of the following options is correct?

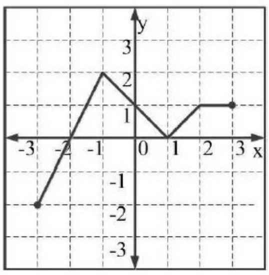
A. Only i
B. Only ii
C. Both i and ii
D. Neither i nor ii

gate2017-me-2 general-aptitude quantitative-aptitude data-interpretation line-graph

Answer key☟

9.32.10 Line Graph: GATE2018 CE-1: GA-8

Which of the following function(s) is an accurate description of the graph for the range(s) indicated?



i. $y = 2x + 4$ for $-3 \leq x \leq -1$
ii. $y = |x - 1|$ for $-1 \leq x \leq 2$
iii. $y = ||x| - 1|$ for $-1 \leq x \leq 2$
iv. $y = 1$ for $2 \leq x \leq 3$

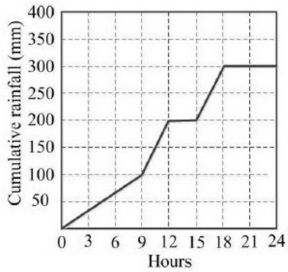
A. i, ii and iii only
B. i, ii and iv only
C. i and iv only
D. ii and iv only

gate2018-ce-1 general-aptitude quantitative-aptitude line-graph

Answer key☟

9.32.11 Line Graph: GATE2018 CE-2: GA-8

The annual average rainfall in a tropical city is 1000 mm. On a particular rainy day (24-hour period), the cumulative rainfall experienced by the city is shown in the graph. Over the 24-hour period, 50% of the rainfall falling on a rooftop, which had an obstruction-free area of $50\ m^2$, was harvested into a tank. What is the total volume of water collected in the tank liters?



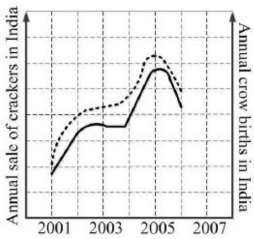
A. $25,000$ B. $18,750$ C. $7,500$ D. 3.125

gate2018-ce-2 general-aptitude quantitative-aptitude data-interpretation line-graph

Answer key

9.32.12 Line Graph: GATE2018 CH: GA-10

In a detailed study of annual crow births in India, it was found that there was relatively no growth during the period 2002 to 2004 and a sudden spike from 2004 to 2005. In another unrelated study, it was found that the revenue from cracker sales in India which remained fairly flat from 2002 to 2004, saw a sudden spike in 2005 before declining again in 2006. The solid line in the graph below refers to annual sale of crackers and the dashed line refers to the annual crow births in India. Choose the most appropriate inference from the above data.



A. There is a strong correlation between crow birth and cracker sales.
B. Cracker usage increases crow birth rate.
C. If cracker sale declines, crow birth will decline.
D. Increased birth rate of crows will cause an increase in the sale of crackers.

gate2018-ch general-aptitude quantitative-aptitude data-interpretation line-graph

Answer key

9.33 Lines (1)

9.33.1 Lines: GATE Civil 2022 Set 2 | GA Question: 8

Consider the following equations of straight lines:

$\text{Line L1}: 2x - 3y = 5$

$\text{Line L2}: 3x + 2y = 8$

$\text{Line L3}: 4x - 6y = 5$

$\text{Line L4}: 6x - 9y = 6$

Which one among the following is the correct statement?

A. L1 is parallel to L2 and L1 is perpendicular to L3
B. L2 is parallel to L4 and L2 is perpendicular to L1
C. L3 is perpendicular to L4 and L3 is parallel to L2
D. L4 is perpendicular to L2 and L4 is parallel to L3

gatecivil-2022-set2 quantitative-aptitude geometry lines

Answer key☟

9.34 Logarithms (14)

9.34.1 Logarithms: GATE CSE 2011 | Question: 57

If $\log(\mathrm{P}) = (1/2)\log(\mathrm{Q}) = (1/3)\log(\mathrm{R})$, then which of the following options is **TRUE**?

A. $\mathrm{P}^2 = \mathrm{Q}^3\mathrm{R}^2$ B. $\mathrm{Q}^2 = \mathrm{PR}$ C. $\mathrm{Q}^2 = \mathrm{R}^3\mathrm{P}$ D. $\mathrm{R} = \mathrm{P}^2\mathrm{Q}^2$

gatecse-2011 quantitative-aptitude normal logarithms

Answer key☟

9.34.2 Logarithms: GATE CSE 2018 | Question: GA-7

If $pqr \neq 0$ and $p^{-x} = \frac{1}{q}, q^{-y} = \frac{1}{r}, r^{-z} = \frac{1}{p}$, what is the value of the product xyz ?

A. -1 B. $\frac{1}{pqr}$ C. 1 D. pqr

gatecse-2018 quantitative-aptitude ratio-proportion two-marks logarithms

Answer key☟

9.34.3 Logarithms: GATE CSE 2024 | Set 1 | GA Question: 5

For positive non-zero real variables p and q, if

$\log\left(p^2+q^2\right) = \log p + \log q + 2\log 3,$

then, the value of $\frac{p^4+q^4}{p^2q^2}$ is

A. 79 B. 81 C. 9 D. 83

gatecse2024-set1 quantitative-aptitude logarithms one-mark

Answer key☟

9.34.4 Logarithms: GATE CSE 2024 | Set 2 | GA Question: 4

For positive non-zero real variables x and y, if

$$\ln\left(\frac{x+y}{2}\right) = \frac{1}{2}[\ln(x) + \ln(y)]$$

then, the value of $\frac{x}{y} + \frac{y}{x}$ is

A. 1 B. $1/2$ C. 2 D. 4

gatecse2024-set2 quantitative-aptitude logarithms one-mark

Answer key☟

9.34.5 Logarithms: GATE Civil 2023 Set 1 | GA Question: 9

Let $a = 30!, b = 50!,$ and $c = 100!$. Consider the following numbers:

$$\log_a c, \quad \log_c a, \quad \log_b a, \quad \log_a b$$

Which one of the following inequalities is $\mathrm{CORRECT}$?

A. $\log_c a < \log_b a < \log_a b < \log_a c$
B. $\log_c a < \log_a b < \log_b a < \log_b c$
C. $\log_c a < \log_b a < \log_a c < \log_a b$
D. $\log_b a < \log_c a < \log_a b < \log_a c$

gatecivil-2023-set1 quantitative-aptitude logarithms

Answer key

9.34.6 Logarithms: GATE Civil 2023 Set 2 | GA Question: 9

If x satisfies the equation $4^{8^x} = 256$, then x is equal to __________.

A. $\frac{1}{2}$ B. $\log_{16} 8$ C. $\frac{2}{3}$ D. $\log_4 8$

gatecivil-2023-set2 quantitative-aptitude logarithms

Answer key

9.34.7 Logarithms: GATE ECE 2024 | GA Question: 4

For a real number $x > 1$,

$$\frac{1}{\log_2 x} + \frac{1}{\log_3 x} + \frac{1}{\log_4 x} = 1$$

The value of x is

A. 4 B. 12 C. 24 D. 36

gateece-2024 logarithms quantitative-aptitude

Answer key

9.34.8 Logarithms: GATE2012 AR: GA-6

A value of x that satisfies the equation $\log x + \log(x-7) = \log(x+11) + \log 2$ is

A. 1 B. 2 C. 7 D. 11

gate2012-ar quantitative-aptitude logarithms

Answer key

9.34.9 Logarithms: GATE2015 EC-1: GA-5

If $\log_x \left(\frac{5}{7}\right) = \frac{-1}{3}$, then the value of x is

A. $343/125$ B. $25/343$ C. $-25/49$ D. $-49/25$

gate2015-ec-1 general-aptitude numerical-methods logarithms

Answer key

9.34.10 Logarithms: GATE2015 EC-3: GA-9

$\log \tan 1^o + \log \tan 2^o + \dots + \log \tan 89^o$ is ...

A. 1 B. $1/\sqrt{2}$ C. 0 D. -1

gate2015-ec-3 quantitative-aptitude logarithms

Answer key

9.34.11 Logarithms: GATE2018 CE-2: GA-5

For non-negative integers, a, b, c, what would be the value of $a+b+c$ if $\log a + \log b + \log c = 0$?

A. 3 B. 1 C. 0 D. -1

gate2018-ce-2 general-aptitude quantitative-aptitude logarithms

Answer key

9.34.12 Logarithms: GATE2018 CE-2: GA-9

Given that $\frac{\log P}{y-z} = \frac{\log Q}{z-x} = \frac{\log R}{x-y} = 10$ for $x \neq y \neq z$, what is the value of the product PQR?

A. 0 B. 1 C. xyz D. 10^{xyz}

gate2018-ce-2 general-aptitude quantitative-aptitude logarithms

Answer key

9.34.13 Logarithms: GATE2018 ME-1: GA-6

For integers, a, b, and c, what would be the minimum and maximum values respectively of $a+b+c$ if $\log \mid a \mid + \log \mid b \mid + \log \mid c \mid = 0$?

A. -3 and 3
B. -1 and 1
C. -1 and 3
D. 1 and 3

gate2018-me-1 general-aptitude quantitative-aptitude logarithms

Answer key

9.34.14 Logarithms: GATE2018 ME-2: GA-5

The value of the expression $\dfrac{1}{1+\log_u vw} + \dfrac{1}{1+\log_v wu} + \dfrac{1}{1+\log_w uv}$ is ________

A. -1
B. 0
C. 1
D. 3

gate2018-me-2 general-aptitude quantitative-aptitude logarithms

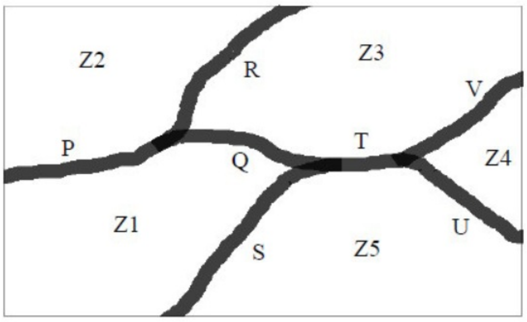
Answer key

9.35 Maps (1)

9.35.1 Maps: GATE CSE 2025 | Set 2 | GA Question: 10

The diagram below shows a river system consisting of 7 segments, marked P, Q, R, S, T, U, and V. It splits the land into 5 zones, marked $Z1, Z2, Z3, Z4, and Z5$. We need to connect these zones using the latest number of bridges, Out of the following options, which one is correct?

Note: The figure shown is representative.



A. Bridges on P, Q and T
B. Bridges on P, Q, S and T
C. Bridges on Q, R, T and V
D. Bridges on P, Q, S, U and V

gatecse2025-set2 quantitative-aptitude maps two-marks

Answer key

9.36 Maxima Minima (4)

9.36.1 Maxima Minima: GATE CSE 2012 | Question: 62

A political party orders an arch for the entrance to the ground in which the annual convention is being held. The profile of the arch follows the equation $y = 2x - 0.1x^2$ where y is the height of the arch in meters. The maximum possible height of the arch is

A. 8 meters
B. 10 meters
C. 12 meters
D. 14 meters

gatecse-2012 quantitative-aptitude normal maxima-minima

Answer key

9.36.2 Maxima Minima: GATE CSE 2017 Set 2 | Question: GA-9

The number of roots of $e^x + 0.5x^2 - 2 = 0$ in the range $[-5, 5]$ is

A. 0 B. 1 C. 2 D. 3

gatecse-2017-set2 quantitative-aptitude normal maxima-minima calculus

Answer key☟

9.36.3 Maxima Minima: GATE2010 TF: GA-5

Consider the function $f(x) = \max(7 - x, x + 3)$. In which range does f take its minimum value?

A. $-6 \leq x < -2$
B. $-2 \leq x < 2$
C. $2 \leq x < 6$
D. $6 \leq x < 10$

general-aptitude quantitative-aptitude gate2010-tf maxima-minima functions

Answer key☟

9.36.4 Maxima Minima: GATE2013 CE: GA-6

X and Y are two positive real numbers such that $2X + Y \leq 6$ and $X + 2Y \leq 8$.For which of the following values of

(X, Y) the function $f(X, Y) = 3X + 6Y$ will give maximum value ?

A. $\left(\frac{4}{3}, \frac{10}{3}\right)$ B. $\left(\frac{8}{3}, \frac{20}{3}\right)$ C. $\left(\frac{8}{3}, \frac{10}{3}\right)$ D. $\left(\frac{4}{3}, \frac{20}{3}\right)$

gate2013-ce quantitative-aptitude maxima-minima

Answer key☟

9.37 Mensuration (1)

9.37.1 Mensuration: GATE DA 2025 | GA Question: 5

A rectangle has a length L and a width W, where $\mathrm{L} > \mathrm{W}$. If the width, W, is increased by 10%, which one of the following statements is correct for all values of L and W?

A. Perimeter increases by 10%.
B. Length of the diagonals increases by 10%.
C. Area increases by 10%.
D. The rectangle becomes a square.

gateda-2025 quantitative-aptitude mensuration one-mark

Answer key☟

9.38 Modular Arithmetic (2)

9.38.1 Modular Arithmetic: GATE Electrical 2023 | GA Question: 9

The digit in the unit's place of the product $3^{999} \times 7^{1000}$ is _________.

A. 7 B. 1 C. 3 D. 9

gate2023-ee quantitative-aptitude modular-arithmetic

Answer key☟

9.38.2 Modular Arithmetic: GATE2012 CY: GA-1

If $(1.001)^{1259} = 3.52$ and $(1.001)^{2062} = 7.85$, then $(1.001)^{3321} =$

A. 2.23 B. 4.33 C. 11.37 D. 27.64

gate2012-cy quantitative-aptitude modular-arithmetic

Answer key☟

9.39 Number Series (10)

9.39.1 Number Series: GATE CSE 2013 | Question: 61

Find the sum of the expression

$\frac{1}{\sqrt{1}+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+\frac{1}{\sqrt{3}+\sqrt{4}}+\ldots\ldots\ldots\ldots+\frac{1}{\sqrt{80}+\sqrt{81}}$

A. 7 B. 8 C. 9 D. 10

gatecse-2013 quantitative-aptitude normal number-series

Answer key☟

9.39.2 Number Series: GATE CSE 2014 Set 2 | Question: GA-5

The value of $\sqrt{12+\sqrt{12+\sqrt{12+\ldots}}}$ is

A. 3.464 B. 3.932 C. 4.000 D. 4.444

gatecse-2014-set2 quantitative-aptitude easy number-series

Answer key☟

9.39.3 Number Series: GATE CSE 2014 Set 3 | Question: GA-4

Which number does not belong in the series below?

$2, 5, 10, 17, 26, 37, 50, 64$

A. 17 B. 37 C. 64 D. 26

gatecse-2014-set3 quantitative-aptitude number-series easy

Answer key☟

9.39.4 Number Series: GATE2010 TF: GA-7

Consider the series $\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\frac{1}{8}+\frac{1}{9}-\frac{1}{16}+\frac{1}{32}+\frac{1}{27}-\frac{1}{64}+\ldots$. The sum of the infinite series above is:

A. ∞ B. $\frac{5}{6}$ C. $\frac{1}{2}$ D. 0

general-aptitude quantitative-aptitude gate2010-tf number-series

Answer key☟

9.39.5 Number Series: GATE2013 EE: GA-10

Find the sum to $'n'$ terms of the series $10+84+734+\ldots$

A. $\frac{9(9^n+1)}{10}+1$ B. $\frac{9(9^n-1)}{8}+1$ C. $\frac{9(9^n-1)}{8}+n$ D. $\frac{9(9^n-1)}{8}+n^2$

gate2013-ee quantitative-aptitude number-series

Answer key☟

9.39.6 Number Series: GATE2014 EC-1: GA-5

What is the next number in the series?

$12 \quad 35 \quad 81 \quad 173 \quad 357$ _______.

gate2014-ec-1 number-series quantitative-aptitude numerical-answers

Answer key☟

9.39.7 Number Series: GATE2014 EC-2: GA-5

Fill in the missing number in the series.

$2 \quad 3 \quad 6 \quad 15$ ___ $157.5 \quad 630$

gate2014-ec-2 number-series quantitative-aptitude numerical-answers

Answer key☟

9.39.8 Number Series: GATE2014 EC-3: GA-4

The next term in the series $81, 54, 36, 24, \ldots$ is_________.

gate2014-ec-3 number-series quantitative-aptitude numerical-answers

Answer key☟

9.39.9 Number Series: GATE2018 CE-1: GA-9

Consider a sequence of numbers $a_1, a_2, a_3, \ldots, a_n$ where $a_n = \frac{1}{n} - \frac{1}{n+2}$, for each integer $n > 0$. Whart is the sum of the first 50 terms?

A. $\left(1+\frac{1}{2}\right) - \frac{1}{50}$
B. $\left(1+\frac{1}{2}\right) + \frac{1}{50}$
C. $\left(1+\frac{1}{2}\right) - \left(\frac{1}{51} + \frac{1}{52}\right)$
D. $1 - \left(\frac{1}{51} + \frac{1}{52}\right)$

gate2018-ce-1 general-aptitude quantitative-aptitude number-series

Answer key☟

9.39.10 Number Series: GATE2018 EC: GA-4

What is the value of $1 + \dfrac{1}{4} + \dfrac{1}{16} + \dfrac{1}{64} + \dfrac{1}{256} + \ldots\ldots?$

A. 2
B. $\dfrac{7}{4}$
C. $\dfrac{3}{2}$
D. $\dfrac{4}{3}$

gate2018-ec general-aptitude quantitative-aptitude number-series

Answer key☟

9.40 Number System (7)

9.40.1 Number System: GATE 2024 Electrical | GA Question: 3

The decimal number system uses the characters $0, 1, 2, \ldots, 8, 9$, and the octal number system uses the characters $0, 1, 2, \ldots, 6, 7$.

For example, the decimal number $12 \left(= 1 \times 10^1 + 2 \times 10^0\right)$ is expressed as $14 \left(= 1 \times 8^1 + 4 \times 8^0\right)$ in the octal number system.

The decimal number 108 in the octal number system is

A. 168
B. 108
C. 150
D. 154

gate2024-ee quantitative-aptitude number-system

Answer key☟

9.40.2 Number System: GATE CSE 2014 Set 3 | Question: GA-10

Consider the equation: $(7526)_8 - (Y)_8 = (4364)_8$, where $(X)_N$ stands for X to the base N. Find Y.

A. 1634
B. 1737
C. 3142
D. 3162

gatecse-2014-set3 quantitative-aptitude number-system normal digital-logic

Answer key☟

9.40.3 Number System: GATE CSE 2017 Set 2 | Question: GA-8

X is a 30 digit number starting with the digit 4 followed by the digit 7. Then the number X^3 will have

A. 90 digits
B. 91 digits
C. 92 digits
D. 93 digits

gatecse-2017-set2 quantitative-aptitude number-system

Answer key☟

9.40.4 Number System: GATE Civil 2020 Set 1 | GA Question: 5

The sum of two positive numbers is 100. After subtracting 5 from each number, the product of the resulting numbers is 0. One of the original numbers is ______.

A. 80
B. 85
C. 90
D. 95

gate2020-ce-1 quantitative-aptitude number-system

Answer key

9.40.5 Number System: GATE2010 MN: GA-9

A positive integer m in base 10 when represented in base 2 has the representation p and in base 3 has the representation q. We get $p - q = 990$ where the subtraction is done in base 10. Which of the following is necessarily true?

A. $m \geq 14$
B. $9 \leq m \leq 13$
C. $6 \leq m \leq 8$
D. $m < 6$

general-aptitude quantitative-aptitude gate2010-mn number-system

Answer key

9.40.6 Number System: GATE2016 CE-2: GA-5

The sum of the digits of a two digit number is 12. If the new number formed by reversing the digits is greater than the original number by 54, find the original number.

A. 39
B. 57
C. 66
D. 93

gate2016-ce-2 quantitative-aptitude number-system

Answer key

9.40.7 Number System: GATE2018 ME-1: GA-5

A number consists of two digits. The sum of the digits is 9. If 45 is subtracted from the number, its digits are interchanged. What is the number?

A. 63
B. 72
C. 81
D. 90

gate2018-me-1 general-aptitude quantitative-aptitude number-system

Answer key

9.41 Number Theory (6)

9.41.1 Number Theory: GATE ECE 2023 | GA Question: 3

What is the smallest number with distinct digits whose digits add up to 45?

A. 123555789
B. 123457869
C. 123456789
D. 99999

gateece-2023 quantitative-aptitude number-theory

Answer key

9.41.2 Number Theory: GATE Electrical 2021 | GA Question: 4

Which one of the following numbers is exactly divisible by $\left(11^{13}+1\right)$?

A. $11^{26}+1$
B. $11^{33}+1$
C. $11^{39}-1$
D. $11^{52}-1$

gateee-2021 quantitative-aptitude number-system number-theory

Answer key

9.41.3 Number Theory: GATE IN 2023 | GA Question: 3

A 'frabjous' number is defined as a 3 digit number with all digits odd, and no two adjacent digits being the same. For example, 137 is a frabjous number, while 133 is not. How many such frabjous numbers exist?

A. 125
B. 720
C. 60
D. 80

gatein-2023 quantitative-aptitude number-theory

Answer key

9.41.4 Number Theory: GATE2012 AE: GA-8

If a prime number on division by 4 gives a remainder of 1, then that number can be expressed as

A. sum of squares of two natural numbers
B. sum of cubes of two natural numbers

C. sum of square roots of two natural numbers
D. sum of cube roots of two natural numbers

gate2012-ae number-theory quantitative-aptitude

Answer key☟

9.41.5 Number Theory: GATE2016 ME-2: GA-9

The binary operation $\square$ is defined as $a\square b = ab + (a + b)$, where a and b are any two real numbers. The value of the identity element of this operation, defined as the number x such that $a\square x = a$, for any a, is

A. 0 B. 1 C. 2 D. 10

gate2016-me-2 quantitative-aptitude number-theory easy

Answer key☟

9.41.6 Number Theory: GATE2017 ME-2: GA-3

If a and b are integers and $a - b$ is even, which of the following must always be even?

A. ab B. $a^2 + b^2 + 1$ C. $a^2 + b + 1$ D. $ab - b$

gate2017-me-2 general-aptitude quantitative-aptitude number-theory

Answer key☟

9.42 Numerical Computation (14)

9.42.1 Numerical Computation: GATE CSE 2014 Set 1 | Question: GA-4

If $\left(z + \frac{1}{z}\right)^2 = 98$, compute $\left(z^2 + \frac{1}{z^2}\right)$.

gatecse-2014-set1 quantitative-aptitude easy numerical-answers numerical-computation

Answer key☟

9.42.2 Numerical Computation: GATE CSE 2014 Set 2 | Question: GA-4

What is the average of all multiples of 10 from 2 to 198?

A. 90 B. 100 C. 110 D. 120

gatecse-2014-set2 quantitative-aptitude easy numerical-computation factors

Answer key☟

9.42.3 Numerical Computation: GATE CSE 2017 Set 1 | Question: GA-4

Find the smallest number y such that $y \times 162$ is a perfect cube.

A. 24 B. 27 C. 32 D. 36

gatecse-2017-set1 general-aptitude quantitative-aptitude numerical-computation

Answer key☟

9.42.4 Numerical Computation: GATE CSE 2017 Set 2 | Question: GA-4

A test has twenty questions worth 100 marks in total. There are two types of questions. Multiple choice questions are worth 3 marks each and essay questions are worth 11 marks each. How many multiple choice questions does the exam have?

A. 12 B. 15 C. 18 D. 19

gatecse-2017-set2 quantitative-aptitude numerical-computation

Answer key☟

9.42.5 Numerical Computation: GATE Chemical 2020 | GA Question: 9

For a matrix $M = [m_{ij}]$; $i, j = 1, 2, 3, 4$, the diagonal elements are all zero and $m_{ij} = -m_{ji}$. The minimum number of elements required to fully specify the matrix is ________

A. 0 B. 6 C. 12 D. 16

gate2020-ch quantitative-aptitude numerical-computation matrix

Answer key☟

9.42.6 Numerical Computation: GATE2010 TF: GA-10

A student is answering a multiple choice examination with 65 questions with a marking scheme as follows: $i)$ 1 marks for each correct answer , $ii)$ $-\frac{1}{4}$ for a wrong answer , $iii)$ $-\frac{1}{8}$ for a question that has not been attempted. If the student gets 37 marks in the test then the least possible number of questions the student has NOT answered is:

A. 6 B. 5 C. 7 D. 4

general-aptitude quantitative-aptitude gate2010-tf numerical-computation

Answer key☟

9.42.7 Numerical Computation: GATE2013 CE: GA-1

A number is as much greater than 75 as it is smaller than 117. The number is:

A. 91 B. 93 C. 89 D. 96

gate2013-ce quantitative-aptitude numerical-computation

Answer key☟

9.42.8 Numerical Computation: GATE2014 EC-2: GA-8

The sum of eight consecutive odd numbers is 656. The average of four consecutive even numbers is 87. What is the sum of the smallest odd number and second largest even number?

gate2014-ec-2 quantitative-aptitude numerical-answers numerical-computation

Answer key☟

9.42.9 Numerical Computation: GATE2014 EC-4: GA-4

Let $f(x, y) = x^n y^m = P$. If x is doubled and y is halved, the new value of f is

A. $2^{n-m}P$ B. $2^{m-n}P$ C. $2(n-m)P$ D. $2(m-n)P$

gate2014-ec-4 quantitative-aptitude easy numerical-computation

Answer key☟

9.42.10 Numerical Computation: GATE2017 ME-1: GA-7

What is the sum of the missing digits in the subtraction problem below?

$$\begin{array}{cccccc} & 5 & \_ & \_ & \_ & \_ \\ - & 4 & 8 & \_ & 8 & 9 \\ \hline & & 1 & 1 & 1 & 1 \end{array}$$

A. 8 B. 10
C. 11 D. Cannot be determined.

gate2017-me-1 general-aptitude quantitative-aptitude numerical-computation

Answer key☟

9.42.11 Numerical Computation: GATE2018 ME-2: GA-9

A house has a number which need to be identified. The following three statements are given that can help in identifying the house number?

i. If the house number is a multiple of 3, then it is a number from 50 to 59.

ii. If the house number is NOT a multiple of 4, then it is a number from 60 to 69.
iii. If the house number is NOT a multiple of 6, then it is a number from 70 to 79.

What is the house number?

A. 54 B. 65 C. 66 D. 76

gate2018-me-2 general-aptitude quantitative-aptitude numerical-computation

Answer key☟

9.42.12 Numerical Computation: GATE2019 EE: GA-9

Given two sets $X = \{1, 2, 3\}$ and $Y = \{2, 3, 4\}$, we construct a set Z of all possible fractions where the numerators belong to set X and the denominators belong to set Y. The product of elements having minimum and maximum values in the set Z is _____.

A. $1/12$ B. $1/8$ C. $1/6$ D. $3/8$

gate2019-ee general-aptitude quantitative-aptitude numerical-computation

Answer key☟

9.42.13 Numerical Computation: GATE2019 ME-1: GA-4

The sum and product of two integers are 26 and 165 respectively. The difference between these two integers is ______

A. 2 B. 3 C. 4 D. 6

gate2019-me-1 general-aptitude quantitative-aptitude numerical-computation

Answer key☟

9.42.14 Numerical Computation: GATE2019 ME-2: GA-4

The product of three integers X, Y and Z is 192. Z is equal to 4 and P is equal to the average of X and Y. What is the minimum possible value of P?

A. 6 B. 7 C. 8 D. 9.5

gate2019-me-2 general-aptitude quantitative-aptitude numerical-computation

Answer key☟

9.43 Percentage (21)

9.43.1 Percentage: GATE CSE 2014 Set 1 | Question: GA-8

Round-trip tickets to a tourist destination are eligible for a discount of 10% on the total fare. In addition, groups of 4 or more get a discount of 5% on the total fare. If the one way single person fare is Rs 100, a group of 5 tourists purchasing round-trip tickets will be charged Rs __________

gatecse-2014-set1 quantitative-aptitude easy numerical-answers percentage

Answer key☟

9.43.2 Percentage: GATE CSE 2014 Set 3 | Question: GA-8

The Gross Domestic Product (GDP) in Rupees grew at 7% during $2012-2013$. For international comparison, the GDP is compared in US Dollars (USD) after conversion based on the market exchange rate. During the period $2012-2013$ the exchange rate for the USD increased from $Rs. 50/USD$ to $Rs. 60/USD$. India's GDP in USD during the period $2012-2013$

A. increased by 5%
B. decreased by 13%
C. decreased by 20%
D. decreased by 11%

gatecse-2014-set3 quantitative-aptitude normal percentage

Answer key☟

9.43.3 Percentage: GATE CSE 2024 | Set 1 | GA Question: 4

The number of coins of ₹1, ₹5, and ₹10 denominations that a person has are in the ratio $5:3:13$. Of the total amount, the percentage of money in ₹5 coins is

A. 21% B. $14\frac{2}{7}\%$ C. 10% D. 30%

gatecse2024-set1 quantitative-aptitude percentage one-mark

Answer key☟

9.43.4 Percentage: GATE Civil 2024 Set 2 | GA Question: 3

A student was supposed to **multiply** a positive real number p with another positive real number q. Instead, the student **divided** p by q. If the percentage error in the student's answer is 80%, the value of q is

A. 5 B. $\sqrt{2}$ C. 2 D. $\sqrt{5}$

gatecivil-2024-set2 quantitative-aptitude percentage

Answer key☟

9.43.5 Percentage: GATE Mechanical 2020 Set 2 | GA Question: 5

There are five levels $\{P, Q, R, S, T\}$ in a linear supply chain before a product reaches customers, as shown in the figure.

P → Q → R → S → T → Customers

At each of the five levels, the price of the product is increased by 25%. If the product is produced at level P at the cost of Rs. 120 per unit, what is the price paid (in rupees) by the customers?

A. 187.50 B. 234.38 C. 292.96 D. 366.21

gateme-2020-set2 quantitative-aptitude percentage

Answer key☟

9.43.6 Percentage: GATE2011 AG: GA-4

There are two candidates P and Q in an election. During the campaign, 40% of the voters promised to vote for P, and rest for Q. However, on the day of election 15% of the voters went back on their promise to vote for P and instead voted for Q. 25% of the voters went back on their promise to vote for Q and instead voted for P. Suppose, P lost by 2 votes, then what was the total number of voters?

A. 100 B. 110 C. 90 D. 95

general-aptitude quantitative-aptitude gate2011-ag percentage

Answer key☟

9.43.7 Percentage: GATE2012 AE: GA-7

The total runs scored by four cricketers P, Q, R and S in years 2009 and 2010 are given in the following table;

Player	2009	2010
P	802	1008
Q	765	912
R	429	619
S	501	701

The player with the lowest percentage increase in total runs is

A. P B. Q C. R D. S

gate2012-ae quantitative-aptitude percentage

Answer key☟

9.43.8 Percentage: GATE2012 CY: GA-8

The data given in the following table summarizes the monthly budget of an average household.

Category	Amount(Rs.)
Food	4000
Rent	2000
Savings	1500
Other expenses	1800
Clothing	1200

The approximate percentage of the monthly budget **NOT** spent on savings is

A. 10% B. 14% C. 81% D. 86%

gate2012-cy quantitative-aptitude percentage

Answer key

9.43.9 Percentage: GATE2013 EE: GA-2

In the summer of 2012, in New Delhi, the mean temperature of Monday to Wednesday was $41^{\circ}C$ and of Tuesday to Thursday was $43^{\circ}C$. If the temperature on Thursday was 15% higher than that of Monday, then the temperature in $^{\circ}C$ on Thursday was

A. 40 B. 43 C. 46 D. 49

gate2013-ee quantitative-aptitude percentage

Answer key

9.43.10 Percentage: GATE2014 AE: GA-9

One percent of the people of country X are taller than 6 ft. Two percent of the people of country Y are taller than 6 ft. There are thrice as many people in country X as in country Y. Taking both countries together, what is the percentage of people taller than 6 ft?

A. 3.0 B. 2.5 C. 1.5 D. 1.25

gate2014-ae percentage quantitative-aptitude

Answer key

9.43.11 Percentage: GATE2014 EC-4: GA-8

Industrial consumption of power doubled from $2000-2001$ to $2010-2011$. Find the annual rate of increase in percent assuming it to be uniform over the years.

A. 5.6 B. 7.2 C. 10.0 D. 12.2

gate2014-ec-4 percentage normal quantitative-aptitude

Answer key

9.43.12 Percentage: GATE2015 CE-2: GA-7

The given question is followed by two statements; select the most appropriate option that solves the question.
Capacity of a solution tank A is 70% of the capacity of tank B. How many gallons of solution are in tank A and tank B?
Statements:

I. Tank A is 80% full and tank B is 40% full.
II. Tank A if full contains $14,000$ gallons of solution.

A. Statement I alone is sufficient.
B. Statement II alone is sufficient.
C. Either statement I or II alone is sufficient.
D. Both the statements I and II together are sufficient.

gate2015-ce-2 quantitative-aptitude general-aptitude percentage

Answer key

9.43.13 Percentage: GATE2016 CE-2: GA-4

$(x\%\text{ of }y) + (y\%\text{ of }x)$ is equivalent to ______ .

A. 2% of xy B. 2% of $(xy/100)$ C. $xy\%$ of 100 D. 100% of xy

gate2016-ce-2 quantitative-aptitude percentage

Answer key

9.43.14 Percentage: GATE2016 EC-1: GA-4

In a huge pile of apples and oranges, both ripe and unripe mixed together, 15% are unripe fruits. Of the unripe fruits, 45% are apples. Of the ripe ones, 66% are oranges. If the pile contains a total of 5692000 fruits, how many of them are apples?

A. 2029198 B. 2467482 C. 2789080 D. 3577422

gate2016-ec-1 percentage quantitative-aptitude

Answer key

9.43.15 Percentage: GATE2017 CE-1: GA-4

If the radius of a right circular cone is increased by 50% its volume increases by

A. 75% B. 100% C. 125% D. 237.5%

gate2017-ce-1 general-aptitude quantitative-aptitude percentage geometry

Answer key

9.43.16 Percentage: GATE2017 EC-1: GA-3

In the summer, water consumption is known to decrease overall by 25%. A Water Board official states that in the summer household consumption decreases by 20%, while other consumption increases by 70%. Which of the following statement is correct?

A. The ratio of household to other consumption is $8/17$
B. The ratio of household to other consumption is $1/17$
C. The ratio of household to other consumption is $17/8$
D. There are errors in the official's statement.

gate2017-ec-1 general-aptitude quantitative-aptitude percentage

Answer key

9.43.17 Percentage: GATE2018 EE: GA-9

A designer uses marbles of four different colours for his designs. The cost of each marble is the same, irrespective of the colour. The table below shows the percentage of marbles of each colour used in the current design. The cost of each marble increased by 25%. Therefore, the designer decided to reduce equal numbers of marbles of each colour to keep the total cost unchanged. What is the percentage of blue marbles in the new design?

Blue	**Black**	**Red**	**Yellow**
40%	25%	20%	15%

A. 35.75 B. 40.25 C. 43.75 D. 46.25

gate2018-ee general-aptitude quantitative-aptitude normal percentage

Answer key

9.43.18 Percentage: GATE2019 CE-2: GA-7

Population of state X increased by x$\%$ and the population of state Y increased by y$\%$ from 2001 to 2011. Assume that x is greater than y. Let P be the ratio of the population of state X to state Y in a given year. The percentage increase in P from 2001 to 2011 is ________

A. $\frac{x}{y}$ B. $x-y$ C. $\frac{100(x-y)}{100+x}$ D. $\frac{100(x-y)}{100+y}$

gate2019-ce-2 general-aptitude quantitative-aptitude percentage

Answer key☟

9.43.19 Percentage: GATE2019 IN: GA-3

The radius as well as the height of a circular cone increases by 10%. The percentage increase in its volume is ________.

A. 17.1 B. 21.0 C. 33.1 D. 72.8

gate2019-in general-aptitude quantitative-aptitude geometry percentage

Answer key☟

9.43.20 Percentage: GATE2019 IN: GA-7

In a country of 1400 million population, 70% own mobile phones. Among the mobile phone owners, only 294 million access the Internet. Among these Internet users, only half buy goods from e-commerce portals. What is the percentage of these buyers in the country?

A. 10.50 B. 14.70 C. 15.00 D. 50.00

gate2019-in general-aptitude quantitative-aptitude percentage

Answer key☟

9.43.21 Percentage: GATE2019 ME-2: GA-6

Fiscal deficit was 4% of the GDP in 2015 and that increased to 5% in 2016. If the GDP increased by 10% from 2015 to 2016, the percentage increase in the actual fiscal deficit is ____

A. 37.50 B. 35.70 C. 25.00 D. 10.00

gate2019-me-2 general-aptitude quantitative-aptitude percentage

Answer key☟

9.44 Permutation and Combination (4)

9.44.1 Permutation and Combination: GATE CSE 2024 | Set 2 | GA Question: 2

Two wizards try to create a spell using all the four elements, *water, air, fire*, and *earth*. For this, they decide to mix all these elements in all possible orders. They also decide to work independently. After trying all possible combination of elements, they conclude that the spell does not work.

How many attempts does each wizard make before coming to this conclusion, independently?

A. 24 B. 48 C. 16 D. 12

gatecse2024-set2 quantitative-aptitude permutation-and-combination one-mark

Answer key☟

9.44.2 Permutation and Combination: GATE CSE 2025 | Set 1 | GA Question: 10

A shop has 4 distinct flavors of ice-cream. One can purchase any number of scoops of any flavor. The order in which the scoops are purchased is inconsequential. If one wants to purchase 3 scoops of ice-cream, in how many ways can one make that purchase?

A. 4 B. 20 C. 24 D. 48

gatecse2025-set1 quantitative-aptitude permutation-and-combination two-marks

Answer key☟

9.44.3 Permutation and Combination: GATE Civil 2024 Set 1 | GA Question: 7

Three distinct sets of indistinguishable twins are to be seated at a circular table that has 8 identical chairs. Unique seating arrangements are defined by the relative positions of the people.

How many unique seating arrangements are possible such that each person is sitting next to their twin?

A. 12 B. 14 C. 10 D. 28

gatecivil-2024-set1 quantitative-aptitude permutation-and-combination

Answer key☟

9.44.4 Permutation and Combination: GATE DS&AI 2024 | GA Question: 3

How many 4-digit positive integers divisible by 3 can be formed using only the digits $\{1,3,4,6,7\}$, such that no digit appears more than once in a number?

A. 24 B. 48 C. 72 D. 12

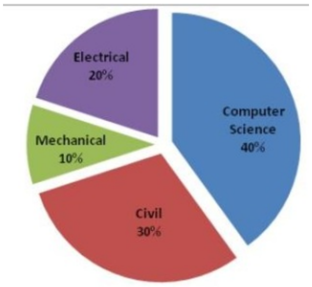
gate-ds-ai-2024 quantitative-aptitude permutation-and-combination one-mark

Answer key☟

9.45 Pie Chart (16)

9.45.1 Pie Chart: GATE CSE 2015 Set 1 | Question: GA-9

The pie chart below has the breakup of the number of students from different departments in an engineering college for the year 2012. The proportion of male to female students in each department is $5:4$. There are 40 males in Electrical Engineering. What is the difference between the numbers of female students in the civil department and the female students in the Mechanical department?

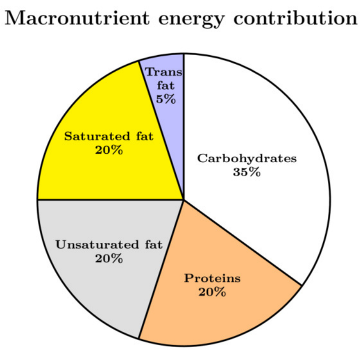


gatecse-2015-set1 quantitative-aptitude data-interpretation numerical-answers pie-chart

Answer key☟

9.45.2 Pie Chart: GATE CSE 2024 | Set 1 | GA Question: 8

The pie chart presents the percentage contribution of different macronutrients to a typical $2,000 \text{ kcal}$ diet of a person.



The typical energy density (kcal/g) of these macronutrients is given in the table.

Macronutrient	Energy density (kcal/g)
Carbohydrates	4
Proteins	4
Unsaturated fat	9
Saturated fat	9

Trans fat	9

The total fat (all three types), in grams, this person consumes is

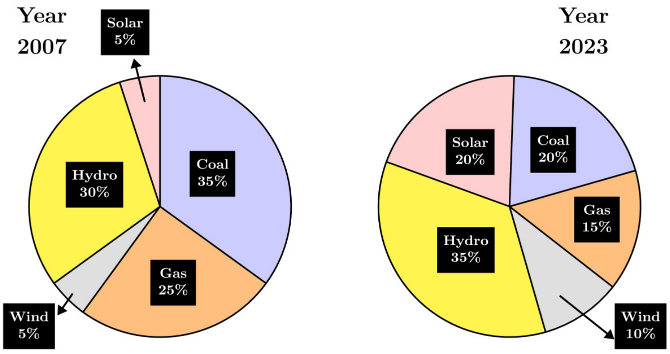
A. 44.4 B. 77.8 C. 100 D. $3,600$

gatecse2024-set1 quantitative-aptitude pie-chart two-marks

Answer key

9.45.3 Pie Chart: GATE CSE 2024 | Set 2 | GA Question: 8

The pie charts depict the shares of various power generation technologies in the total electricity generation of a country for the years 2007 and 2023.



The renewable sources of electricity generation consist of Hydro, Solar and Wind. Assuming that the total electricity generated remains the same from 2007 to 2023, what is the percentage increase in the share of the renewable sources of electricity generation over this period?

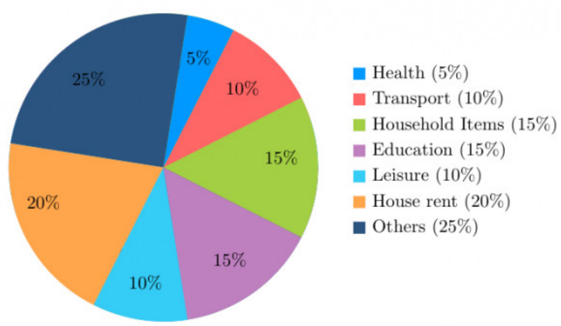
A. 25% B. 50% C. 77.5% D. 62.5%

gatecse2024-set2 quantitative-aptitude pie-chart two-marks

Answer key

9.45.4 Pie Chart: GATE Civil 2020 Set 1 | GA Question: 10

The total expenditure of a family, on different activities in a month, is shown in the pie-chart. The extra money spent on education as compared to transport (in percent) is ______.



A. 5 B. 33.3 C. 50 D. 100

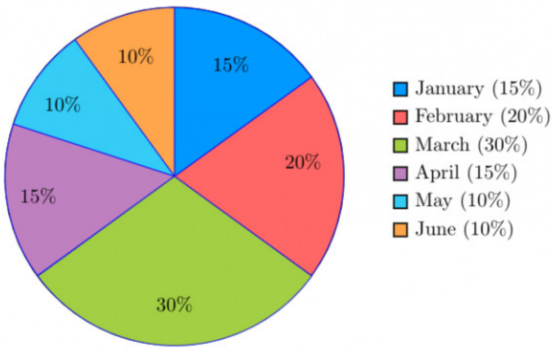
gate2020-ce-1 quantitative-aptitude data-interpretation pie-chart

Answer key

9.45.5 Pie Chart: GATE Civil 2020 Set 2 | GA Question: 10

The monthly distribution of 9 Watt LED bulbs sold by two firms X and Y from January to June 2018 is shown in the pie-chart and the corresponding table. If the total number of LED bulbs sold by the two firms during April-June 2018 is 50000, then the number of LED bulbs sold by the firm Y during April-June 2018 is ________.

Percentage of 9 Watt LED bulbs sold by the firms X and Y from January 2018 to June,2018



Month	Ratio of LED bulbs sold by two firms $(X : Y)$
January	$7:8$
February	$2:3$
March	$2:1$
April	$3:2$
May	$1:4$
June	$9:11$

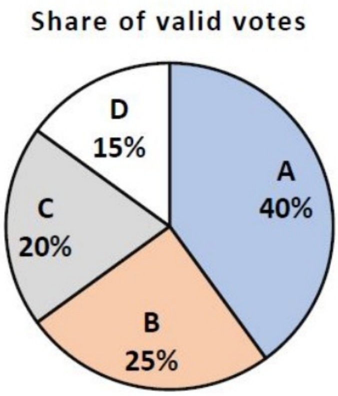
A. 11250 B. 9750 C. 8750 D. 8250

gate2020-ce-2 quantitative-aptitude data-interpretation pie-chart

Answer key

9.45.6 Pie Chart: GATE DS&AI 2024 | GA Question: 5

In an election, the share of valid votes received by the four candidates $\mathrm{A}, \mathrm{B}, \mathrm{C}$, and D is represented by the pie chart shown. The total number of votes cast in the election were $1,15,000$, out of which $5,000$ were invalid.



Based on the data provided, the total number of valid votes received by the candidates B and C is

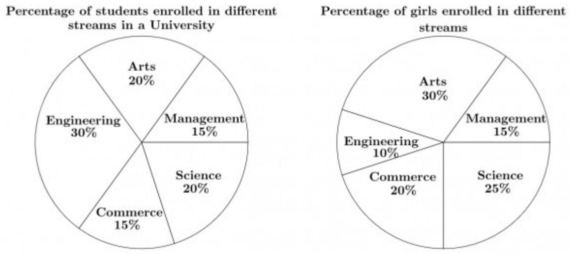
A. $45,000$ B. $49,500$ C. $51,750$ D. $54,000$

gate-ds-ai-2024 quantitative-aptitude pie-chart one-mark

Answer key

9.45.7 Pie Chart: GATE Mechanical 2020 Set 2 | GA Question: 10

The two pie-charts given below show the data of total students and only girls registered in different streams in a university. If the total number of students registered in the university is 5000, and the total number of the registered girls is 1500; then, the ratio of boys enrolled in Arts to the girls enrolled in Management is ___________

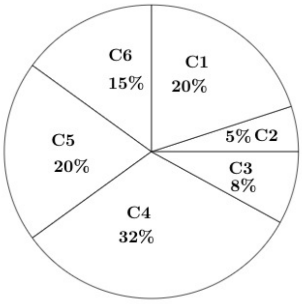


A. $2:1$ B. $9:22$ C. $11:9$ D. $22:9$

gateme-2020-set2 quantitative-aptitude data-interpretation pie-chart

Answer key

9.45.8 Pie Chart: GATE Mechanical 2021 Set 1 | GA Question: 9



Company	Ratio
$C1$	$3:2$
$C2$	$1:4$
$C3$	$5:3$
$C4$	$2:3$
$C5$	$9:1$
$C6$	$3:4$

The distribution of employees at the rank of executives, across different companies $C1, C2, \ldots, C6$ is presented in the chart given above. The ratio of executives with a management degree to those without a management degree in each of these companies is provided in the table above. The total number of executives across all companies is $10,000$.

The total number of management degree holders among the executives in companies $C2$ and $C5$ together is ________.

A. 225 B. 600 C. 1900 D. 2500

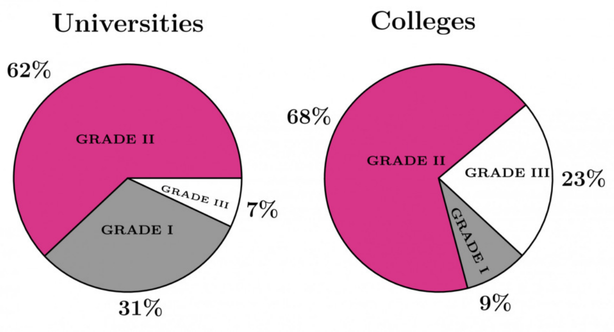
gateme-2021-set1 quantitative-aptitude data-interpretation pie-chart

Answer key

9.45.9 Pie Chart: GATE Mechanical 2023 | GA Question: 3

A certain country has 504 universities and 25951 colleges. These are categorised into Grades I, II, and III as shown in the given pie charts.

What is the percentage, correct to one decimal place, of higher education institutions (colleges and universities) that fall into Grade III?



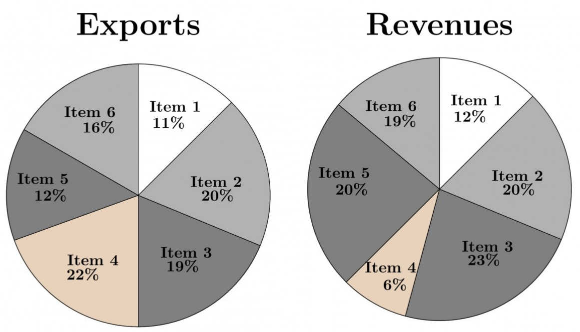
A. 22.7 B. 23.7 C. 15.0 D. 66.8

gateme-2023 quantitative-aptitude pie-chart

Answer key

9.45.10 Pie Chart: GATE2014 AG: GA-8

The total exports and revenues from the exports of a country are given in the two pie charts below. The pie chart for exports shows the quantity of each item as a percentage of the total quantity of exports. The pie chart for the revenues shows the percentage of the total revenue generated through export of each item. The total quantity of exports of all the items is 5 lakh tonnes and the total revenues are 250 crore rupees. What is the ratio of the revenue generated through export of Item 1 per kilogram to the revenue generated through export of Item 4 per kilogram?



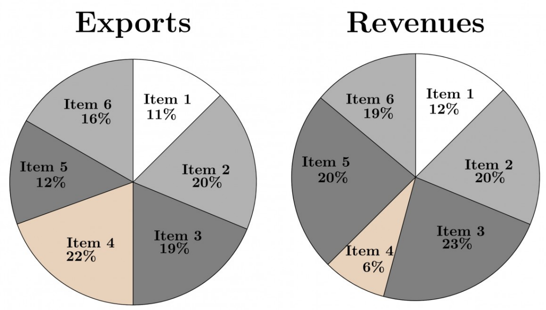
A. $1:2$ B. $2:1$ C. $1:4$ D. $4:1$

gate2014-ag quantitative-aptitude data-interpretation pie-chart ratio-proportion normal

Answer key

9.45.11 Pie Chart: GATE2014 EC-2: GA-9

The total exports and revenues from the exports of a country are given in the two charts shown below. The pie chart for exports shows the quantity of each item exported as a percentage of the total quantity of exports. The pie chart for the revenues shows the percentage of the total revenue generated through export of each item. The total quantity of exports of all the items is 500 thousand tonnes and the total revenues are 250 crore rupees. Which item among the following has generated the maximum revenue per kg?



A. Item 2 B. Item 3 C. Item 6 D. Item 5

gate2014-ec-2 quantitative-aptitude data-interpretation pie-chart normal

Answer key

9.45.12 Pie Chart: GATE2014 EC-3: GA-7

The multi-level hierarchical pie chart shows the population of animals in a reserve forest. The correct conclusions from this information are:

(i) Butterflies are birds

(ii) There are more tigers in this forest than red ants

(iii) All reptiles in this forest are either snakes or crocodiles

(iv) Elephants are the largest mammals in this forest

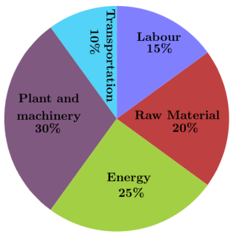
A. (i) and (ii) only
B. (i), (ii), (iii) and (iv)
C. (i), (iii) and (iv) only
D. (i), (ii) and (iii) only

gate2014-ec-3 quantitative-aptitude data-interpretation pie-chart normal

Answer key

9.45.13 Pie Chart: GATE2014 EC-3: GA-9

A firm producing air purifiers sold 200 units in 2012. The following pie chart presents the share of raw material, labour, energy, plant & machinery, and transportation costs in the total manufacturing cost of the firm in 2012. The expenditure on labour in 2012 is Rs.$4,50,000$. In 2013, the raw material expenses increased by 30% and all other expenses increased by 20%. If the company registered a profit of Rs.10 lakhs in 2012, at what price (in Rs) was each air purifier sold?

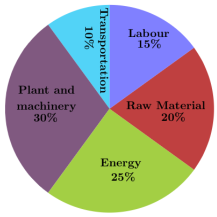


gate2014-ec-3 quantitative-aptitude data-interpretation pie-chart numerical-answers

Answer key

9.45.14 Pie Chart: GATE2014 EC-4: GA-9

A firm producing air purifiers sold 200 units in 2012. The following pie chart presents the share of raw material, labour, energy, plant $\&$ machinery, and transportation costs in the total manufacturing cost of the firm in 2012. The expenditure on labour in 2012 is Rs. $4,50,000$. In 2013, the raw material expenses increased by 30% and all other expenses increased by 20%. What is the percentage increase in total cost for the company in 2013?



gate2014-ec-4 quantitative-aptitude data-interpretation pie-chart numerical-answers

Answer key

9.45.15 Pie Chart: GATE2017 EC-1: GA-4

40% of deaths on city roads may be attributed to drunken driving. The number of degrees needed to represent this as a slice of a pie chart is

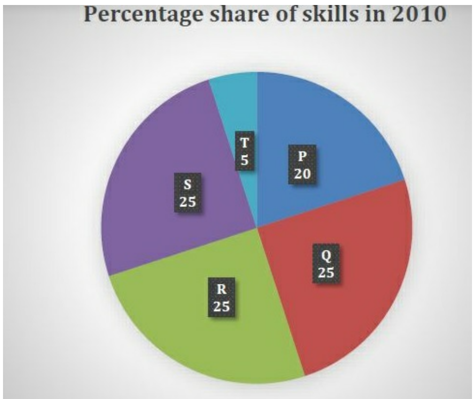
A. 120 B. 144 C. 160 D. 212

gate2017-ec-1 general-aptitude quantitative-aptitude percentage pie-chart

Answer key

9.45.16 Pie Chart: GATE2019 ME-1: GA-9

A firm hires employees at five different skill levels P, Q, R, S, T. The shares of employment at these skills levels of total employment in 2010 is given in the pie chart as shown. There were a total of 600 employees in 2010 and the total employment increased by 15% from 2010 to 2016. The total employment at skill levels P, Q and R remained unchanged during this period. If the employment at skill level S increased by 40% from 2010 to 2016, how many employees were there at skill level T in 2016?



A. 30 B. 35 C. 60 D. 72

gate2019-me-1 general-aptitude quantitative-aptitude data-interpretation pie-chart

Answer key

9.46 Polynomials (4)

9.46.1 Polynomials: GATE CH 2023 | GA Question: 9

The coefficient of x^4 in the polynomial $(x-1)^3(x-2)^3$ is equal to____________.

A. 33 B. -3 C. 30 D. 21

gatech-2023 quantitative-aptitude polynomials

Answer key

9.46.2 Polynomials: GATE CSE 2016 Set 1 | Question: GA09

If $f(x) = 2x^7 + 3x - 5$, which of the following is a factor of $f(x)$?

A. (x^3+8) B. $(x-1)$ C. $(2x-5)$ D. $(x+1)$

gatecse-2016-set1 quantitative-aptitude polynomials normal

Answer key☟

9.46.3 Polynomials: GATE2018 CH: GA-9

If $x^2+x-1=0$ what is the value of $x^4+\dfrac{1}{x^4}$?

A. 1 B. 5 C. 7 D. 9

gate2018-ch quantitative-aptitude easy polynomials

Answer key☟

9.46.4 Polynomials: GATE2018 EE: GA-3

The three roots of the equation $f(x)=0$ are $x=\{-2,0,3\}$. What are the three values of x for which $f(x-3)=0$?

A. $-5,-3,0$ B. $-2,0,3$ C. $0,6,8$ D. $1,3,6$

gate2018-ee general-aptitude quantitative-aptitude easy polynomials

Answer key☟

9.47 Powers (1)

9.47.1 Powers: GATE DA 2025 | GA Question: 9

If a real variable x satisfies $3^{x^2}=27\times 9^x$, then the value of $\frac{2^{x^2}}{(2^x)^2}$ is:

A. 2^{-1} B. 2^0 C. 2^3 D. 2^{15}

gateda-2025 quantitative-aptitude powers two-marks

Answer key☟

9.48 Prime Numbers (1)

9.48.1 Prime Numbers: GATE CSE 2025 | Set 2 | GA Question: 5

Let p_1 and p_2 denote two arbitrary prime numbers. Which one of the following statemetns is correct for all values of p_1 and p_2?

A. p_1+p_2 not a prime number
B. p_1p_2 is not a prime number
C. p_1+p_2+1 is a prime number
D. p_1p_2+1 is a prime number

gatecse2025-set2 quantitative-aptitude prime-numbers one-mark

Answer key☟

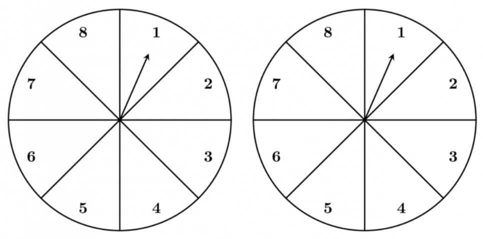
9.49 Probability (29)

9.49.1 Probability: GATE CH 2022 | GA Question: 7

A game consists of spinning an arrow around a stationary disk as shown below. When the arrow comes to rest, there are eight equally likely outcomes. It could come to rest in any one of the sectors numbered $1,2,3,4,5,6,7$ or 8 as shown.

Two such disks are used in a game where their arrows are independently spun.

What is the probability that the sum of the numbers on the resulting sectors upon spinning the two disks is equal to 8 after the arrows come to rest?



A. $\dfrac{1}{16}$
B. $\dfrac{5}{64}$
C. $\dfrac{3}{32}$
D. $\dfrac{7}{64}$

gatech-2022 quantitative-aptitude probability

Answer key☟

9.49.2 Probability: GATE CSE 2015 Set 1 | Question: GA-10

The probabilities that a student passes in mathematics, physics and chemistry are m, p and c respectively. Of these subjects, the student has 75% chance of passing in at least one, a 50% chance of passing in at least two and a 40% chance of passing in exactly two. Following relations are drawn in m, p, c:

I. $p + m + c = 27/20$
II. $p + m + c = 13/20$
III. $(p) \times (m) \times (c) = 1/10$

A. Only relation I is true.
B. Only relation II is true.
C. Relations II and III are true.
D. Relations I and III are true.

gatecse-2015-set1 quantitative-aptitude probability

Answer key☟

9.49.3 Probability: GATE CSE 2015 Set 1 | Question: GA-3

Given Set $A = 2, 3, 4, 5$ and Set $B = 11, 12, 13, 14, 15$, two numbers are randomly selected, one from each set. What is the probability that the sum of the two numbers equals 16?

A. 0.20
B. 0.25
C. 0.30
D. 0.33

gatecse-2015-set1 quantitative-aptitude probability normal

Answer key☟

9.49.4 Probability: GATE CSE 2017 Set 1 | Question: GA-5

The probability that a k-digit number does NOT contain the digits $0, 5,$ or 9 is

A. 0.3^k
B. 0.6^k
C. 0.7^k
D. 0.9^k

gatecse-2017-set1 general-aptitude quantitative-aptitude probability easy

Answer key☟

9.49.5 Probability: GATE CSE 2017 Set 2 | Question: GA-5

There are 3 red socks, 4 green socks and 3 blue socks.You choose 2 socks. The probability that they are of the same colour is

A. $\dfrac{1}{5}$
B. $\dfrac{7}{30}$
C. $\dfrac{1}{4}$
D. $\dfrac{4}{15}$

gatecse-2017-set2 quantitative-aptitude probability

Answer key

9.49.6 Probability: GATE CSE 2018 | Question: GA-10

A six sided unbiased die with four green faces and two red faces is rolled seven times. Which of the following combinations is the most likely outcome of the experiment?

A. Three green faces and four red faces.
B. Four green faces and three red faces.
C. Five green faces and two red faces.
D. Six green faces and one red face

gatecse-2018 quantitative-aptitude probability normal two-marks

Answer key

9.49.7 Probability: GATE CSE 2021 Set 1 | GA Question: 8

There are five bags each containing identical sets of ten distinct chocolates. One chocolate is picked from each bag.

The probability that at least two chocolates are identical is __________

A. 0.3024
B. 0.4235
C. 0.6976
D. 0.8125

gatecse-2021-set1 quantitative-aptitude probability two-marks

Answer key

9.49.8 Probability: GATE CSE 2022 | GA Question: 8

A box contains five balls of same size and shape. Three of them are green coloured balls and two of them are orange coloured balls. Balls are drawn from the box one at a time. If a green ball is drawn, it is not replaced. If an orange ball is drawn, it is replaced with another orange ball.

First ball is drawn. What is the probability of getting an orange ball in the next draw?

A. $\frac{1}{2}$
B. $\frac{8}{25}$
C. $\frac{19}{50}$
D. $\frac{23}{50}$

gatecse-2022 quantitative-aptitude probability two-marks

Answer key

9.49.9 Probability: GATE CSE 2025 | Set 1 | GA Question: 8

A fair six-faced dice, with the faces labelled $'1', '2', '3', '4', '5'$, and $'6'$, is rolled thrice. What is the probability of rolling $'6'$ exactly once?

A. $\frac{75}{216}$
B. $\frac{1}{6}$
C. $\frac{1}{18}$
D. $\frac{25}{216}$

gatecse2025-set1 quantitative-aptitude probability easy two-marks

Answer key

9.49.10 Probability: GATE Civil 2021 Set 2 | GA Question: 3

Two identical cube shaped dice each with faces numbered 1 to 6 are rolled simultaneously. The probability that an even number is rolled out on each dice is:

A. $\frac{1}{36}$
B. $\frac{1}{12}$
C. $\frac{1}{8}$
D. $\frac{1}{4}$

gatecivil-2021-set2 quantitative-aptitude probability

Answer key

9.49.11 Probability: GATE DS&AI 2024 | GA Question: 7

The probability of a boy or a girl being born is $1/2$. For a family having only three children, what is the probability of having two girls and one boy?

A. $3/8$
B. $1/8$
C. $1/4$
D. $1/2$

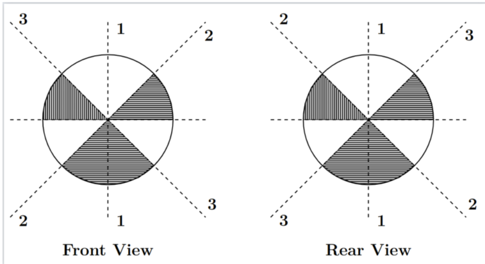
gate-ds-ai-2024 quantitative-aptitude probability two-marks

Answer key

9.49.12 Probability: GATE Electrical 2022 | GA Question: 5

The figure below shows the front and rear view of a disc, which is shaded with identical patterns. The disc is flipped once with respect to any one of the fixed axes $1-1$, $2-2$ or $3-3$ chosen uniformly at random.

What is the probability that the disc DOES NOT retain the same front and rear views after the flipping operation?



A. 0 B. $\frac{1}{3}$ C. $\frac{2}{3}$ D. 1

gate2022-ee quantitative-aptitude probability

Answer key

9.49.13 Probability: GATE Electrical 2022 | GA Question: 7

There are two identical dice with a single letter on each of the faces. The following six letters: Q, R, S, T, U and V, one on each of the faces. Any of the six outcomes are equally likely.

The two dice are thrown once independently at random.

What is the probability that the outcomes on the dice were composed only of any combination of the following possible outcomes: Q, U and V?

A. $\frac{1}{4}$ B. $\frac{3}{4}$ C. $\frac{1}{6}$ D. $\frac{5}{36}$

gate2022-ee quantitative-aptitude probability

Answer key

9.49.14 Probability: GATE Electrical 2023 | GA Question: 3

Given a fair six-faced dice where the faces are labelled '1', '2', '3', '4', '5', and '6',
what is the probability of getting a '1' on the first roll of the dice and a '4' on the second roll?

A. $\frac{1}{36}$ B. $\frac{1}{6}$
C. $\frac{5}{6}$ D. $\frac{1}{3}$

gate2023-ee quantitative-aptitude probability

Answer key

9.49.15 Probability: GATE Mechanical 2024 | GA Question: 5

Take two long dice (rectangular parallelepiped), each having four rectangular faces labelled as $2, 3, 5$, and 7. If thrown, the long dice cannot land on the square faces and has $1/4$ probability of landing on any of the four rectangular faces. The label on the top face of the dice is the score of the throw.

If thrown together, what is the probability of getting the sum of the two long dice scores greater than 11?

A. $3/8$ B. $1/8$ C. $1/16$ D. $3/16$

gateme-2024 quantitative-aptitude probability

Answer key

9.49.16 Probability: GATE2012 AE: GA-6

Two policemen, A and B, fire once each at the same time at an escaping convict. The probability that A hits the convict is three times the probability that B hits the convict. If the probability of the convict not getting injured is 0.5, the probability that B hits the convict is

A. 0.14 B. 0.22 C. 0.33 D. 0.40

gate2012-ae quantitative-aptitude probability

Answer key☟

9.49.17 Probability: GATE2012 AR: GA-9

A smuggler has 10 capsules in which five are filled with narcotic drugs and the rest contain the original medicine. All the 10 capsules are mixed in a single box, from which the customs officials picked two capsules at random and tested for the presence of narcotic drugs. The probability that the smuggler will be caught is

A. 0.50 B. 0.67 C. 0.78 D. 0.82

gate2012-ar quantitative-aptitude probability

Answer key☟

9.49.18 Probability: GATE2012 CY: GA-7

A and B are friends. They decide to meet between 1:00 pm and 2:00 pm on a given day. There is a condition that whoever arrives first will not wait for the other for more than 15 minutes. The probability that they will meet on that day is

A. $1/4$ B. $1/16$ C. $7/16$ D. $9/16$

gate2012-cy quantitative-aptitude probability

Answer key☟

9.49.19 Probability: GATE2013 EE: GA-6

What is the chance that a leap year, selected at random, will contain 53 Saturdays?

A. $2/7$ B. $3/7$ C. $1/7$ D. $5/7$

gate2013-ee quantitative-aptitude probability

Answer key☟

9.49.20 Probability: GATE2014 AG: GA-4

In any given year, the probability of an earthquake greater than Magnitude 6 occurring in the Garhwal Himalayas is 0.04. The average time between successive occurrences of such earthquakes is ____ years.

gate2014-ag quantitative-aptitude probability numerical-answers normal

Answer key☟

9.49.21 Probability: GATE2014 EC-2: GA-4

A regular die has six sides with numbers 1 to 6 marked on its sides. If a very large number of throws show the following frequencies of occurrence: $1 \rightarrow 0.167; 2 \rightarrow 0.167; 3 \rightarrow 0.152; 4 \rightarrow 0.166; 5 \rightarrow 0.168; 6 \rightarrow 0.180$. We call this die:

A. Irregular B. Biased C. Gaussian D. Insufficient

gate2014-ec-2 quantitative-aptitude probability normal

Answer key☟

9.49.22 Probability: GATE2014 EC-3: GA-10

A batch of one hundred bulbs is inspected by testing four randomly chosen bulbs. The batch is rejected if even one of the bulbs is defective. A batch typically has five defective bulbs. The probability that the current batch is accepted is__________.

gate2014-ec-3 quantitative-aptitude probability numerical-answers normal

Answer key☟

9.49.23 Probability: GATE2015 CE-2: GA-5

Four cards are randomly selected from a pack of 52 cards. If the first two cards are kings, what is the probability that the third card is a king?

A. $4/52$ B. $2/50$
C. $(1/52) \times (1/52)$ D. $(1/52) \times (1/51) \times (1/50)$

gate2015-ce-2 quantitative-aptitude probability easy

Answer key

9.49.24 Probability: GATE2015 EC-2: GA- 5

Ram and Ramesh appeared in an interview for two vacancies in the same department. The probability of Ram's selection is $1/6$ and that of Ramesh is $1/8$. What is the probability that only one of them will be selected?

A. $47/48$ B. $1/4$ C. $13/48$ D. $35/48$

gate2015-ec-2 quantitative-aptitude probability

Answer key

9.49.25 Probability: GATE2016 EC-1: GA-6

A person moving through a tuberculosis prone zone has a 50% probability of becoming infected. However, only 30% of infected people develop the disease. What percentage of people moving through a tuberculosis prone zone remains infected but does not show symptoms of disease?

A. 15 B. 33 C. 35 D. 37

gate2016-ec-1 quantitative-aptitude probability

Answer key

9.49.26 Probability: GATE2017 CE-2: GA-5

Two dice are thrown simultaneously. The probability that the product of the numbers appearing on the top faces of the dice is a perfect square is

A. $\frac{1}{9}$ B. $\frac{2}{9}$ C. $\frac{1}{3}$ D. $\frac{4}{9}$

gate2017-ce-2 quantitative-aptitude probability

Answer key

9.49.27 Probability: GATE2017 ME-2: GA-4

A couple has 2 children. The probability that both children are boys if the older one is a boy is

A. $1/4$ B. $1/3$ C. $1/2$ D. 1

gate2017-me-2 quantitative-aptitude probability

Answer key

9.49.28 Probability: GATE2018 EE: GA-8

A class of twelve children has two more boys than girls. A group of three children are randomly picked from this class to accompany the teacher on a field trip. What is the probability that the group accompanying the teacher contains more girls than boys?

A. 0 B. $\dfrac{325}{864}$ C. $\dfrac{525}{864}$ D. $\dfrac{5}{12}$

gate2018-ee quantitative-aptitude probability

Answer key

9.49.29 Probability: GATE2018 ME-2: GA-10

An unbiased coin is tossed six times in a row and four different such trials are conducted. One trial implies six tosses of the coin. If H stands for head ans T stands for tail, the following are the observations from the four trials.

1. HTHTHT
2. TTHHHT
3. HTTHHT
4. HHHT_ _

Which statement describing the last two coin tosses of the fourth trial has the highest probability of being correct?

A. Two T will occur.
B. One H and one T will occur.
C. Two H will occur.
D. One H will be followed by one T.

gate2018-me-2 quantitative-aptitude probability

Answer key☟

9.50 Probability Density Function (1)

9.50.1 Probability Density Function: GATE Electrical 2021 | GA Question: 8

Let X be a continuous random variable denoting the temperature measured. The range of temperature is $[0, 100]$ degree Celsius and let the probability density function of X be $f(x) = 0.01$ for $0 \leq X \leq 100$.

The mean of X is __________

A. 2.5 B. 5.0 C. 25.0 D. 50.0

gateee-2021 quantitative-aptitude probability probability-density-function expectation

Answer key☟

9.51 Profit Loss (6)

9.51.1 Profit Loss: GATE CSE 2021 Set 1 | GA Question: 7

Items	**Cost** (₹)	**Profit %**	**Marked Price** (₹)
P	$5,400$	$---$	$5,860$
Q	$---$	25	$10,000$

Details of prices of two items P and Q are presented in the above table. The ratio of cost of item P to cost of item Q is $3:4$. Discount is calculated as the difference between the marked price and the selling price. The profit percentage is calculated as the ratio of the difference between selling price and cost, to the cost

$$(\text{Profit}\% = \frac{\text{Selling price} - \text{Cost}}{\text{Cost}} \times 100)$$

The discount on item Q, as a percentage of its marked price, is _______

A. 25 B. 12.5 C. 10 D. 5

gatecse-2021-set1 quantitative-aptitude profit-loss two-marks

Answer key☟

9.51.2 Profit Loss: GATE CSE 2024 | Set 2 | GA Question: 7

A person sold two different items at the same price. He made 10% profit in one item, and 10% loss in the other item. In selling these two items, the person made a total of

A. 1% profit B. 2% profit C. 1% loss D. 2% loss

gatecse2024-set2 quantitative-aptitude profit-loss two-marks

Answer key☟

9.51.3 Profit Loss: GATE Civil 2022 Set 1 | GA Question: 7

P invested ₹5000 per month for 6 months of a year and Q invested ₹x per month for 8 months of the year in a partnership business. The profit is shared in proportion to the total investment made in that year.

If at the end of that investment year, Q receives $\dfrac{4}{9}$ of the total profit, what is the value of x (in ₹)?

A. 2500 B. 3000 C. 4687 D. 8437

gatecivil-2022-set1 quantitative-aptitude profit-loss

Answer key☟

9.51.4 Profit Loss: GATE2013 CE: GA-9

A firm is selling its product at Rs. 60 per unit. The total cost of production is Rs. 100 and firm is earning total profit of Rs. 500. Later, the total cost increased by 30%. By what percentage the price should be increased to maintained the same profit level.

A. 5 B. 10 C. 15 D. 30

quantitative-aptitude gate2013-ce profit-loss

Answer key☟

9.51.5 Profit Loss: GATE2018 CE-1: GA-6

A fruit seller sold a basket of fruits at 12.5% loss. Had he sold it for Rs. 108 more, he would have made a 10% gain. What is the loss in Rupees incurred by the fruit seller?

A. 48 B. 52 C. 60 D. 108

gate2018-ce-1 general-aptitude quantitative-aptitude profit-loss

Answer key☟

9.51.6 Profit Loss: GATE2019 ME-1: GA-7

A person divided an amount of Rs. $100,000$ into two parts and invested in two different schemes. In one he got 10% profit and in the other he got 12%. If the profit percentages are interchanged with these investments he would have got Rs. 120 less. Find the ratio between his investments in the two schemes.

A. $9:16$ B. $11:14$ C. $37:63$ D. $47:53$

gate2019-me-1 general-aptitude quantitative-aptitude ratio-proportion profit-loss

Answer key☟

9.52 Quadratic Equations (11)

9.52.1 Quadratic Equations: GATE CSE 2014 Set 1 | Question: GA-5

The roots of $ax^2+bx+c=0$ are real and positive. a, b and c are real. Then $ax^2+b\mid x\mid+c=0$ has

A. no roots B. 2 real roots C. 3 real roots D. 4 real roots

gatecse-2014-set1 quantitative-aptitude quadratic-equations normal

Answer key☟

9.52.2 Quadratic Equations: GATE CSE 2016 Set 2 | Question: GA-05

In a quadratic function, the value of the product of the roots (α, β) is 4. Find the value of

$$\frac{\alpha^n+\beta^n}{\alpha^{-n}+\beta^{-n}}$$

A. n^4 B. 4^n C. 2^{2n-1} D. 4^{n-1}

gatecse-2016-set2 quantitative-aptitude quadratic-equations normal

Answer key☟

9.52.3 Quadratic Equations: GATE CSE 2021 Set 2 | GA Question: 4

If $\left(x-\frac{1}{2}\right)^2-\left(x-\frac{3}{2}\right)^2=x+2$, then the value of x is:

A. 2 B. 4 C. 6 D. 8

gatecse-2021-set2 quantitative-aptitude quadratic-equations one-mark

Answer key☟

9.52.4 Quadratic Equations: GATE CSE 2022 | GA Question: 3

Let r be a root of the equation $x^2+2x+6=0$.
Then the value of the expression $(r+2)(r+3)(r+4)(r+5)$ is

A. 51 B. -51 C. 126 D. -126

gatecse-2022 quantitative-aptitude quadratic-equations one-mark

Answer key

9.52.5 Quadratic Equations: GATE ECE 2020 | GA Question: 9

a, b, c are real numbers. The quadratic equation $ax^2 - bx + c = 0$ has equal roots, which is β, then

A. $\beta = b/a$
B. $\beta^2 = ac$
C. $\beta^3 = bc/\left(2a^2\right)$
D. $\beta^2 \neq 4ac$

gate2020-ec quantitative-aptitude quadratic-equations

Answer key

9.52.6 Quadratic Equations: GATE2011 MN: GA-62

A student attempted to solve a quadratic equation in x twice. However, in the first attempt, he incorrectly wrote the constant term and ended up with the roots as $(4, 3)$. In the second attempt, he incorrectly wrote down the coefficient of x and got the roots as $(3, 2)$. Based on the above information, the roots of the correct quadratic equation are

A. $(-3, 4)$
B. $(3, -4)$
C. $(6, 1)$
D. $(4, 2)$

gate2011-mn quadratic-equations quantitative-aptitude

Answer key

9.52.7 Quadratic Equations: GATE2013 EE: GA-8

The set of values of p for which the roots of the equation $3x^2 + 2x + p(p-1) = 0$ are of opposite sign is

A. $(-\infty, 0)$
B. $(0, 1)$
C. $(1, \infty)$
D. $(0, \infty)$

gate2013-ee quantitative-aptitude quadratic-equations

Answer key

9.52.8 Quadratic Equations: GATE2015 EC-2: GA-9

if $a^2 + b^2 + c^2 = 1$ then $ab + bc + ac$ lies in the interval

A. $[1, 2/3]$
B. $[-1/2, 1]$
C. $[-1, 1/2]$
D. $[2, -4]$

gate2015-ec-2 numerical-answers quantitative-aptitude quadratic-equations

Answer key

9.52.9 Quadratic Equations: GATE2016 EC-2: GA-4

Given $(9 \text{ inches})^{\frac{1}{2}} = (0.25 \text{ yards})^{\frac{1}{2}}$, which one of the following statements is TRUE?

A. 3 inches = 0.5 yards
B. 9 inches = 1.5 yards
C. 9 inches = 0.25 yards
D. 81 inches = 0.0625 yards

gate2016-ec-2 quantitative-aptitude quadratic-equations

Answer key

9.52.10 Quadratic Equations: GATE2018 EE: GA-4

For what values of k given below is $\dfrac{(k+2)^2}{(k-3)}$ an integer?

A. $4, 8, 18$
B. $4, 10, 16$
C. $4, 8, 28$
D. $8, 26, 28$

gate2018-ee general-aptitude quantitative-aptitude easy quadratic-equations

Answer key

9.52.11 Quadratic Equations: GATE2018 ME-1: GA-7

Given that a and b are integers and $a + a^2b^3$ is odd, which of the following statements is correct?

A. a and b are both odd
B. a and b are both even
C. a is even and b is odd
D. a is odd and b is even

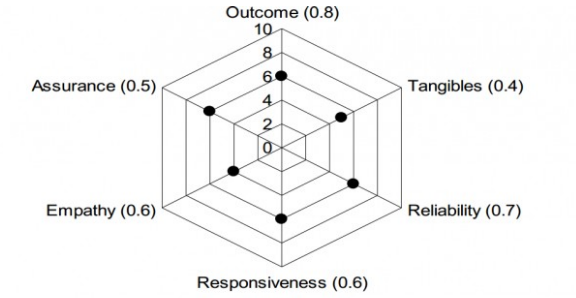
gate2018-me-1 general-aptitude quantitative-aptitude quadratic-equations system-of-equations

Answer key

9.53 Radar Chart (1)

9.53.1 Radar Chart: GATE2011 GG: GA-9

The quality of services delivered by a company consists of six factors as shown below in the radar diagram. The dots in the figure indicate the score for each factor on a scale of 0 to 10. The standardized coefficient for each factor is given in the parentheses. The contribution of each factor to the overall service quality is directly proportional to the factor score and its standardized coefficient.



The lowest contribution among all the above factors to the overall quality of services delivered by the company is

A. 10%
B. 20%
C. 24%
D. 40%

gate2011-gg difficult quantitative-aptitude data-interpretation radar-chart

Answer key

9.54 Ratio Proportion (26)

9.54.1 Ratio Proportion: GATE 2024 Electrical | GA Question: 5

If, for non-zero real variables x, y, and real parameter $a > 1$,

$$x : y = (a+1) : (a-1),$$

then, the ratio $\left(x^2 - y^2\right) : \left(x^2 + y^2\right)$ is

A. $2a : \left(a^2 + 1\right)$
B. $a : \left(a^2 + 1\right)$
C. $2a : \left(a^2 - 1\right)$
D. $a : \left(a^2 - 1\right)$

gate2024-ee quantitative-aptitude ratio-proportion

Answer key

9.54.2 Ratio Proportion: GATE CSE 2016 Set 1 | Question: GA10

In a process, the number of cycles to failure decreases exponentially with an increase in load. At a load of 80 units, it takes 100 cycles for failure. When the load is halved, it takes 10000 cycles for failure.The load for which the failure will happen in 5000 cycles is ____________.

A. 40.00
B. 46.02
C. 60.01
D. 92.02

gatecse-2016-set1 quantitative-aptitude ratio-proportion normal

Answer key

9.54.3 Ratio Proportion: GATE CSE 2018 | Question: GA-8

In a party, 60% of the invited guests are male and 40% are female.If 80% of the invited guests attended the party and if all the invited female guests attended, what would be the ratio of males to females among the attendees in the party?

A. $2:3$
B. $1:1$
C. $3:2$
D. $2:1$

gatecse-2018 quantitative-aptitude ratio-proportion two-marks

Answer key

9.54.4 Ratio Proportion: GATE CSE 2021 Set 1 | GA Question: 1

The ratio of boys to girls in a class is 7 to 3.

Among the options below, an acceptable value for the total number of students in the class is:

A. 21 B. 37 C. 50 D. 73

gatecse-2021-set1 quantitative-aptitude ratio-proportion one-mark

Answer key

9.54.5 Ratio Proportion: GATE CSE 2021 Set 2 | GA Question: 8

The number of students in three classes is in the ratio $3:13:6$. If 18 students are added to each class, the ratio changes to $15:35:21$.

The total number of students in all the three classes in the beginning was:

A. 22 B. 66 C. 88 D. 110

gatecse-2021-set2 ratio-proportion quantitative-aptitude two-marks

Answer key

9.54.6 Ratio Proportion: GATE CSE 2024 | Set 1 | GA Question: 2

If two distinct non-zero real variables x and y are such that $(x+y)$ is proportional to $(x-y)$ then the value of $\frac{x}{y}$

A. depends on xy
B. depends only on x and not on y
C. depends only on y and not on x
D. is a constant

gatecse2024-set1 quantitative-aptitude ratio-proportion one-mark

Answer key

9.54.7 Ratio Proportion: GATE Civil 2020 Set 2 | GA Question: 8

The ratio of 'the sum of the odd positive integers from 1 to 100' to 'the sum of the even positive integers from 150 to 200' is ________.

A. $45:95$ B. $1:2$ C. $50:91$ D. $1:1$

gate2020-ce-2 quantitative-aptitude ratio-proportion

Answer key

9.54.8 Ratio Proportion: GATE Civil 2022 Set 1 | GA Question: 3

If

$p:q=1:2$

$q:r=4:3$

$r:s=4:5$

and u is 50% more than s, what is the ratio $p:u$?

A. $2:15$ B. $16:15$ C. $1:5$ D. $16:45$

gatecivil-2022-set1 quantitative-aptitude ratio-proportion

Answer key

9.54.9 Ratio Proportion: GATE Civil 2022 Set 2 | GA Question: 2

$x:y:z=\frac{1}{2}:\frac{1}{3}:\frac{1}{4}$.

What is the value of $\frac{x+z-y}{y}$?

A. 0.75 B. 1.25 C. 2.25 D. 3.25

gatecivil-2022-set2 quantitative-aptitude ratio-proportion

Answer key

9.54.10 Ratio Proportion: GATE Civil 2022 Set 2 | GA Question: 7

In a partnership business, the monthly investment by three friends for the first six months is in the ratio $3:4:5$. After six months, they had to increase their monthly investments by $10\%, 15\%, \text{ and } 20\%$, respectively, of their initial monthly investment. The new investment ratio was kept constant for the next six months.

What is the ratio of their shares in the total profit (in the same order) at the end of the year such that the share is proportional to their individual total investment over the year?

A. $22:23:24$
B. $22:33:50$
C. $33:46:60$
D. $63:86:110$

gatecivil-2022-set2 quantitative-aptitude ratio-proportion

Answer key

9.54.11 Ratio Proportion: GATE Civil 2024 Set 2 | GA Question: 5

The ratio of the number of girls to boys in class VIII is the same as the ratio of the number of boys to girls in class IX. The total number of students (boys and girls) in classes VIII and IX is 450 and 360, respectively. If the number of girls in classes VIII and IX is the same, then the number of girls in each class is

A. 150
B. 200
C. 250
D. 175

gatecivil-2024-set2 quantitative-aptitude ratio-proportion

Answer key

9.54.12 Ratio Proportion: GATE ECE 2022 | GA Question: 2

A sum of money is to be distributed among P, Q, R, and S in the proportion $5:2:4:3$, respectively.

If R gets ₹ 1000 more than S, what is the share of Q (in ₹)?

A. 500
B. 1000
C. 1500
D. 2000

gateece-2022 quantitative-aptitude ratio-proportion

Answer key

9.54.13 Ratio Proportion: GATE ECE 2024 | GA Question: 3

Five years ago, the ratio of Aman's age to his father's age was $1:4$, and five years from now, the ratio will be $2:5$. What was his father's age when Aman was born?

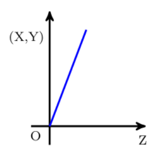
A. 28 years
B. 30 years
C. 35 years
D. 32 years

gateece-2024 quantitative-aptitude ratio-proportion

Answer key

9.54.14 Ratio Proportion: GATE Mechanical 2020 Set 2 | GA Question: 9

An engineer measures THREE quantities, X, Y and Z in an experiment. She finds that they follow a relationship that is represented in the figure below:(the product of X and Y linearly varies with Z)



Then, which of the following statements is FALSE?

A. For fixed Z; X is proportional to Y
B. For fixed Y; X is proportional to Z
C. For fixed X; Z is proportional to Y
D. XY/Z is constant

gateme-2020-set2 quantitative-aptitude ratio-proportion

Answer key

9.54.15 Ratio Proportion: GATE Mechanical 2021 Set 1 | GA Question: 8

The number of hens, ducks and goats in farm P are $65, 91$ and 169, respectively. The total number of hens, ducks and goats in a nearby farm Q is 416. The ratio of hens : ducks : goats in farm Q is $5:14:13$. All the hens, ducks and goats are sent from farm Q to farm P.

The new ratio of hens : ducks : goats in farm P is ________

A. $5:7:13$ B. $5:14:13$ C. $10:21:26$ D. $21:10:26$

gateme-2021-set1 quantitative-aptitude ratio-proportion

Answer key

9.54.16 Ratio Proportion: GATE Mechanical 2024 | GA Question: 4

The real variables x, y, z, and the real constants p, q, r satisfy

$$\frac{x}{pq-r^2} = \frac{y}{qr-p^2} = \frac{z}{rp-q^2}$$

Given that the denominators are non-zero, the value of $px+qy+rz$ is

A. 0 B. 1 C. pqr D. $p^2+q^2+r^2$

gateme-2024 quantitative-aptitude ratio-proportion

Answer key

9.54.17 Ratio Proportion: GATE2011 AG: GA-8

Three friends, R, S and T shared toffee from a bowl. R took $\frac{1}{3}^{\text{rd}}$ of the toffees, but returned four to the bowl. S took $\frac{1}{4}^{\text{th}}$ of what was left but returned three toffees to the bowl. T took half of the remainder but returned two back into the bowl. If the bowl had 17 toffees left, how may toffees were originally there in the bowl?

A. 38 B. 31 C. 48 D. 41

general-aptitude quantitative-aptitude gate2011-ag ratio-proportion

Answer key

9.54.18 Ratio Proportion: GATE2011 GG: GA-4

If m students require a total of m pages of stationery in m days, then 100 students will require 100 pages of stationery in

A. 100 days B. $m/100$ days C. $100/m$ days D. m days

gate2011-gg quantitative-aptitude ratio-proportion

Answer key

9.54.19 Ratio Proportion: GATE2013 AE: GA-1

If $3 \leq X \leq 5$ and $8 \leq Y \leq 11$ then which of the following options is TRUE?

A. $\left(\frac{3}{5} \leq \frac{X}{Y} \leq \frac{8}{5}\right)$

B. $\left(\frac{3}{11} \leq \frac{X}{Y} \leq \frac{5}{8}\right)$

C. $\left(\frac{3}{11} \leq \frac{X}{Y} \leq \frac{8}{5}\right)$

D. $\left(\frac{3}{5} \leq \frac{X}{Y} \leq \frac{8}{11}\right)$

gate2013-ae quantitative-aptitude ratio-proportion normal

Answer key

9.54.20 Ratio Proportion: GATE2014 AE: GA-8

The smallest angle of a triangle is equal to two thirds of the smallest angle of a quadrilateral. The ratio between the angles of the quadrilateral is $3:4:5:6$. The largest angle of the triangle is twice its smallest angle. What is the sum, in degrees, of the second largest angle of the triangle and the largest angle of the quadrilateral?

gate2014-ae quantitative-aptitude ratio-proportion numerical-answers

Answer key

9.54.21 Ratio Proportion: GATE2015 EC-1: GA-9

A cube of side 3 units is formed using a set of smaller cubes of side 1 unit. Find the proportion of the number of faces of the smaller cubes visible to those which are NOT visible.

A. $1:4$ B. $1:3$ C. $1:2$ D. $2:3$

gate2015-ec-1 general-aptitude quantitative-aptitude geometry ratio-proportion

Answer key☟

9.54.22 Ratio Proportion: GATE2017 CE-2: GA-4

What is the value of x when $81 \times \left(\frac{16}{25}\right)^{x+2} \div \left(\frac{3}{5}\right)^{2x+4} = 144$?

A. 1
B. -1
C. -2
D. Can not be determined

gate2017-ce-2 ratio-proportion quantitative-aptitude

Answer key☟

9.54.23 Ratio Proportion: GATE2018 CE-1: GA-7

The price of a wire made of a super alloy material is proportional to the square of its length. The price of $10m$ length of the wire is Rs. 1600. What would be the total price (in Rs.) of two wires of length $4m$ and $6m$?

A. 768 B. 832 C. 1440 D. 1600

gate2018-ce-1 general-aptitude quantitative-aptitude ratio-proportion

Answer key☟

9.54.24 Ratio Proportion: GATE2018 CE-2: GA-6

In manufacturing industries, loss is usually taken to be proportional to the square of the deviation from a target. If the loss is Rs. 4900 for a deviation of 7 units, what would be the loss in Rupees for a deviation of 4 units from the target?

A. 400 B. 1200 C. 1600 D. 2800

gate2018-ce-2 general-aptitude quantitative-aptitude ratio-proportion

Answer key☟

9.54.25 Ratio Proportion: GATE2018 EC: GA-7

Two alloys A and B contain gold and copper in the ratios of $2:3$ and $3:7$ by mass, respectively. Equal masses of alloys A and B are melted to make an alloy C. The ratio of gold to copper in alloy C is ______.

A. $5:10$ B. $7:13$ C. $6:11$ D. $9:13$

gate2018-ec general-aptitude quantitative-aptitude normal ratio-proportion

Answer key☟

9.54.26 Ratio Proportion: GATE2019 EE: GA-7

The ratio of the number of boys and girls who participated in an examination is $4:3$. The total percentage of candidates who passed the examination is 80 and the percentage of girls who passed the exam is 90. The percentage of boys who passed is _______.

A. 55.50 B. 72.50 C. 80.50 D. 90.00

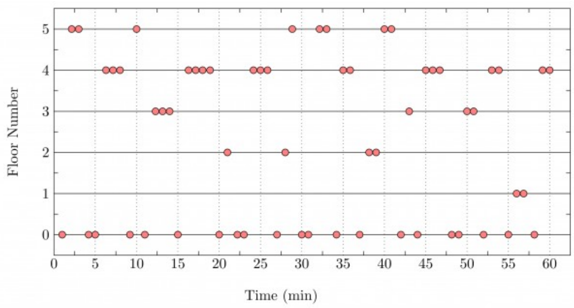
gate2019-ee general-aptitude quantitative-aptitude ratio-proportion percentage

Answer key☟

9.55 Scatter Plot (1)

9.55.1 Scatter Plot: GATE2017 CE-2: GA-10

The points in the graph below represent the halts of a lift for a durations of 1 minute, over a period of 1 hour.



Which of the following statements are correct?

i. The elevator moves directly from any non-ground floor to another non-ground floor over the one hour period.
ii. The elevator stays on the fourth floor for the longest duration over the one hour period.

A. Only i
B. Only ii
C. Both i and ii
D. Neither i nor ii

gate2017-ce-2 data-interpretation scatter-plot

Answer key

9.56 Seating Arrangement (4)

9.56.1 Seating Arrangement: GATE Civil 2021 Set 1 | GA Question: 5

Four persons $\text{P}, \text{Q}, \text{R}$ and S are to be seated in a row, all facing the same direction, but not necessarily in the same order. P and R cannot sit adjacent to each other. S should be seated to the right of Q. The number of distinct seating arrangements possible is:

A. 2
B. 4
C. 6
D. 8

gatecivil-2021-set1 quantitative-aptitude permutation-and-combination seating-arrangement

Answer key

9.56.2 Seating Arrangement: GATE Civil 2021 Set 2 | GA Question: 5

Four persons P, Q, R and S are to be seated in a row. R should not be seated at the second position from the left end of the row. The number of distinct seating arrangements possible is:

A. 6
B. 9
C. 18
D. 24

gatecivil-2021-set2 quantitative-aptitude permutation-and-combination seating-arrangement

Answer key

9.56.3 Seating Arrangement: GATE Mechanical 2021 Set 1 | GA Question: 10

Five persons $\text{P}, \text{Q}, \text{R}, \text{S}$ and T are sitting in a row not necessarily in the same order. Q and R are separated by one person, and S should not be seated adjacent to Q.

The number of distinct seating arrangements possible is:

A. 4
B. 8
C. 10
D. 16

gateme-2021-set1 quantitative-aptitude permutation-and-combination seating-arrangement

Answer key

9.56.4 Seating Arrangement: GATE Mechanical 2021 Set 2 | GA Question: 1

Five persons $\text{P}, \text{Q}, \text{R}, \text{S}$ and T are to be seated in a row, all facing the same direction, but not necessarily in the same order. P and T cannot be seated at either end of the row. P should not be seated adjacent to S. R is to be seated at the second position from the left end of the row. The number of distinct seating arrangements possible is:

A. 2
B. 3
C. 4
D. 5

gateme-2021-set2 quantitative-aptitude permutation-and-combination seating-arrangement

Answer key

9.57 Sequence Series (10)

9.57.1 Sequence Series: GATE CSE 2012 | Question: 65

Given the sequence of terms, AD CG FK JP, the next term is

A. OV B. OW C. PV D. PW

gatecse-2012 quantitative-aptitude sequence-series easy

Answer key☟

9.57.2 Sequence Series: GATE CSE 2018 | Question: GA-5

What is the missing number in the following sequence?

$$2,\ 12,\ 60,\ 240,\ 720,\ 1440,\ \_\_\_\_,0$$

A. 2880 B. 1440 C. 720 D. 0

gatecse-2018 quantitative-aptitude sequence-series easy one-mark

Answer key☟

9.57.3 Sequence Series: GATE CSE 2023 | GA Question: 3

A series of natural numbers $F_1, F_2, F_3, F_4, F_5, F_6, F_7, \dots$ obeys $F_{n+1} = F_n + F_{n-1}$ for all integers $n \geq 2$. If $F_6 = 37$, and $F_7 = 60$, then what is F_1?

A. 4 B. 5 C. 8 D. 9

gatecse-2023 quantitative-aptitude sequence-series one-mark

Answer key☟

9.57.4 Sequence Series: GATE DS&AI 2024 | GA Question: 4

The sum of the following infinite series is

$$2+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{8}+\frac{1}{9}+\frac{1}{16}+\frac{1}{27}+\cdots$$

A. $11/3$ B. $7/2$ C. $13/4$ D. $9/2$

gate-ds-ai-2024 quantitative-aptitude sequence-series one-mark

Answer key☟

9.57.5 Sequence Series: GATE Mechanical 2024 | GA Question: 3

In the following series, identify the number that needs to be changed to form the Fibonacci series.

$1, 1, 2, 3, 6, 8, 13, 21, \dots$

A. 8 B. 21 C. 6 D. 13

gateme-2024 quantitative-aptitude sequence-series

Answer key☟

9.57.6 Sequence Series: GATE2014 EC-4: GA-5

In a sequence of 12 consecutive odd numbers, the sum of the first 5 numbers is 425. What is the sum of the last 5 numbers in the sequence?

gate2014-ec-4 quantitative-aptitude sequence-series normal numerical-answers

Answer key☟

9.57.7 Sequence Series: GATE2014 EC-4: GA-6

Find the next term in the sequence: 13M, 17Q, 19S,________.

A. 21W B. 21V C. 23W D. 23V

gate2014-ec-4 quantitative-aptitude sequence-series normal

Answer key

9.57.8 Sequence Series: GATE2016 ME-2: GA-8

Find the missing sequence in the letter series. B, FH, LNP, _______.

A. $SUWY$ B. $TUVW$ C. $TVXZ$ D. $TWXZ$

gate2016-me-2 sequence-series quantitative-aptitude

Answer key

9.57.9 Sequence Series: GATE2019 EE: GA-3

The missing number in the given sequence $343, 1331$, _____, 4913 is

A. 3375 B. 2744 C. 2197 D. 4096

gate2019-ee general-aptitude quantitative-aptitude sequence-series

Answer key

9.57.10 Sequence Series: GATE2019 ME-2: GA-3

If IMHO=JNIP; IDK=JEL; and SO=TP, then IDC=______

A. JDE B. JED C. JDC D. JCD

gate2019-me-2 general-aptitude quantitative-aptitude sequence-series easy

Answer key

9.58 Set Theory (1)

9.58.1 Set Theory: GATE Mechanical 2024 | GA Question: 7

How many combinations of non-null sets $\mathrm{A}, \mathrm{B}, \mathrm{C}$ are possible from the subsets of $\{2, 3, 5\}$ satisfying the conditions: (i) A is a subset of B, and (ii) B is a subset of C ?

A. 28 B. 27 C. 18 D. 19

gateme-2024 quantitative-aptitude set-theory

Answer key

9.59 Speed Time Distance (19)

9.59.1 Speed Time Distance: GATE CSE 2013 | Question: 64

A tourist covers half of his journey by train at 60 km/h, half of the remainder by bus at 30 km/h and the rest by cycle at 10 km/h. The average speed of the tourist in km/h during his entire journey is

A. 36 B. 30 C. 24 D. 18

gatecse-2013 quantitative-aptitude easy speed-time-distance

Answer key

9.59.2 Speed Time Distance: GATE CSE 2019 | Question: GA-3

Two cars at the same time from the same location and go in the same direction. The speed of the first car is 50 km/h and the speed of the second car is 60 km/h. The number of hours it takes for the distance between the two cars to be 20 km is _____.

A. 1 B. 2 C. 3 D. 6

gatecse-2019 general-aptitude quantitative-aptitude speed-time-distance one-mark

Answer key

9.59.3 Speed Time Distance: GATE Chemical 2020 | GA Question: 8

The distance between Delhi and Agra is 233 km. A car P started travelling from Delhi to Agra and another car Q started from Agra to Delhi along the same road 1 hour after the car P started. The two cars crossed each other 75 minutes after the car Q started. Both cars were travelling at constant speed. The speed of car P was 10 km/hr more than the speed of car Q. How many kilometers the car Q had travelled when the cars crossed each other?

A. 66.6 B. 75.2 C. 88.2 D. 116.5

gate2020-ch quantitative-aptitude speed-time-distance

Answer key☟

9.59.4 Speed Time Distance: GATE Civil 2023 Set 2 | GA Question: 10

Consider a spherical globe rotating about an axis passing through its poles. There are three points P, Q, and R situated respectively on the equator, the north pole, and midway between the equator and the north pole in the northern hemisphere. Let P, Q, and R move with speeds v_P, v_Q, and v_R, respectively.

Which one of the following options is CORRECT?

A. $v_P < v_R < v_Q$
B. $v_P < v_Q < v_R$
C. $v_P > v_R > v_Q$
D. $v_P = v_R \neq v_Q$

gatecivil-2023-set2 quantitative-aptitude speed-time-distance

Answer key☟

9.59.5 Speed Time Distance: GATE Electrical 2022 | GA Question: 2

In a $500\, m$ race, P and Q have speeds in the ratio of $3:4$. Q starts the race when P has already covered $140\, m$.

What is the distance between P and Q (in m) when P wins the race?

A. 20 B. 40 C. 60 D. 140

gate2022-ee quantitative-aptitude speed-time-distance

Answer key☟

9.59.6 Speed Time Distance: GATE Mechanical 2022 Set 1 | GA Question: 3

A person travelled $80\,\text{km}$ in 6 hours. If the person travelled the first part with a uniform speed of $10\,\text{kmph}$ and the remaining part with a uniform speed of $18\,\text{kmph}$.

What percentage of the total distance is travelled at a uniform speed of $10\,\text{kmph}$?

A. 28.25 B. 37.25 C. 43.75 D. 50.00

gateme-2022-set1 quantitative-aptitude speed-time-distance

Answer key☟

9.59.7 Speed Time Distance: GATE2013 AE: GA-6

Velocity of an object fired directly in upward direction is given by $V = 80 - 32t,$ where t (time) is in seconds. When will the velocity be between $32\ m/sec$ and $64\ m/sec$?

A. $\left(1, \frac{3}{2}\right)$
B. $\left(\frac{1}{2}, 1\right)$
C. $\left(\frac{1}{2}, \frac{3}{2}\right)$
D. $(1,3)$

gate2013-ae quantitative-aptitude speed-time-distance

Answer key☟

9.59.8 Speed Time Distance: GATE2013 EE: GA-9

A car travels $8\ km$ in the first quarter of an hour, $6\ km$ in the second quarter and $16\ km$ in the third quarter. The average speed of the car in km per hour over the entire journey is

A. 30 B. 36 C. 40 D. 24

gate2013-ee speed-time-distance quantitative-aptitude

Answer key☟

9.59.9 Speed Time Distance: GATE2014 EC-1: GA-8

A train that is 280 metres long, travelling at a uniform speed, crosses a platform in 60 seconds and passes a man standing on the platform in 20 seconds. What is the length of the platform in metres?

gate2014-ec-1 quantitative-aptitude speed-time-distance normal numerical-answers

Answer key☟

9.59.10 Speed Time Distance: GATE2014 EC-2: GA-10

It takes 30 minutes to empty a half-full tank by draining it at a constant rate. It is decided to simultaneously pump water into the half-full tank while draining it. What is the rate at which water has to be pumped in so that it gets fully filled in 10 minutes?

A. 4 times the draining rate
B. 3 times the draining rate
C. 2.5 times the draining rate
D. 2 times the draining rate

gate2014-ec-2 quantitative-aptitude speed-time-distance normal

Answer key☟

9.59.11 Speed Time Distance: GATE2014 EC-3: GA-8

A man can row at 8 km per hour in still water. If it takes him thrice as long to row upstream, as to row downstream, then find the stream velocity in km per hour.

gate2014-ec-3 quantitative-aptitude speed-time-distance normal numerical-answers

Answer key☟

9.59.12 Speed Time Distance: GATE2015 EC-2: GA- 8

A tiger is 50 leaps of its own behind a deer. The tiger takes 5 leaps per minute to the deer's $4.$ If the tiger and the deer cover 8 meter and 5 meter per leap respectively, what distance in meters will the tiger have to run before it catches the deer$?$

gate2015-ec-2 quantitative-aptitude numerical-answers speed-time-distance

Answer key☟

9.59.13 Speed Time Distance: GATE2016 EC-3: GA-5

It takes 10 s and 15 s, respectively, for two trains travelling at different constant speeds to completely pass a telegraph post. The length of the first train is 120 m and that of the second train is 150 m. The magnitude of the difference in the speeds of the two trains (in m/s) is ____________.

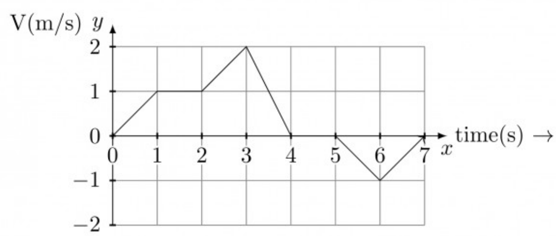
A. 2.0
B. 10.0
C. 12.0
D. 22.0

gate2016-ec-3 speed-time-distance quantitative-aptitude

Answer key☟

9.59.14 Speed Time Distance: GATE2016 EC-3: GA-6

The velocity V of a vehicle along a straight line is measured in m/s and plotted as shown with respect to time in seconds. At the end of the 7 seconds, how much will the odometer reading increase by (in m)?



A. 0
B. 3
C. 4
D. 5

gate2016-ec-3 quantitative-aptitude speed-time-distance data-interpretation

Answer key☟

9.59.15 Speed Time Distance: GATE2017 CE-2: GA-9

Budhan covers a distance of 19 km in 2 hours by cycling one fourth of the time and walking the rest. The next day he cycles (at the same speed as before) for half the time and walks the rest (at the same speed as before) and covers 26 km in 2 hours. The speed in km/h at which Budhan walk is

A. 1 B. 4 C. 5 D. 6

gate2017-ce-2 speed-time-distance quantitative-aptitude

Answer key☟

9.59.16 Speed Time Distance: GATE2017 EC-1: GA-8

Trucks (10 m long) and cars (5 m long) go on a single lane bridge. There must be a gap of at least 20 m after each truck and a gap of at least 15 m after each car. Trucks and cars travel at a speed of 36 km/h. If cars and trucks go alternately, what is the maximum number of vehicles that can use the bridge in one hour?

A. 1440 B. 1200 C. 720 D. 600

gate2017-ec-1 general-aptitude quantitative-aptitude speed-time-distance

Answer key☟

9.59.17 Speed Time Distance: GATE2018 CH: GA-6

An automobile travels from city A to city B and returns to city A by the same route. The speed of the vehicle during the onward and return journeys were constant at $60 km/h$ and $90 km/h$, respectively. What is the average speed in km/h for the entire journey?

A. $72 \; km/h$
B. $73 \; km/h$
C. $74 \; km/h$
D. $75 \; km/h$

gate2018-ch general-aptitude quantitative-aptitude normal speed-time-distance

Answer key☟

9.59.18 Speed Time Distance: GATE2018 ME-1: GA-8

From the time the front of a train enters a platform, it takes 25 seconds for the back of the train to leave the platform, while traveling at a constant speed of 54 km/h. At the same speed, it takes 14 seconds to pass a man running at 9 km/h in the same direction as the train. What is the length of the train and that of the platform in meters, respectively?

A. 210 and 140 B. 162.5 and 187.5 C. 245 and 130 D. 175 and 200

gate2018-me-1 general-aptitude quantitative-aptitude speed-time-distance

Answer key☟

9.59.19 Speed Time Distance: GATE2019 IN: GA-9

Two trains started at 7AM from the same point. The first train travelled north at a speed of 80km/h and the second train travelled south at a speed of 100km/h. The time at which they were 540 km apart is ________ AM.

A. 9 B. 10 C. 11 D. 11.30

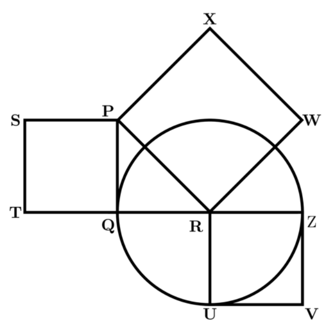
gate2019-in general-aptitude quantitative-aptitude speed-time-distance

Answer key☟

9.60 Squares (4)

9.60.1 Squares: GATE Civil 2022 Set 1 | GA Question: 5

In the following diagram, the point R is the center of the circle. The lines PQ and ZV are tangential to the circle. The relation among the areas of the squares, PXWR, RUVZ and SPQT is

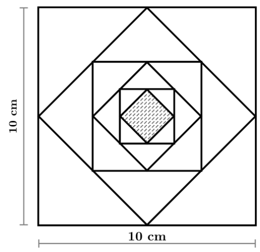


A. Area of SPQT = Area of RUVZ – Area of PXWR
B. Area of SPQT = Area of PXWR – Area of RUVZ
C. Area of PXWR = Area of SPQT – Area of RUVZ
D. Area of PXWR = Area of RUVZ – Area of SPQT

gatecivil-2022-set1 quantitative-aptitude geometry squares area

Answer key

9.60.2 Squares: GATE Electrical 2021 | GA Question: 7



In the figure shown above, each inside square is formed by joining the midpoints of the sides of the next larger square. The area of the smallest square (shaded) as shown, in cm^2 is:

A. 12.50 B. 6.25 C. 3.125 D. 1.5625

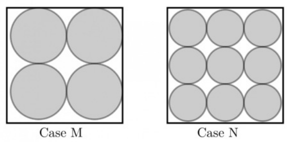
gateee-2021 quantitative-aptitude geometry squares area

Answer key

9.60.3 Squares: GATE Mechanical 2022 Set 2 | GA Question: 10

Equal sized circular regions are shaded in a square sheet of paper of 1 cm side length. Two cases, case M and case N, are considered as shown in the figures below. In the case M, four circles are shaded in the square sheet and in the case N, nine circles are shaded in the square sheet as shown.

What is the ratio of the areas of unshaded regions of case M to that of case N?



A. $2:3$ B. $1:1$ C. $3:2$ D. $2:1$

gateme-2022-set2 quantitative-aptitude geometry circle squares area

Answer key

9.60.4 Squares: GATE2019 CE-1: GA-9

A square has side 5 cm smaller than the sides of a second square. The area of the larger square is four times the area of the smaller square. The side of the larger square is _______ cm.

A. 18.50 B. 15.10 C. 10.00 D. 8.50

gate2019-ce-1 general-aptitude quantitative-aptitude geometry squares

Answer key

9.61 Statistics (9)

9.61.1 Statistics: GATE CSE 2012 | Question: 64

Which of the following assertions are **CORRECT**?

- P: Adding 7 to each entry in a list adds 7 to the mean of the list

- Q: Adding 7 to each entry in a list adds 7 to the standard deviation of the list
- R: Doubling each entry in a list doubles the mean of the list
- S: Doubling each entry in a list leaves the standard deviation of the list unchanged

A. P, Q B. Q, R C. P, R D. R, S

gatecse-2012 quantitative-aptitude statistics normal

Answer key

9.61.2 Statistics: GATE CSE 2024 | Set 1 | GA Question: 3

Consider the following sample of numbers:

$9, 18, 11, 14, 15, 17, 10, 69, 11, 13$

The median of the sample is

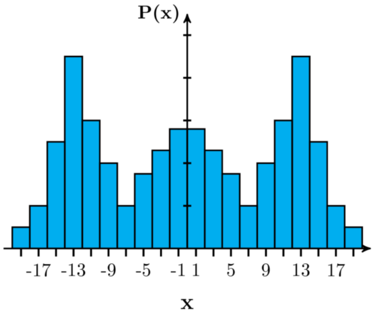
A. 13.5 B. 14 C. 11 D. 18.7

gatecse2024-set1 quantitative-aptitude statistics one-mark

Answer key

9.61.3 Statistics: GATE Civil 2023 Set 1 | GA Question: 7

Which one of the options can be inferred about the mean, median, and mode for the given probability distribution (i.e. probability mass function), $P(x)$, of a variable x?



A. mean $=$ median $\neq$ mode
B. mean $=$ median $=$ mode
C. mean $\neq$ median $=$ mode
D. mean $\neq$ mode $=$ median

gatecivil-2023-set1 quantitative-aptitude statistics

Answer key

9.61.4 Statistics: GATE IN 2023 | GA Question: 4

Which one among the following statements must be TRUE about the mean and the median of the scores of all candidates appearing for GATE 2023?

A. The median is at least as large as the mean.
B. The mean is at least as large as the median.
C. At most half the candidates have a score that is larger than the median
D. At most half the candidates have a score that is larger than the mean.

gatein-2023 quantitative-aptitude statistics

Answer key

9.61.5 Statistics: GATE2012 AE: GA-5

The arithmetic mean of five different natural numbers is 12. The largest possible value among the numbers is

A. 12 B. 40 C. 50 D. 60

gate2012-ae statistics quantitative-aptitude

Answer key☟

9.61.6 Statistics: GATE2014 EC-1: GA-4

The statistics of runs scored in a series by four batsmen are provided in the following table. Who is the most consistent batsman of these four?

Batsman	Average	Standard deviation
K	31.2	5.21
L	46.0	6.35
M	54.4	6.22
N	17.9	5.90

A. K B. L C. M D. N

gate2014-ec-1 statistics quantitative-aptitude

Answer key☟

9.61.7 Statistics: GATE2016 ME-2: GA-6

Students taking an exam are divided into two groups, **P** and **Q** such that each group has the same number of students. The performance of each of the students in a test was evaluated out of 200 marks. It was observed that the mean of group **P** was 105, while that of group **Q** was 85. The standard deviation of group **P** was 25, while that of group **Q** was 5. Assuming that the marks were distributed on a normal distribution, which of the following statements will have the highest probability of being **TRUE**?

A. No student in group **Q** scored less marks than any student in group **P**.
B. No student in group **P** scored less marks than any student in group **Q**.
C. Most students of group **Q** scored marks in a narrower range than students in group **P**.
D. The median of the marks of group **P** is 100.

gate2016-me-2 probability statistics

Answer key☟

9.61.8 Statistics: GATE2017 CE-1: GA-5

The following sequence of numbers is arranged in increasing order: $1, x, x, x, y, y, 9, 16, 18$. Given that the mean and median are equal, and are also equal to twice the mode, the value of y is

A. 5 B. 6 C. 7 D. 8

gate2017-ce-1 general-aptitude quantitative-aptitude statistics

Answer key☟

9.61.9 Statistics: GATE2017 ME-1: GA-4

In a company with 100 employees, 45 earn $Rs. 20,000$ per month, 25 earn $Rs. 30000$, 20 earn $Rs. 40000$, 8 earn $Rs. 60000$, and 2 earn $Rs. 150,000$. The median of the salaries is

A. $Rs. 20,000$ B. $Rs. 30,000$ C. $Rs. 32,300$ D. $Rs. 40,000$

gate2017-me-1 general-aptitude quantitative-aptitude statistics

Answer key☟

9.62 System of Equations (1)

9.62.1 System of Equations: GATE2011 GG: GA-6

The number of solutions for the following system of inequalities is

- $X_1 \geq 0$
- $X_2 \geq 0$
- $X_1 + X_2 \leq 10$
- $2X_1 + 2X_2 \geq 22$

A. 0 B. infinite C. 1 D. 2

gate2011-gg quantitative-aptitude system-of-equations

Answer key

9.63 Tables (1)

9.63.1 Tables: GATE CSE 2025 | Set 2 | GA Question: 8

Which one of the following options is correct for the given data in the table?

Iteration (i)	0	1	2	3
Input (I)	20	-4	10	15
Output (X)	20	16	26	41
Output (Y)	20	-80	-800	-12000

A. $X(i) = X(i-1) + I(i);\ Y(i) = Y(i-1)I(i);\ i > 0$
B. $X(i) = X(i-1)I(i);\ Y(i) = Y(i-1) + I(i);\ i > 0$
C. $X(i) = X(i-1)I(i);\ Y(i) = Y(i-1)I(i);\ i > 0$
D. $X(i) = X(i-1) + I(i);\ Y(i) = Y(i-1)I(i-1);\ i > 0$

gatecse2025-set2 quantitative-aptitude tables two-marks

Answer key

9.64 Tabular Data (10)

9.64.1 Tabular Data: GATE CSE 2014 Set 1 | Question: GA-9

In a survey, 300 respondents were asked whether they own a vehicle or not. If yes, they were further asked to mention whether they own a car or scooter or both. Their responses are tabulated below. What percent of respondents do not own a scooter?

		Men	Women
Own vehicle	Car	40	34
Own vehicle	Scooter	30	20
Own vehicle	Both	60	46
Do not own vehicle		20	50

gatecse-2014-set1 quantitative-aptitude normal numerical-answers data-interpretation tabular-data

Answer key

9.64.2 Tabular Data: GATE CSE 2014 Set 3 | Question: GA-5

The table below has question-wise data on the performance of students in an examination. The marks for each question are also listed. There is no negative or partial marking in the examination.

Q No.	Marks	Answered Correctly	Answered Wrongly	Not Attempted
1	2	21	17	6
2	3	15	27	2
3	2	23	18	3

What is the average of the marks obtained by the class in the examination?

A. 1.34 B. 1.74 C. 3.02 D. 3.91

gatecse-2014-set3 quantitative-aptitude normal data-interpretation tabular-data

Answer key

9.64.3 Tabular Data: GATE CSE 2015 Set 1 | Question: GA-6

The number of students in a class who have answered correctly, wrongly, or not attempted each question in an exam, are listed in the table below. The marks for each question are also listed. There is no negative or partial marking.

Q No.	Marks	Answered Correctly	Answered Wrongly	Not Attempted
1	2	21	17	6
2	3	15	27	2
3	1	11	29	4
4	2	23	18	3
5	5	31	12	1

What is the average of the marks obtained by the class in the examination?

A. 2.290 B. 2.970 C. 6.795 D. 8.795

gatecse-2015-set1 quantitative-aptitude easy data-interpretation tabular-data

Answer key☟

9.64.4 Tabular Data: GATE CSE 2016 Set 1 | Question: GA06

A shaving set company sells 4 different types of razors- Elegance, Smooth, Soft and Executive.

Elegance sells at $\text{Rs.}\ 48$, Smooth at $\text{Rs.}\ 63$, Soft at $\text{Rs.}\ 78$ and Executive at $\text{Rs.}\ 173$ per piece. The table below shows the numbers of each razor sold in each quarter of a year.

Quarter/Product	Elegance	Smooth	Soft	Executive
Q1	27300	20009	17602	9999
Q2	25222	19392	18445	8942
Q3	28976	22429	19544	10234
Q4	21012	18229	16595	10109

Which product contributes the greatest fraction to the revenue of the company in that year?

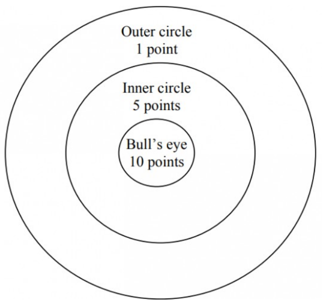
A. Elegance B. Executive C. Smooth D. Soft

gatecse-2016-set1 quantitative-aptitude data-interpretation easy tabular-data

Answer key☟

9.64.5 Tabular Data: GATE2011 MN: GA-64

Four archers P, Q, R, and S try to hit a bull's eye during a tournament consisting of seven rounds. As illustrated in the figure below, a player receives 10 points for hitting the bull's eye, 5 points for hitting within the inner circle and 1 point for hitting within the outer circle.



The final scores received by the players during the tournament are listed in the table below.

Round	P	Q	R	S
1	1	5	1	10
2	5	10	10	1
3	1	1	1	5
4	10	10	1	1
5	1	5	5	10
6	10	5	1	1
7	5	10	1	1

The most accurate and the most consistent players during the tournament are respectively

A. P and S
B. Q and R
C. Q and Q
D. R and Q

gate2011-mn data-interpretation quantitative-aptitude tabular-data

Answer key

9.64.6 Tabular Data: GATE2013 AE: GA-7

Following table gives data on tourist from different countries visiting India in the year 2011

Country	Number of tourists
USA	2000
England	3500
Germany	1200
Italy	1100
Japan	2400
Australia	2300
France	1000

Which two countries contributed to the one third of the total number of tourists who visited India in 2011?

A. USA and Japan
B. USA and Australia
C. England and France
D. Japan and Australia

gate2013-ae quantitative-aptitude data-interpretation normal tabular-data

Answer key

9.64.7 Tabular Data: GATE2013 CE: GA-8

Following table provides figures(in rupees) on annual expenditure of a firm for two years -2010 and 2011.

Category	2010	2011
Raw material	5200	6240
Power & fuel	7000	9450
Salary & wages	9000	12600
Plant & machinery	20000	25000
Advertising	15000	19500
Research & Development	22000	26400

In 2011, which of the two categories have registered increase by same percentage?

A. Raw material and Salary & wages.
B. Salary & wages and Advertising.
C. Power & fuel and Advertising.
D. Raw material and research & Development.

quantitative-aptitude gate2013-ce data-interpretation normal tabular-data

Answer key

9.64.8 Tabular Data: GATE2015 CE-2: GA-9

Read the following table giving sales data of five types of batteries for years 2006 to 2012:

Year	Type I	Type II	Type III	Type IV	Type V
2006	75	144	114	102	108
2007	90	126	102	84	126
2008	96	114	75	105	135
2009	105	90	150	90	75
2010	90	75	135	75	90
2011	105	60	165	45	120
2012	115	85	160	100	145

Out of the following , which type of battery achieved highest growth between the years 2006 and 2012?

A. Type V B. Type III C. Type II D. Type I

gate2015-ce-2 general-aptitude quantitative-aptitude data-interpretation tabular-data

Answer key☟

9.64.9 Tabular Data: GATE2015 EC-2: GA-4

An electric bus has onboard instruments that report the total electricity consumed since the start of the trip, as well as the total distance, covered. During a single day of operation, the bus travels on stretches M, N, O, and P, in that order. The cumulative distances travelled and the corresponding electricity consumption are shown in the Table below:

Stretch	Cumulative distance (km)	Electricity used (kWh)
M	20	12
N	45	25
O	75	45
P	100	57

The stretch where the electricity consumption per km is minimum is

A. M B. N C. O D. P

gate2015-ec-2 quantitative-aptitude data-interpretation tabular-data

Answer key☟

9.64.10 Tabular Data: GATE2019 ME-2: GA-9

Mola is a digital platform for taxis in a city. It offers three types of rides – Pool, Mini and Prime. The table below presents the number of rides for the past four months. The platform earns one US dollar per ride. What is the percentage share of the revenue contributed by Prime to the total revenues of Mola, for the entire duration?

Type	Month			
	January	February	March	April
Pool	170	320	215	190
Mini	110	220	180	70
Prime	75	180	120	90

A. 16.24 B. 23.97 C. 25.86 D. 38.74

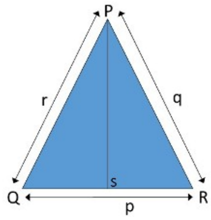
gate2019-me-2 general-aptitude quantitative-aptitude data-interpretation tabular-data

Answer key☟

9.65 Triangles (11)

9.65.1 Triangles: GATE CSE 2015 Set 2 | Question: GA-8

In a triangle PQR, PS is the angle bisector of $\angle QPR$ and $\angle QPS = 60^\circ$. What is the length of PS ?



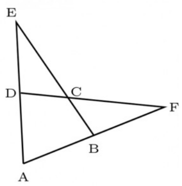
A. $\left(\frac{(q+r)}{qr}\right)$
B. $\left(\frac{qr}{q+r}\right)$
C. $\sqrt{(q^2+r^2)}$
D. $\left(\frac{(q+r)^2}{qr}\right)$

gatecse-2015-set2 quantitative-aptitude geometry difficult triangles

Answer key☟

9.65.2 Triangles: GATE CSE 2018 | Question: GA-9

In the figure below, $\angle DEC + \angle BFC$ is equal to _____



A. $\angle BCD - \angle BAD$
B. $\angle BAD + \angle BCF$
C. $\angle BAD + \angle BCD$
D. $\angle CBA + \angle ADC$

gatecse-2018 quantitative-aptitude geometry normal triangles two-marks

Answer key☟

9.65.3 Triangles: GATE CSE 2022 | GA Question: 9

The corners and mid-points of the sides of a triangle are named using the distinct letters P, Q, R, S, T and U, but not necessarily in the same order. Consider the following statements:

- The line joining P and R is parallel to the line joining Q and S.
- P is placed on the side opposite to the corner T.
- S and U cannot be placed on the same side.

Which one of the following statements is correct based on the above information?

A. P cannot be placed at a corner
B. S cannot be placed at a corner
C. U cannot be placed at a mid-point
D. R cannot be placed at a corner

gatecse-2022 quantitative-aptitude geometry triangles two-marks

Answer key☟

9.65.4 Triangles: GATE Civil 2021 Set 2 | GA Question: 10

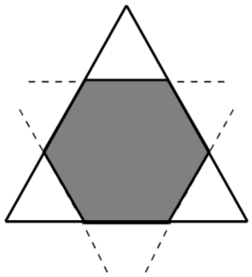
In an equilateral triangle PQR, side PQ is divided into four equal parts, side QR is divided into six equal parts and side PR is divided into eight equals parts. The length of each subdivided part in cm is an integer. The minimum area of the triangle PQR possible, in cm^2, is

A. 18
B. 24
C. $48\sqrt{3}$
D. $144\sqrt{3}$

gatecivil-2021-set2 quantitative-aptitude geometry triangles

Answer key☟

9.65.5 Triangles: GATE ECE 2021 | GA Question: 10



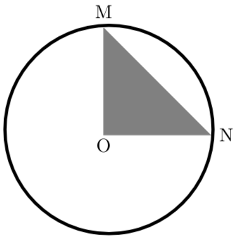
Corners are cut from an equilateral triangle to produce a regular convex hexagon as shown in the figure above. The ratio of the area of the regular convex hexagon to the area of the original equilateral triangle is

A. $2:3$ B. $3:4$ C. $4:5$ D. $5:6$

gateec-2021 quantitative-aptitude geometry triangles area

Answer key

9.65.6 Triangles: GATE Mechanical 2021 Set 1 | GA Question: 3



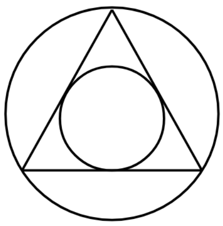
In the above figure, O is the center of the circle and, M and N lie on the circle. The area of the right triangle MON is 50 cm^2. What is the area of the circle in cm^2?

A. 2π B. 50π C. 75π D. 100π

gateme-2021-set1 quantitative-aptitude geometry triangles circle area

Answer key

9.65.7 Triangles: GATE Mechanical 2021 Set 2 | GA Question: 8



The ratio of the area of the inscribed circle to the area of the circumscribed circle of an equilateral triangle is _________

A. $\frac{1}{8}$
B. $\frac{1}{6}$
C. $\frac{1}{4}$
D. $\frac{1}{2}$

gateme-2021-set2 quantitative-aptitude geometry triangles

Answer key

9.65.8 Triangles: GATE Mechanical 2022 Set 1 | GA Question: 8

An equilateral triangle, a square and a circle have equal areas.
What is the ratio of the perimeters of the equilateral triangle to square to circle?

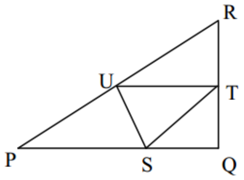
A. $3\sqrt{3} : 2 : \sqrt{\pi}$
B. $\sqrt{(3\sqrt{3})} : 2 : \sqrt{\pi}$
C. $\sqrt{(3\sqrt{3})} : 4 : 2\sqrt{\pi}$
D. $\sqrt{(3\sqrt{3})} : 2 : 2\sqrt{\pi}$

gateme-2022-set1 quantitative-aptitude geometry triangles

Answer key

9.65.9 Triangles: GATE2015 ME-3: GA-8

In the given figure angle Q is a right angle, $PS : QS = 3 : 1, RT : QT = 5 : 2$ and $PU : UR = 1 : 1$. If area of triangle QTS is $20cm^2$, then the area of triangle PQR in cm^2 is ______



gate2015-me-3 quantitative-aptitude numerical-answers triangles

Answer key

9.65.10 Triangles: GATE2015 ME-3: GA-9

Right triangle PQR is to be constructed in the xy - plane so that the right angle is at P and line PR is parallel to the x-axis. The x and y coordinates of $P, Q,$ and R are to be integers that satisfy the inequalities: $-4 \leq x \leq 5$ and $6 \leq y \leq 16$. How many different triangles could be constructed with these properties?

A. 110
B. $1,100$
C. $9,900$
D. $10,000$

gate2015-me-3 quantitative-aptitude triangles

Answer key

9.65.11 Triangles: GATE2018 CH: GA-4

The area of an equilateral triangle is $\sqrt{3}$. What is the perimeter of the triangle?

A. 2
B. 4
C. 6
D. 8

gate2018-ch general-aptitude quantitative-aptitude easy geometry triangles

Answer key

9.66 Trigonometry (1)

9.66.1 Trigonometry: GATE2018 CH: GA-3

For $0 \leq x \leq 2\pi$, $\sin x$ and $\cos x$ are both decreasing functions in the interval _________ .

A. $\left(0, \frac{\pi}{2}\right)$
B. $\left(\frac{\pi}{2}, \pi\right)$
C. $\left(\pi, \frac{3\pi}{2}\right)$
D. $\left(\frac{3\pi}{2}, 2\pi\right)$

gate2018-ch quantitative-aptitude functions trigonometry

Answer key

9.67 Unit Digit (2)

9.67.1 Unit Digit: GATE Civil 2020 Set 1 | GA Question: 9

The unit's place in $26591749^{\ 110016}$ is ______.

A. 1
B. 3
C. 6
D. 9

gate2020-ce-1 quantitative-aptitude number-system unit-digit

Answer key

9.67.2 Unit Digit: GATE2017 CE-1: GA-8

The last digit of $(2171)^7 + (2172)^9 + (2173)^{11} + (2174)^{13}$ is

A. 2 B. 4 C. 6 D. 8

gate2017-ce-1 modular-arithmetic quantitative-aptitude unit-digit

Answer key

9.68 Venn Diagram (16)

9.68.1 Venn Diagram: GATE CSE 2010 | Question: 59

25 persons are in a room. 15 of them play hockey, 17 of them play football and 10 of them play both hockey and football. Then the number of persons playing neither hockey nor football is:

A. 2 B. 17 C. 13 D. 3

gatecse-2010 quantitative-aptitude easy set-theory&algebra venn-diagram

Answer key

9.68.2 Venn Diagram: GATE CSE 2016 Set 2 | Question: GA-06

Among 150 faculty members in an institute, 55 are connected with each other through Facebook and 85 are connected through Whatsapp. 30 faculty members do not have Facebook or Whatsapp accounts. The numbers of faculty members connected only through Facebook accounts is _______.

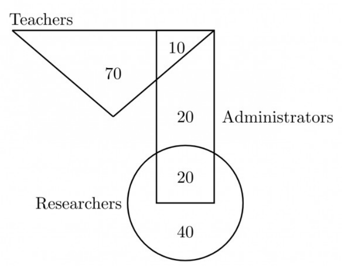
A. 35 B. 45 C. 65 D. 90

gatecse-2016-set2 quantitative-aptitude venn-diagram easy

Answer key

9.68.3 Venn Diagram: GATE CSE 2019 | Question: GA-7

In the given diagram, teachers are represented in the triangle, researchers in the circle and administrators in the rectangle. Out of the total number of the people, the percentage of administrators shall be in the range of _______



A. 0 to 15 B. 16 to 30 C. 31 to 45 D. 46 to 60

gatecse-2019 general-aptitude quantitative-aptitude venn-diagram two-marks

Answer key

9.68.4 Venn Diagram: GATE CSE 2019 | Question: GA-9

In a college, there are three student clubs, 60 students are only in the Drama club, 80 students are only in the Dance club, 30 students are only in Maths club, 40 students are in both Drama and Dance clubs, 12 students are in both Dance and Maths clubs, 7 students are in both Drama and Maths clubs, and 2 students are in all clubs. If 75% of the students in the college are not in any of these clubs, then the total number of students in the college is _____.

A. 1000 B. 975 C. 900 D. 225

gatecse-2019 general-aptitude quantitative-aptitude venn-diagram two-marks

Answer key

9.68.5 Venn Diagram: GATE CSE 2024 | Set 2 | GA Question: 3

In an engineering college of $10,000$ students, $1,500$ like neither their core branches nor other branches. The number of students who like their core branches is $1/4^{\text{th}}$ of the number of students who like other branches. The number of students who like both their core and other branches is 500.

The number of students who like their core branches is

A. $1,800$ B. $3,500$ C. $1,600$ D. $1,500$

gatecse2024-set2 quantitative-aptitude venn-diagram one-mark

Answer key

9.68.6 Venn Diagram: GATE Civil 2020 Set 2 | GA Question: 9

In a school of 1000 students, 300 students play chess and 600 students play football. If 50 students play both chess and football, the number of students who play neither is ________.

A. 200 B. 150 C. 100 D. 50

gate2020-ce-2 quantitative-aptitude venn-diagram

Answer key

9.68.7 Venn Diagram: GATE Civil 2021 Set 1 | GA Question: 3

In a company, 35% of the employees drink coffee, 40% of the employees drink tea and 10% of the employees drink both tea and coffee. What $\%$ of employees drink neither tea nor coffee?

A. 15 B. 25 C. 35 D. 40

gatecivil-2021-set1 quantitative-aptitude venn-diagram

Answer key

9.68.8 Venn Diagram: GATE Civil 2022 Set 2 | GA Question: 4

A survey of 450 students about their subjects of interest resulted in the following outcome.

- 150 students are interested in Mathematics.
- 200 students are interested in Physics.
- 175 students are interested in Chemistry.
- 50 students are interested in Mathematics and Physics.
- 60 students are interested in Physics and Chemistry.
- 40 students are interested in Mathematics and Chemistry.
- 30 students are interested in Mathematics, Physics, and Chemistry.
- Remaining students are interested in Humanities.

Based on the above information, the number of students interested in Humanities is

A. 10 B. 30 C. 40 D. 45

gatecivil-2022-set2 quantitative-aptitude venn-diagram

Answer key

9.68.9 Venn Diagram: GATE ECE 2023 | GA Question: 4

In a class of 100 students,

i. there are 30 students who neither like romantic movies nor comedy movies,
ii. the number of students who like romantic movies is twice the number of students who like comedy movies, and
iii. the number of students who like both romantic movies and comedy movies is 20.

How many students in the class like romantic movies?

A. 40 B. 20 C. 60 D. 30

gateece-2023 quantitative-aptitude venn-diagram

Answer key

9.68.10 Venn Diagram: GATE2010 TF: GA-8

A gathering of 50 linguists discovered that 4 knew Kannada, Telugu and Tamil , 7 knew only Telugu and Tamil , 5 knew only Kannada and Tamil , 6 knew only Telugu and Kannada. If the number of linguists who knew Tamil is 24 and those who knew Kannada is also 24, how many linguists knew only Telugu?

A. 9 B. 10 C. 11 D. 8

general-aptitude quantitative-aptitude gate2010-tf venn-diagram

Answer key

9.68.11 Venn Diagram: GATE2011 GG: GA-7

In a class of 300 students in an M.Tech programme, each student is required to take at least one subject from the following three:

- M600: Advanced Engineering Mathematics
- C600: Computational Methods for Engineers
- E600: Experimental Techniques for Engineers

The registration data for the M.Tech class shows that 100 students have taken M600, 200 students have taken C600, and 60 students have taken E600. What is the maximum possible number of students in the class who have taken all the above three subjects?

A. 20 B. 30 C. 40 D. 50

gate2011-gg quantitative-aptitude set-theory&algebra venn-diagram

Answer key

9.68.12 Venn Diagram: GATE2015 CE-2: GA-10

There are 16 teachers who can teach Thermodynamics (TD), 11 who can teach Electrical Sciences (ES), and 5 who can teach both TD and Engineering Mechanics (EM). There are a total of 40 teachers. 6 cannot teach any of the three subjects, i.e. EM, ES or TD. 6 can teach only ES. 4 can teach all three subjects, i.e. EM, ES and TD. 4 can teach ES and TD. How many can teach both ES and EM but not TD?

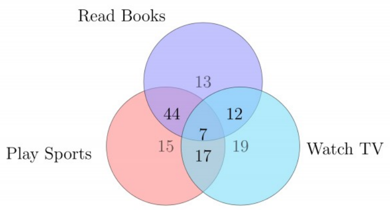
A. 1 B. 2 C. 3 D. 4

gate2015-ce-2 quantitative-aptitude venn-diagram

Answer key

9.68.13 Venn Diagram: GATE2016 EC-2: GA-6

The Venn diagram shows the preference of the student population for leisure activities.



From the data given, the number of students who like to read books or play sports is _______.

A. 44 B. 51 C. 79 D. 108

gate2016-ec-2 venn-diagram

Answer key

9.68.14 Venn Diagram: GATE2017 EC-2: GA-5

500 students are taking one or more course out of Chemistry, Physics, and Mathematics. Registration records indicate course enrollment as follows: Chemistry (329), Physics (186), and Mathematics (295). Chemistry and Physics (83), Chemistry and Mathematics (217), and Physics and Mathematics (63). How many students are taking all 3 subjects?

A. 37 B. 43 C. 47 D. 53

gate2017-ec-2 general-aptitude quantitative-aptitude venn-diagram

Answer key

9.68.15 Venn Diagram: GATE2018 ME-2: GA-6

Forty students watched films A, B and C over a week. Each student watched either only one film or all three. Thirteen students watched film A, sixteen students watched film B and nineteen students watched film C. How many students watched all three films?

A. 0　B. 2　C. 4　D. 8

gate2018-me-2 general-aptitude quantitative-aptitude venn-diagram

Answer key

9.68.16 Venn Diagram: GATE2019 CE-1: GA-7

In a sports academy of 300 peoples, 105 play only cricket, 70 play only hockey, 50 play only football, 25 play both cricket and hockey, 15 play both hockey and football and 30 play both cricket and football. The rest of them play all three sports. What is the percentage of people who play at least two sports?

A. 23.30　B. 25.00　C. 28.00　D. 50.00

gate2019-ce-1 general-aptitude quantitative-aptitude venn-diagram easy

Answer key

9.69 Volume (2)

9.69.1 Volume: GATE CH 2022 | GA Question: 2

A sphere of radius r cm is packed in a box of cubical shape.

What should be the minimum volume (in cm^3) of the box that can enclose the sphere?

A. $\frac{r^3}{8}$　B. r^3　C. $2r^3$　D. $8r^3$

gatech-2022 quantitative-aptitude mensuration volume

Answer key

9.69.2 Volume: GATE CSE 2021 Set 1 | GA Question: 6

We have 2 rectangular sheets of paper, M and N, of dimensions $6 \text{ cm} \times 1 \text{ cm}$ each. Sheet M is rolled to form an open cylinder by bringing the short edges of the sheet together. Sheet N is cut into equal square patches and assembled to form the largest possible closed cube. Assuming the ends of the cylinder are closed, the ratio of the volume of the cylinder to that of the cube is _________.

A. $\frac{\pi}{2}$　B. $\frac{3}{\pi}$　C. $\frac{9}{\pi}$　D. 3π

gatecse-2021-set1 quantitative-aptitude mensuration volume two-marks

Answer key

9.70 Work Time (17)

9.70.1 Work Time: GATE CH 2022 | GA Question: 3

Pipes P and Q can fill a storage tank in full with water in 10 and 6 minutes, respectively. Pipe R draws the water out from the storage tank at a rate of 34 litres per minute. P, Q and R operate at a constant rate.

If it takes one hour to completely empty a full storage tank with all the pipes operating simultaneously, what is the capacity of the storage tank (in litres)?

A. 26.8　B. 60.0　C. 120.0　D. 127.5

gatech-2022 quantitative-aptitude work-time

Answer key

9.70.2 Work Time: GATE CSE 2010 | Question: 64

5 skilled workers can build a wall in 20 days; 8 semi-skilled workers can build a wall in 25 days; 10 unskilled workers can build a wall in 30 days. If a team has 2 skilled, 6 semi-skilled and 5 unskilled workers, how long it will take to build the wall?

A. 20 days B. 18 days C. 16 days D. 15 days

gatecse-2010 quantitative-aptitude normal work-time

Answer key

9.70.3 Work Time: GATE CSE 2011 | Question: 64

A transporter receives the same number of orders each day. Currently, he has some pending orders (backlog) to be shipped. If he uses 7 trucks, then at the end of the 4^{th} day he can clear all the orders. Alternatively, if he uses only 3 trucks, then all the orders are cleared at the end of the 10^{th} day. What is the minimum number of trucks required so that there will be no pending order at the end of 5^{th} day?

A. 4 B. 5 C. 6 D. 7

gatecse-2011 quantitative-aptitude normal work-time

Answer key

9.70.4 Work Time: GATE CSE 2013 | Question: 65

The current erection cost of a structure is Rs. $13,200$. If the labour wages per day increase by $1/5$ of the current wages and the working hours decrease by $1/24$ of the current period, then the new cost of erection in Rs. is

A. $16,500$ B. $15,180$ C. $11,000$ D. $10,120$

gatecse-2013 quantitative-aptitude normal work-time

Answer key

9.70.5 Work Time: GATE Mechanical 2020 Set 2 | GA Question: 8

IT was estimated that 52 men can complete a strip in a newly constructed highway connecting cities P and Q in 10 days, Due to an emergency, 12 men were sent to another project. How many number of days, more than the original estimate, will be required to complete the strip?

A. 3 days B. 5 days C. 10 days D. 13 days

gateme-2020-set2 quantitative-aptitude work-time

Answer key

9.70.6 Work Time: GATE2016 CE-2: GA-10

Ananth takes 6 hours and Bharath takes 4 hours to read a book. Both started reading copies of the book at the same time. After how many hours is the number of pages to be read by Ananth, twice that to be read by Bharath? Assume Ananth and Bharath read all the pages with constant pace.

A. 1 B. 2 C. 3 D. 4

gate2016-ce-2 work-time quantitative-aptitude

Answer key

9.70.7 Work Time: GATE2016 EC-1: GA-10

P, Q, R and **S** are working on a project. **Q** can finish the task in 25 days, working alone for 12 hours a day. **R** can finish the task in 50 days, working alone for 12 hours per day. **Q** worked 12 hours a day but took sick leave in the beginning for two days. **R** worked 18 hours a day on all days. What is the ratio of work done by **Q** and **R** after 7 days from the start of the project?

A. $10:11$ B. $11:10$ C. $20:21$ D. $21:20$

gate2016-ec-1 quantitative-aptitude work-time

Answer key

9.70.8 Work Time: GATE2016 EC-2: GA-5

S, M, E and **F** are working in shifts in a team to finish a project. **M** works with twice the efficiency of others but for half as many days as **E** worked. **S** and **M** have 6 hour shifts in a day, whereas **E** and **F** have 12 hours shifts. What is the ratio of contribution of **M** to contribution of **E** in the project?

A. $1:1$ B. $1:2$ C. $1:4$ D. $2:1$

gate2016-ec-2 quantitative-aptitude work-time

Answer key☟

9.70.9 Work Time: GATE2017 CE-1: GA-9

Two machine $M1$ and $M2$ are able to execute any of four jobs P, Q, R and S. The machines can perform one job on one object at a time. Jobs P, Q, R and S take 30 minutes, 20 minutes, 60 minutes and 15 minutes each respectively. There are 10 objects each requiring exactly 1 job. Job P is to be performed on 2 objects. Job Q on 3 objects, Job R on 1 object and Job S on 4 objects. What is the minimum time needed to complete all the jobs?

A. 2 hours B. 2.5 hours C. 3 hours D. 3.5 hours

gate2017-ce-1 general-aptitude quantitative-aptitude work-time

Answer key☟

9.70.10 Work Time: GATE2017 EC-2: GA-8

1200 men and 500 women can build a bridge in 2 weeks. 900 men and 250 women will take 3 weeks to build the same bridge. How many men will be needed to build the bridge in one week?

A. 3000 B. 3300 C. 3600 D. 3900

gate2017-ec-2 general-aptitude quantitative-aptitude work-time

Answer key☟

9.70.11 Work Time: GATE2017 ME-2: GA-7

X bullocks and Y tractors take 8 days to plough a field. If we have half the number of bullocks and double the number of tractors, it takes 5 days to plough the same field. How many days will it take X bullocks alone to plough the field?

A. 30 B. 35 C. 40 D. 45

gate2017-me-2 general-aptitude quantitative-aptitude work-time

Answer key☟

9.70.12 Work Time: GATE2018 ME-1: GA-3

Seven machines take 7 minutes to make 7 identical toys. At the same rate, how many minutes would it take for 100 machine to make 100 toys?

A. 1 B. 7 C. 100 D. 700

gate2018-me-1 general-aptitude quantitative-aptitude work-time

Answer key☟

9.70.13 Work Time: GATE2018 ME-2: GA-8

A contract is to be completed in 52 days and 125 identical robots were employed, each operational for 7 hours a day. After 39 days, five-seventh of the work was completed. How many additional robots would be required to complete the work on time, if each robot is now operational for 8 hours a day?

A. 50 B. 89 C. 146 D. 175

gate2018-me-2 general-aptitude quantitative-aptitude work-time

Answer key☟

9.70.14 Work Time: GATE2019 CE-2: GA-9

An oil tank can be filled by pipe X in 5 hours and pipe Y in 4 hours, each pump working on its own. When the oil tank is full and the drainage hole is open, the oil is drained in 20 hours. If initially the tank was empty and someone started the two pipes together but left the drainage hole open, how many hours will it take for the tank to be filled? (Assume that the rate of drainage is independent of the Head)

A. 1.50 B. 2.00 C. 2.50 D. 4.00

gate2019-ce-2 general-aptitude quantitative-aptitude work-time

Answer key☟

9.70.15 Work Time: GATE2019 EC: GA-3

It would take one machine 4 hours to complete a production order and another machine 2 hours to complete the same order. If both machines work simultaneously at their respective constant rates, the time taken to complete the same order is ________ hours.

A. $2/3$
B. $3/4$
C. $4/3$
D. $7/3$

gate2019-ec general-aptitude quantitative-aptitude work-time

Answer key☟

9.70.16 Work Time: GATE2019 EE: GA-4

It takes two hours for a person X to mow the lawn. Y can mow the same lawn in four hours. How long (in minutes) will it take X and Y, if they work together to mow the lawn?

A. 60
B. 80
C. 90
D. 120

gate2019-ee general-aptitude quantitative-aptitude work-time

Answer key☟

9.70.17 Work Time: GATE2019 ME-2: GA-7

Two pipes P and Q can fill a tank in 6 hours and 9 hours respectively, while a third pipe R can empty the tank in 12 hours. Initially, P and R are open for 4 hours, Then P is closed and Q is opened. After 6 more hours R is closed. The total time taken to fill the tank (in hours) is ____

A. 13.50
B. 14.50
C. 15.50
D. 16.50

gate2019-me-2 general-aptitude quantitative-aptitude work-time

Answer key☟

Answer Keys

9.0.1	B	9.0.2	A	9.1.1	D	9.1.2	B	9.1.3	A
9.1.4	D	9.1.5	B	9.1.6	B	9.1.7	B	9.2.1	D
9.3.1	B	9.3.2	D	9.3.3	B	9.3.4	A	9.3.5	B
9.3.6	B	9.4.1	A	9.4.2	D	9.4.3	B	9.5.1	A
9.5.2	A	9.6.1	C	9.6.2	B	9.6.3	B	9.6.4	C
9.6.5	A	9.6.6	A	9.6.7	D	9.6.8	C	9.6.9	C
9.7.1	C	9.7.2	A	9.7.3	C	9.8.1	C	9.8.2	2006
9.8.3	C	9.8.4	A	9.8.5	B	9.8.6	B	9.8.7	A
9.8.8	A	9.8.9	B	9.8.10	C	9.8.11	C	9.8.12	C
9.8.13	D	9.8.14	C	9.8.15	C	9.8.16	D	9.8.17	C
9.8.18	A	9.9.1	C	9.9.2	A	9.10.1	B	9.10.2	C
9.10.3	A	9.10.4	C	9.10.5	C	9.10.6	C	9.10.7	C
9.10.8	A	9.10.9	B	9.10.10	C	9.11.1	D	9.11.2	A
9.11.3	C	9.11.4	A	9.11.5	C	9.11.6	A	9.11.7	B
9.12.1	A	9.12.2	C	9.12.3	B	9.12.4	A	9.12.5	X
9.12.6	A	9.12.7	D	9.12.8	B	9.12.9	B	9.12.10	D
9.13.1	B	9.13.2	C	9.13.3	D	9.13.4	C	9.13.5	D
9.13.6	B	9.13.7	4536	9.13.8	B	9.13.9	C	9.13.10	C

9.13.11	D	9.13.12	B	9.14.1	B	9.14.2	C	9.14.3	D
9.14.4	A	9.14.5	B	9.15.1	B	9.15.2	B	9.15.3	C
9.15.4	0.4895 : 0.4897	9.15.5	B	9.15.6	B	9.16.1	A	9.17.1	C
9.17.2	C	9.17.3	C	9.17.4	B	9.18.1	C	9.18.2	A
9.18.3	A	9.18.4	C	9.18.5	B	9.18.6	1300	9.19.1	A
9.19.2	C	9.19.3	A	9.19.4	A	9.19.5	D	9.19.6	C
9.20.1	D	9.20.2	B	9.21.1	A	9.22.1	B	9.22.2	A
9.23.1	D	9.23.2	B	9.23.3	B	9.23.4	C	9.23.5	C
9.24.1	C	9.25.1	D	9.25.2	C	9.25.3	C	9.25.4	B
9.26.1	B	9.26.2	C	9.26.3	A	9.26.4	C	9.27.1	8
9.27.2	C	9.27.3	B	9.27.4	C	9.27.5	B	9.27.6	C
9.27.7	B	9.27.8	A	9.27.9	A	9.27.10	B	9.27.11	A
9.27.12	D	9.27.13	A	9.27.14	A	9.27.15	C	9.27.16	A
9.27.17	C	9.27.18	A	9.28.1	A	9.28.2	X	9.28.3	B
9.28.4	6	9.28.5	A	9.28.6	D	9.28.7	A	9.28.8	B
9.28.9	B	9.28.10	A	9.28.11	A	9.28.12	B	9.28.13	D
9.28.14	A	9.28.15	2.06	9.28.16	D	9.28.17	B	9.28.18	B
9.28.19	D	9.28.20	22.2 : 22.3	9.28.21	D	9.28.22	C	9.28.23	B
9.28.24	B	9.28.25	A	9.28.26	C	9.28.27	C	9.28.28	A
9.28.29	C	9.29.1	A	9.30.1	A	9.30.2	C	9.30.3	C
9.30.4	A	9.31.1	A	9.32.1	140	9.32.2	C	9.32.3	C
9.32.4	C	9.32.5	B	9.32.6	B	9.32.7	D	9.32.8	A
9.32.9	B	9.32.10	B	9.32.11	C	9.32.12	A	9.33.1	D
9.34.1	B	9.34.2	C	9.34.3	A	9.34.4	C	9.34.5	A
9.34.6	C	9.34.7	C	9.34.8	D	9.34.9	A	9.34.10	C
9.34.11	A	9.34.12	B	9.34.13	A	9.34.14	C	9.35.1	C
9.36.1	B	9.36.2	C	9.36.3	C	9.36.4	A	9.37.1	C
9.38.1	A	9.38.2	D	9.39.1	B	9.39.2	C	9.39.3	C
9.39.4	B	9.39.5	D	9.39.6	725	9.39.7	45	9.39.8	16
9.39.9	C	9.39.10	D	9.40.1	D	9.40.2	C	9.40.3	A
9.40.4	D	9.40.5	B	9.40.6	A	9.40.7	B	9.41.1	C
9.41.2	D	9.41.3	D	9.41.4	X	9.41.5	A	9.41.6	D
9.42.1	96	9.42.2	B	9.42.3	D	9.42.4	B	9.42.5	B
9.42.6	A	9.42.7	D	9.42.8	163	9.42.9	A	9.42.10	D
9.42.11	D	9.42.12	D	9.42.13	C	9.42.14	X	9.43.1	850
9.43.2	D	9.43.3	C	9.43.4	D	9.43.5	D	9.43.6	A
9.43.7	B	9.43.8	D	9.43.9	C	9.43.10	D	9.43.11	B
9.43.12	D	9.43.13	A	9.43.14	A	9.43.15	C	9.43.16	D
9.43.17	C	9.43.18	D	9.43.19	C	9.43.20	A	9.43.21	A
9.44.1	A	9.44.2	B	9.44.3	A	9.44.4	B	9.45.1	32
9.45.2	C	9.45.3	D	9.45.4	C	9.45.5	X	9.45.6	B
9.45.7	D	9.45.8	C	9.45.9	A	9.45.10	D	9.45.11	D

9.45.12	D	9.45.13	20000	9.45.14	22	9.45.15	B	9.45.16	C
9.46.1	A	9.46.2	B	9.46.3	C	9.46.4	D	9.47.1	C
9.48.1	B	9.49.1	D	9.49.2	D	9.49.3	A	9.49.4	C
9.49.5	D	9.49.6	C	9.49.7	C	9.49.8	D	9.49.9	A
9.49.10	D	9.49.11	A	9.49.12	C	9.49.13	A	9.49.14	A
9.49.15	D	9.49.16	A	9.49.17	C	9.49.18	C	9.49.19	A
9.49.20	25	9.49.21	B	9.49.22	0.81	9.49.23	B	9.49.24	B
9.49.25	C	9.49.26	B	9.49.27	C	9.49.28	X	9.49.29	B
9.50.1	D	9.51.1	C	9.51.2	C	9.51.3	B	9.51.4	A
9.51.5	C	9.51.6	D	9.52.1	D	9.52.2	B	9.52.3	B
9.52.4	D	9.52.5	C	9.52.6	C	9.52.7	B	9.52.8	B
9.52.9	C	9.52.10	C	9.52.11	D	9.53.1	A	9.54.1	A
9.54.2	B	9.54.3	B	9.54.4	C	9.54.5	C	9.54.6	D
9.54.7	C	9.54.8	D	9.54.9	B	9.54.10	D	9.54.11	B
9.54.12	D	9.54.13	B	9.54.14	A	9.54.15	C	9.54.16	A
9.54.17	C	9.54.18	D	9.54.19	B	9.54.20	180	9.54.21	C
9.54.22	B	9.54.23	B	9.54.24	C	9.54.25	B	9.54.26	B
9.55.1	D	9.56.1	C	9.56.2	C	9.56.3	D	9.56.4	B
9.57.1	A	9.57.2	B	9.57.3	A	9.57.4	B	9.57.5	C
9.57.6	495	9.57.7	C	9.57.8	C	9.57.9	C	9.57.10	B
9.58.1	X	9.59.1	C	9.59.2	B	9.59.3	B	9.59.4	C
9.59.5	A	9.59.6	C	9.59.7	C	9.59.8	C	9.59.9	560
9.59.10	A	9.59.11	4	9.59.12	800	9.59.13	A	9.59.14	D
9.59.15	D	9.59.16	A	9.59.17	A	9.59.18	D	9.59.19	B
9.60.1	B	9.60.2	C	9.60.3	B	9.60.4	C	9.61.1	C
9.61.2	A	9.61.3	A	9.61.4	C	9.61.5	C	9.61.6	A
9.61.7	C	9.61.8	D	9.61.9	B	9.62.1	A	9.63.1	A
9.64.1	48	9.64.2	C	9.64.3	C	9.64.4	B	9.64.5	B
9.64.6	C	9.64.7	D	9.64.8	D	9.64.9	D	9.64.10	B
9.65.1	B	9.65.2	A	9.65.3	B	9.65.4	D	9.65.5	A
9.65.6	D	9.65.7	C	9.65.8	B	9.65.9	280	9.65.10	C
9.65.11	C	9.66.1	B	9.67.1	A	9.67.2	B	9.68.1	D
9.68.2	A	9.68.3	C	9.68.4	C	9.68.5	A	9.68.6	B
9.68.7	C	9.68.8	D	9.68.9	C	9.68.10	C	9.68.11	B
9.68.12	A	9.68.13	D	9.68.14	D	9.68.15	C	9.68.16	B
9.69.1	D	9.69.2	C	9.70.1	C	9.70.2	D	9.70.3	C
9.70.4	B	9.70.5	A	9.70.6	C	9.70.7	C	9.70.8	B
9.70.9	A	9.70.10	C	9.70.11	A	9.70.12	B	9.70.13	X
9.70.14	C	9.70.15	C	9.70.16	B	9.70.17	B		